Model Structure (26.2_FAA):

This is a Fleets−as−Areas model (do_spatial == 1).
There are 2 sexes, 2 fishing fleets, 5 survey fleets, and 1 regions being modeled.
The model has 67 years, starting in 1960 and ending in 2026.
There are 30 ages, where recruitment occurs at age−2 and the plus group occurs at age−31.
There are 30 length bins, which start at 41cm and end at 99cm.

Growth Estimation:

This model estimates new growth curves for the post−1996 time block (upd_growth == 1, use_new_bio == 2) and uses the 2021 SAFE growth estimates for the pre−1996 time block (use_hist_grwth_blk == 1, do_rec_growth == 0). A new weight curve is estimated with a single block (do_rec_WAA == 0).
For fishery composition data aggregation catch−weighting, the newly estimated sex−disaggregated Weight parameters are used to convert to catch in numbers (sex_disagg_comp_wt == 1).

Aleutian Islands (AI) Data Use:

This model uses data from the Aleutian Islands (AI) −− no data is being removed (drop_AI == 0).

Longline Survey Index Construction:

There was a longline survey in 2026, so there is a terminal year LLS RPN value (LLS_active == 1).

The extrapolated values for the western AI longline survey stations were NOT included in the LLS RPN index (drop_WAI_LLS == 1).

The longline survey did NOT use off−year extrapolations for the BS (even years) or AI (odd years).
Only the non−extrapolated area RPNs were included in the index (drop_LLS_extrap == 1 & do_LLS_carry == 0).
The FAA model is not using a carryover version of the LLS RPN index (do_spatial == 1 & do_carry_FAA == 0).

The SPoRC model is being passed the index SEs, no conversions needed (do_cont_SE == 1).

Composition Data Structure and ISS:

Composition proportions by age/length are aggregated across regions after weighting by catch in numbers or RPNs.

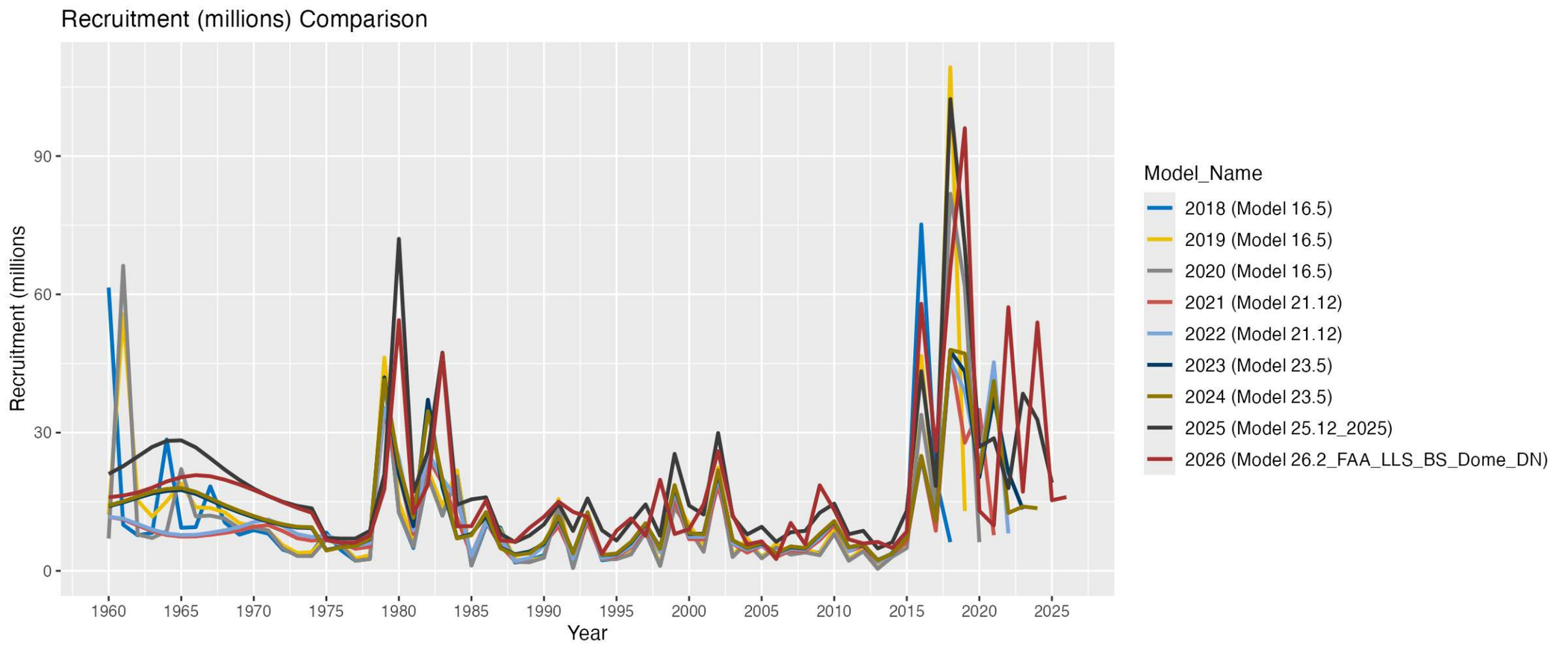
Age and length compositions are fit 'joint' by sex, but lengths are dropped for years when ages exist (JPN LLS lengths are kept for all years) (do_spatial > 0 & sex_age_comps == 5).
The Input Sample Size (ISS) for composition data is scaled to a maximum value across years/sexes/fleets to address interannual variability (do_spatial == 1 & do_scale_ISS == 1).

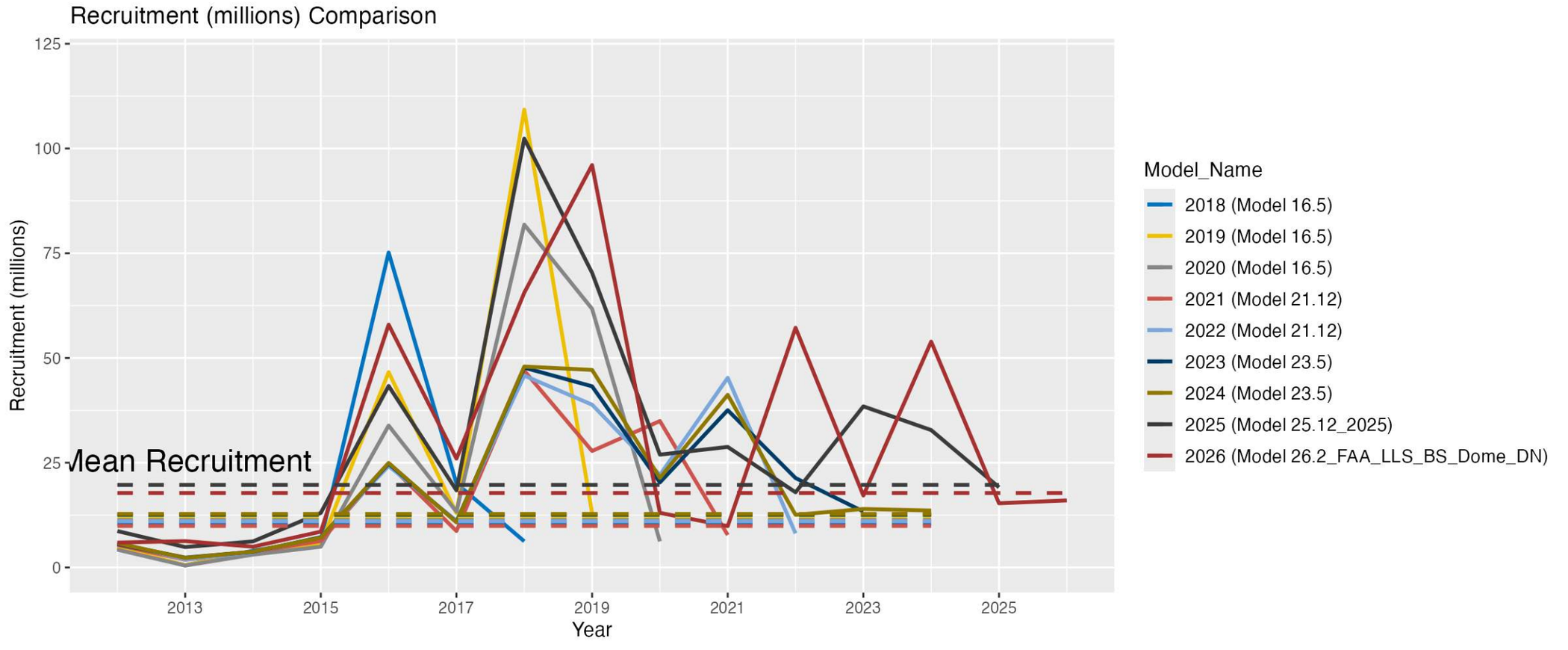
Length Composition Data Use:

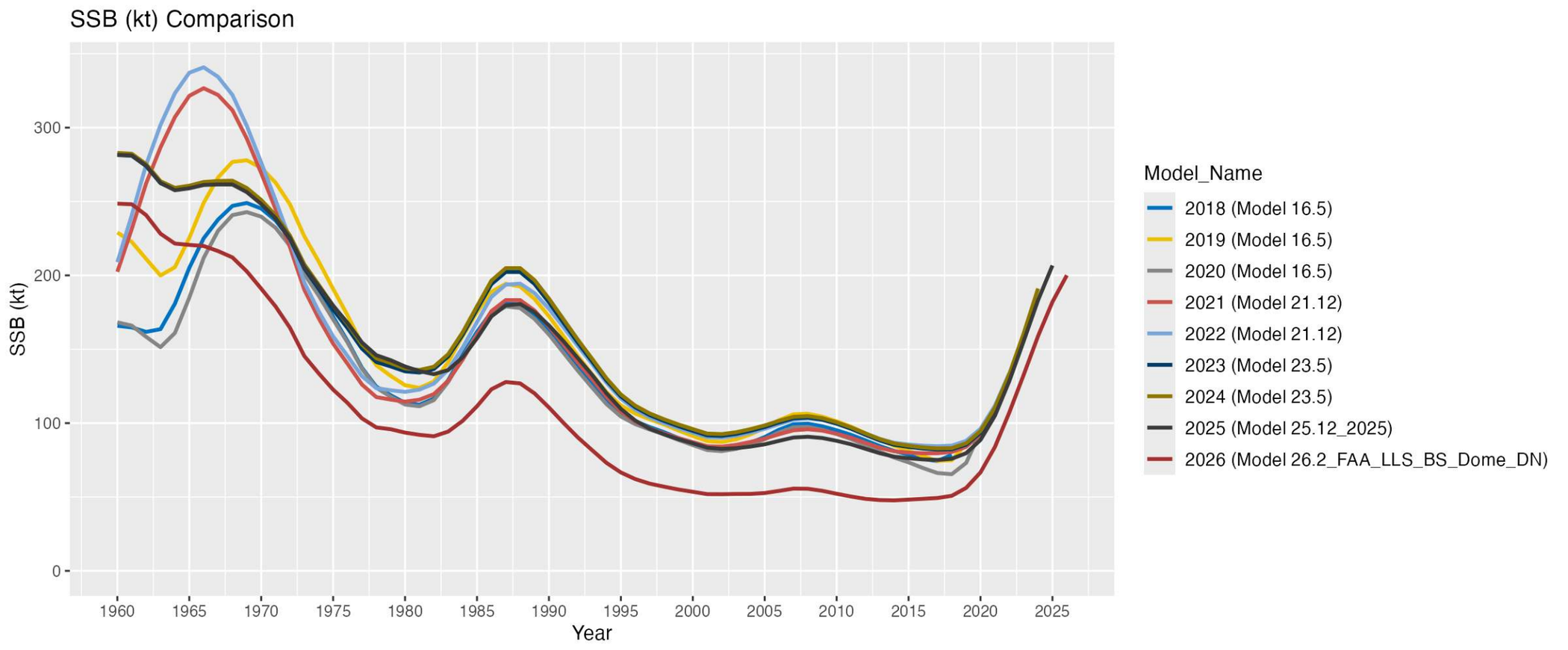
The longline survey length composition data is weighted/calculated using the raw length data and weighted by area RPNs consistent with the age composition data (LLS_length_comp_source == 1).
No longline survey length data are fit in the model (this does not apply to JPN LLS lengths) −− only age data are used (fit_LLS_len == 0).
No fixed gear fishery length data are fit in the model (this does not apply to trawl fishery lengths) −− only age data are used (fit_LLF_len == 0).

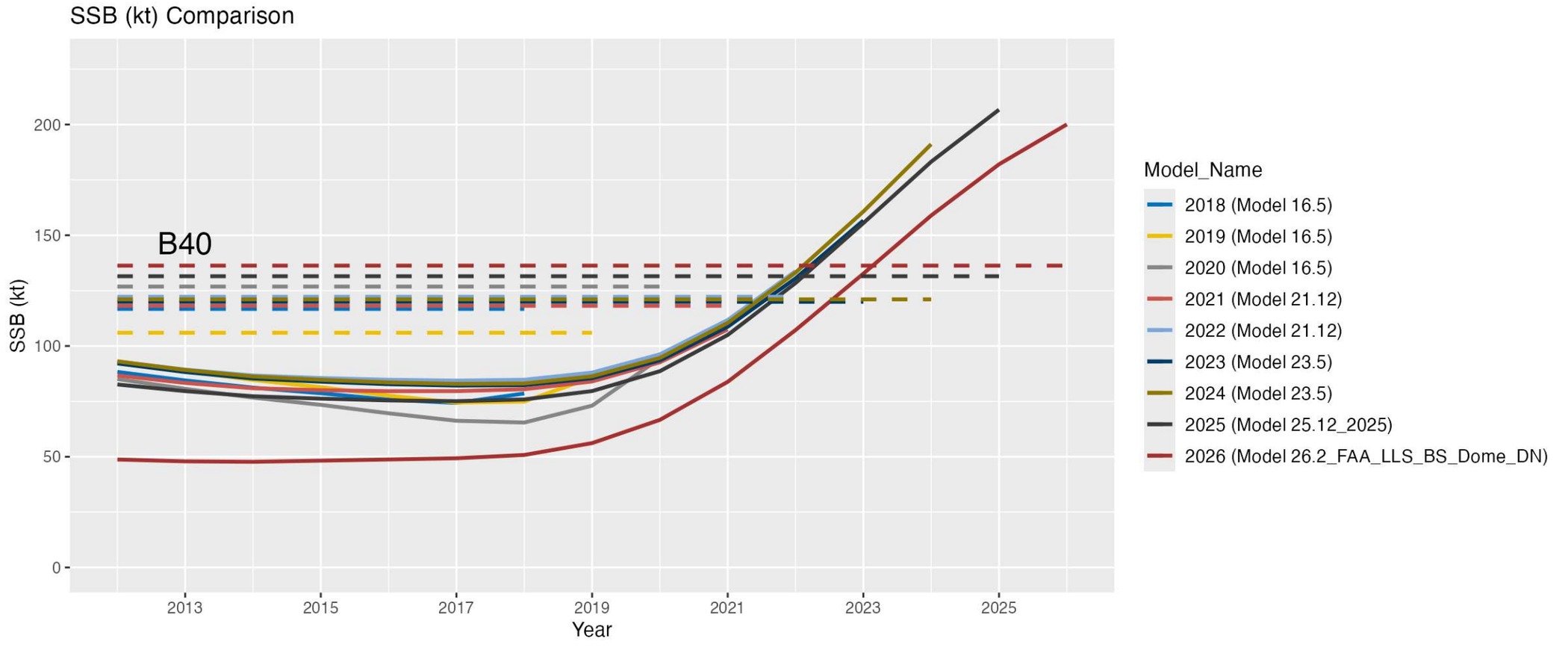
General Data Fitting Notes:

Fishery whale depredation estimates were fixed/static at the previous year's values (whale_dep_static == 1).
The GOA trawl survey data was NOT fit in this model (use_trwl_srvy == 0).
The fishery CPUE index was NOT fit in this model (use_CPUE == 0).
Non−fishery catch (e.g., scientific removals) WAS included in the model as fixed gear catch, where only LLS removals were available for terminal year (add_non_fish_catch == 1 & non_fsh_catch_updated == 0).


Recruitment (millions) Comparison
Recruitment (millions
0
30
60
90
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_LLS_BS_Dome_DN)


Recruitment (millions) Comparison
Recruitment (millions)
0
25
50
75
100
125
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_LLS_BS_Dome_DN)


SSB (kt) Comparison
SSB (kt)
300
200
100
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_LLS_BS_Dome_DN)


SSB (kt) Comparison
B40
SSB (kt)
200
150
100
50
0
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_LLS_BS_Dome_DN)

colour
First Iter
Last Iter
Fixed Gear Fishery
Trawl Fishery
Fishery Ages
Fishery Lengths
Mean Composition Bin
14
12
10
8
6
4
65
60
55
1990
2000
2010
2020
Year

colour — First Iter — Last Iter

Japan LL Survey | NOAA LL AI Survey | NOAA LL BS Survey | NOAA LL GOA Survey

Survey Ages

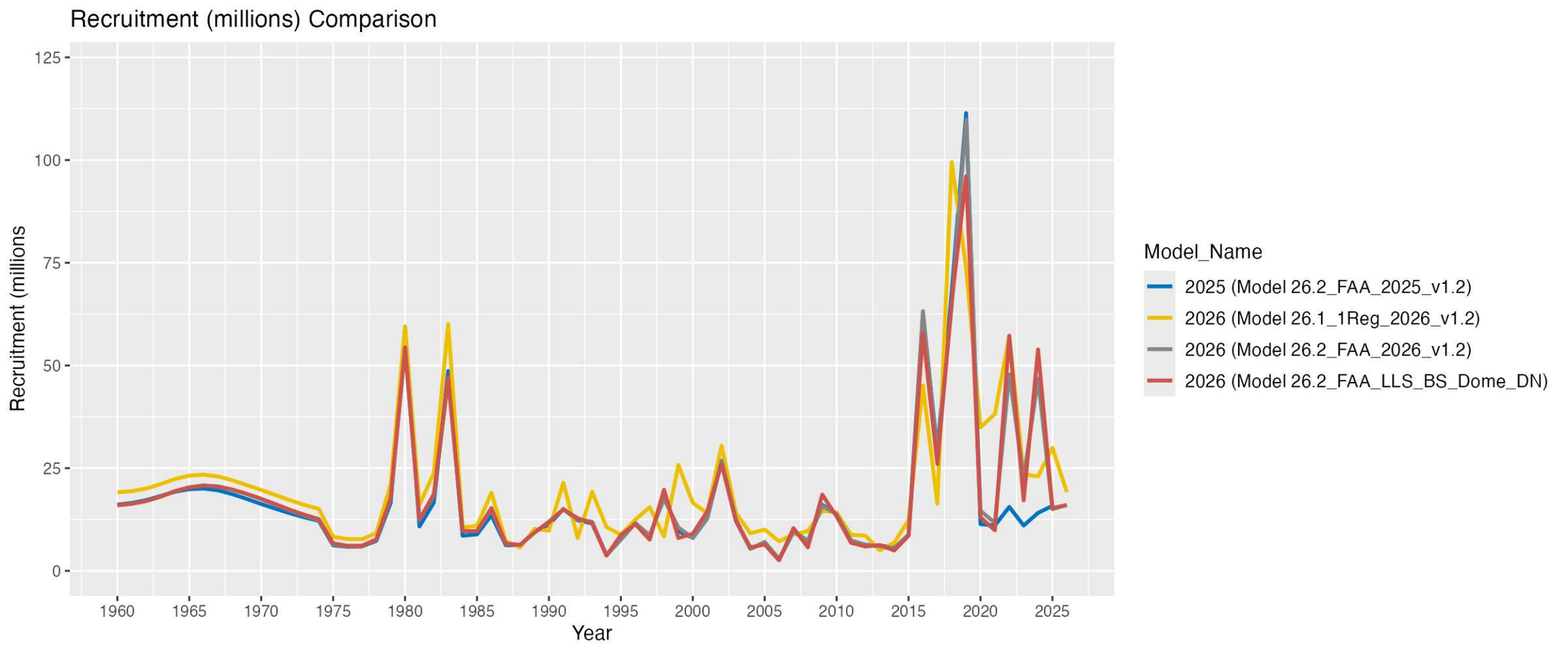
Survey Lengths

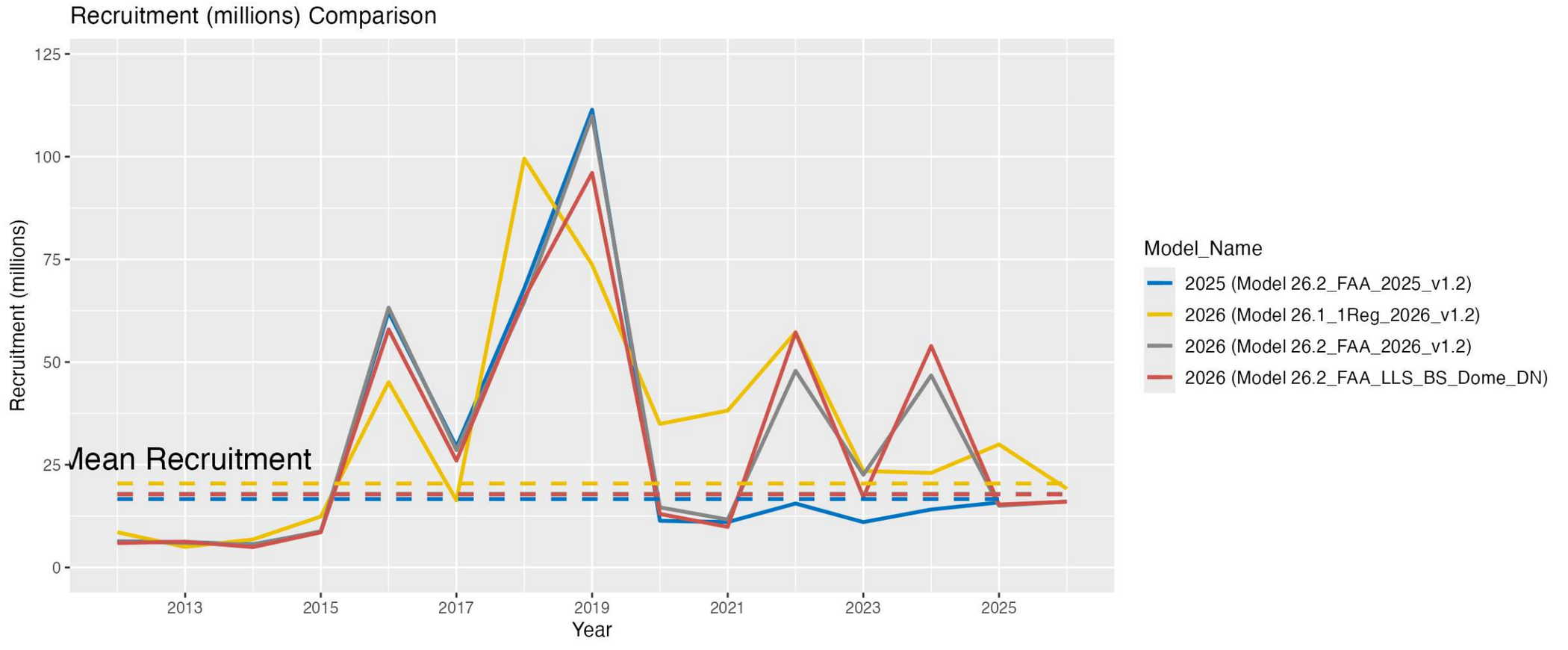
Mean Composition Bin

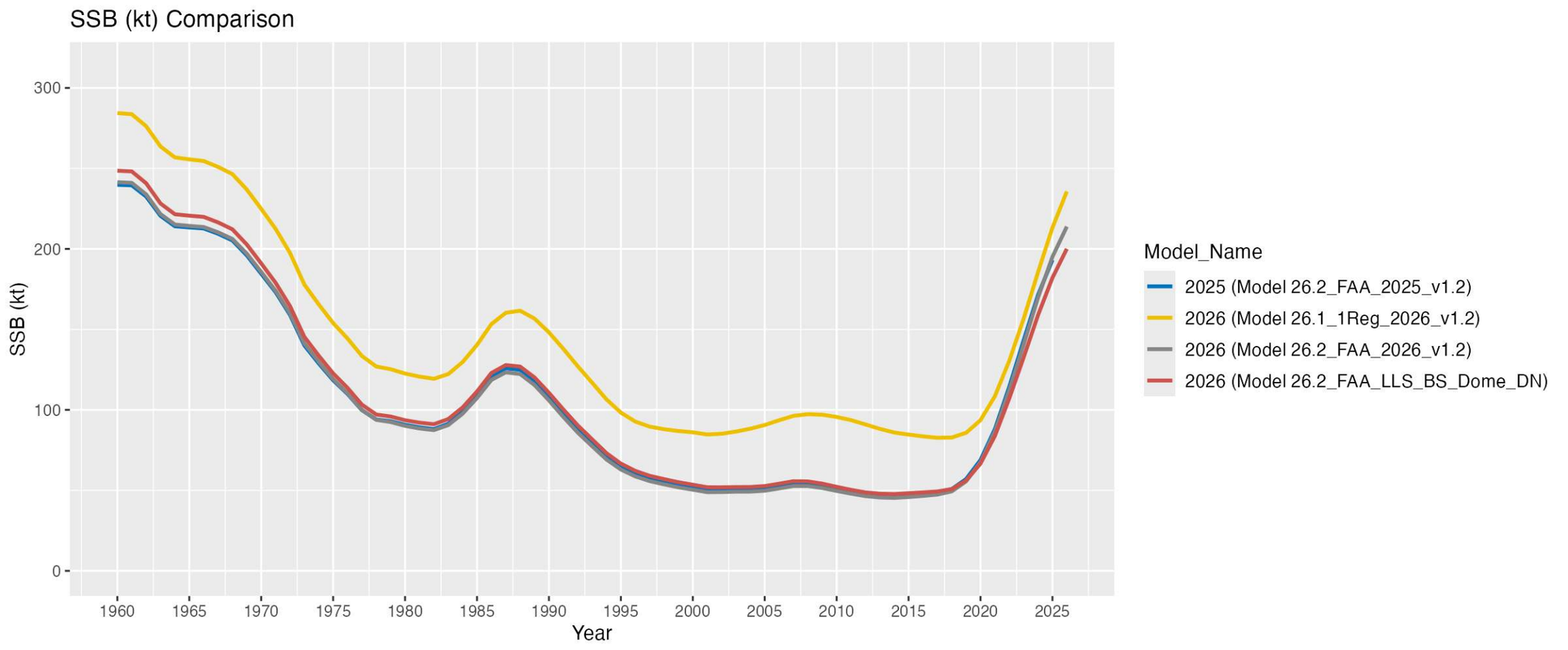
10.0
7.5
5.0
67.5
65.0
62.5

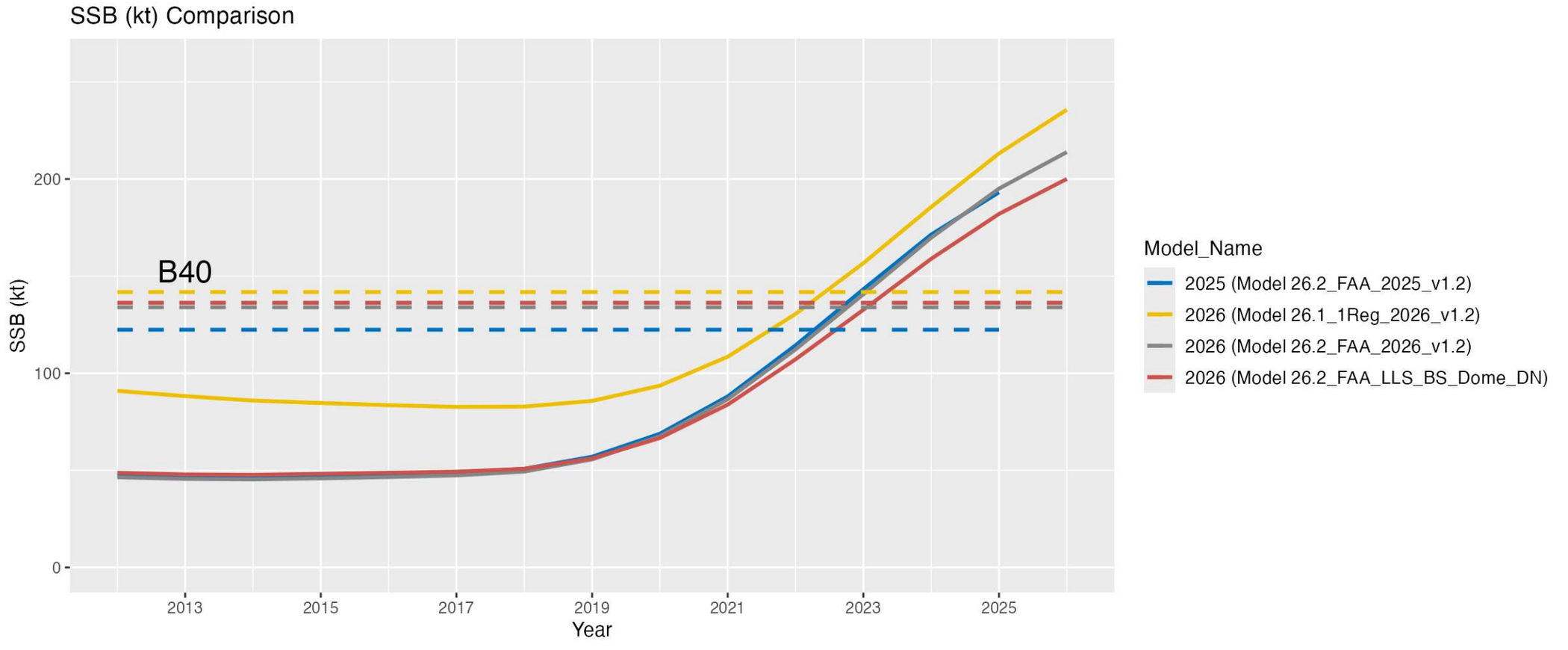
1980 1990 2000 2010 2020

Year


Recruitment (millions) Comparison
Recruitment (millions)
125
100
75
50
25
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_LLS_BS_Dome_DN)


Recruitment (millions) Comparison
Recruitment (millions)
125
100
75
50
25
0
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_LLS_BS_Dome_DN)


SSB (kt) Comparison
SSB (kt)
300
200
100
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_LLS_BS_Dome_DN)


SSB (kt) Comparison
B40
SSB (kt)
200
100
0
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_2026_v1.2)
2026 (Model 26.2_FAA_LLS_BS_Dome_DN)

AI LL Survey Age Comps Female AK

1996 1998 2000 2002
2004 2006 2008 2010
2012 2014 2016 2018
2020 2022

Proportion

0.00 0.05 0.10 0.15 0.20

0 10 20 30

Age

AI LL Survey Age Comps Male AK

1996 1998 2000 2002
2004 2006 2008 2010
2012 2014 2016 2018
2020 2022

Proportion
0.15 0.10 0.05 0.00
0 10 20 30
Age

AI LL Survey Age Comps OSA AK

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

AK

Age

Year

AK

SDNR=0.97
(0.95-1.05)

Sample quantiles

Theoretical quantiles

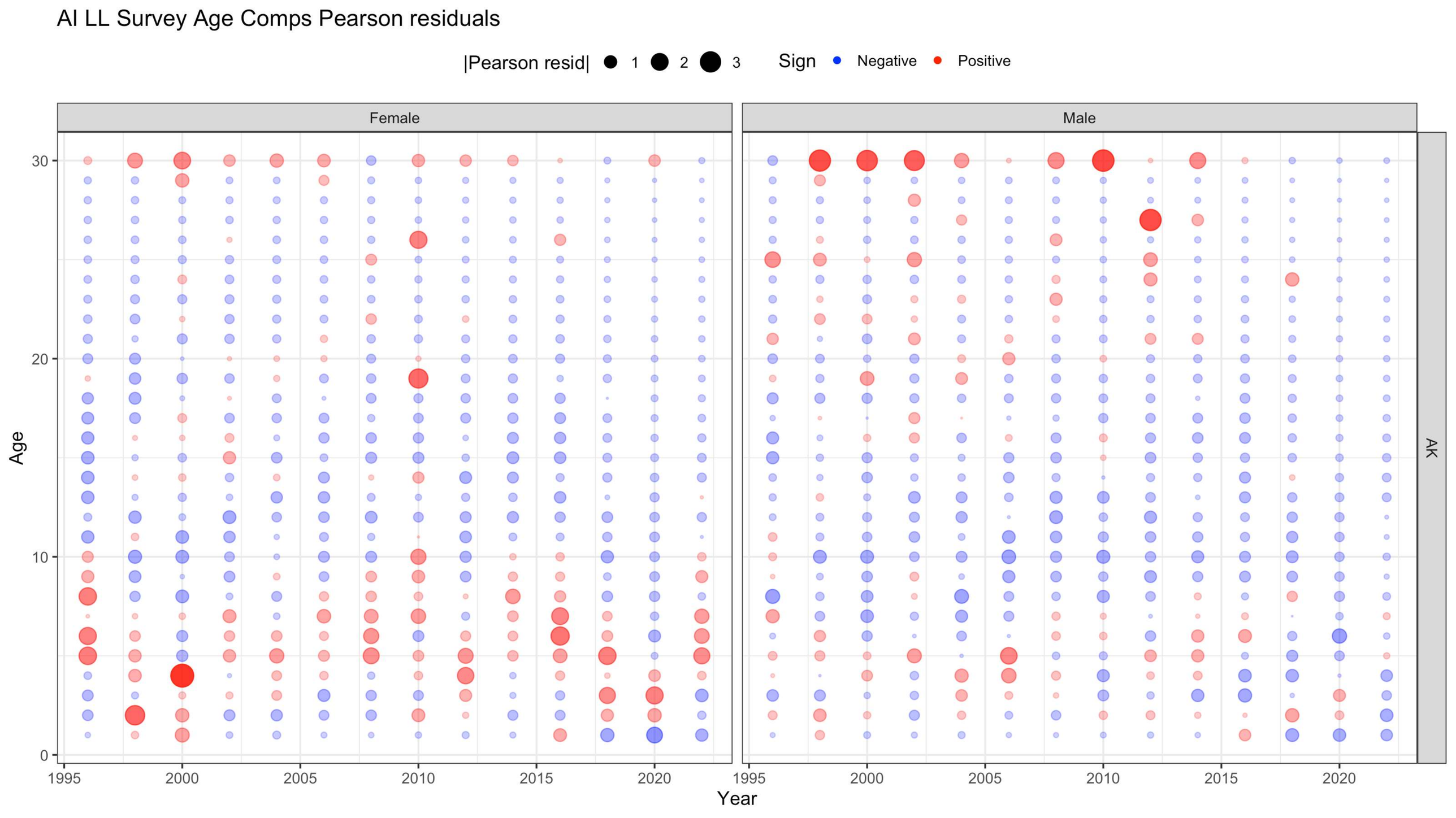
Female

Male

AK

Proportions

Age


AI LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
1995
2000
2005
2010
2015
2020

Model 26.2_FAA_LLS_BS_Dome_DN Population Population 1

Female | Male | AK

Spawning Weight-at-Age (kg)

0.075
0.050
0.025
0.000

0 10 20 30

Age

Model 26.2_FAA_LLS_BS_Dome_DN Population Population 1

Female

Male

AK

Maturity at Age

1.00

0.75

0.50

0.25

0.00

0

10

20

30

Age

Model 26.2_FAA_LLS_BS_Dome_DN Population Population 1

Female

Male

AK

Spawning Weight at Age

6

4

2

0

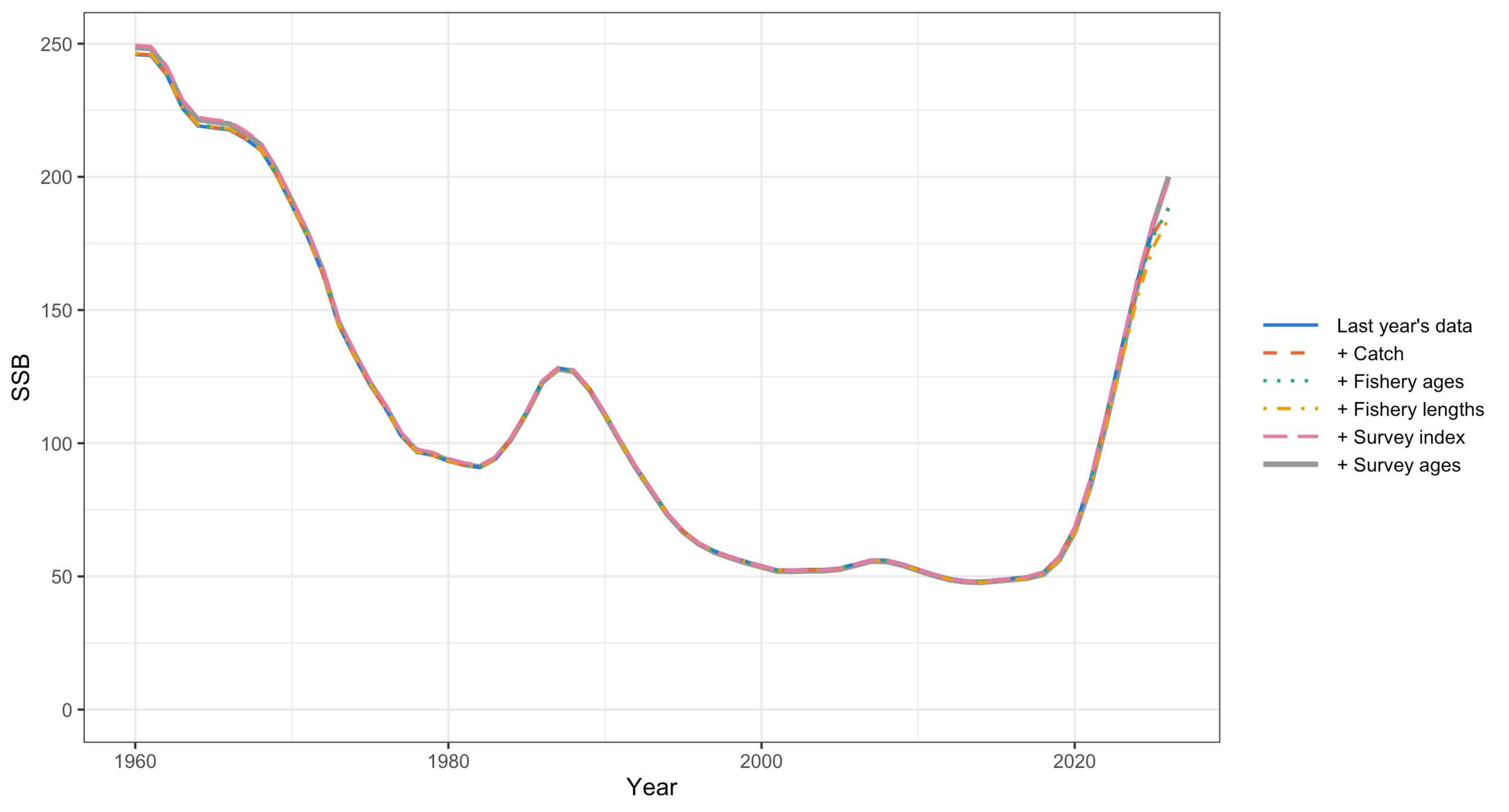
0

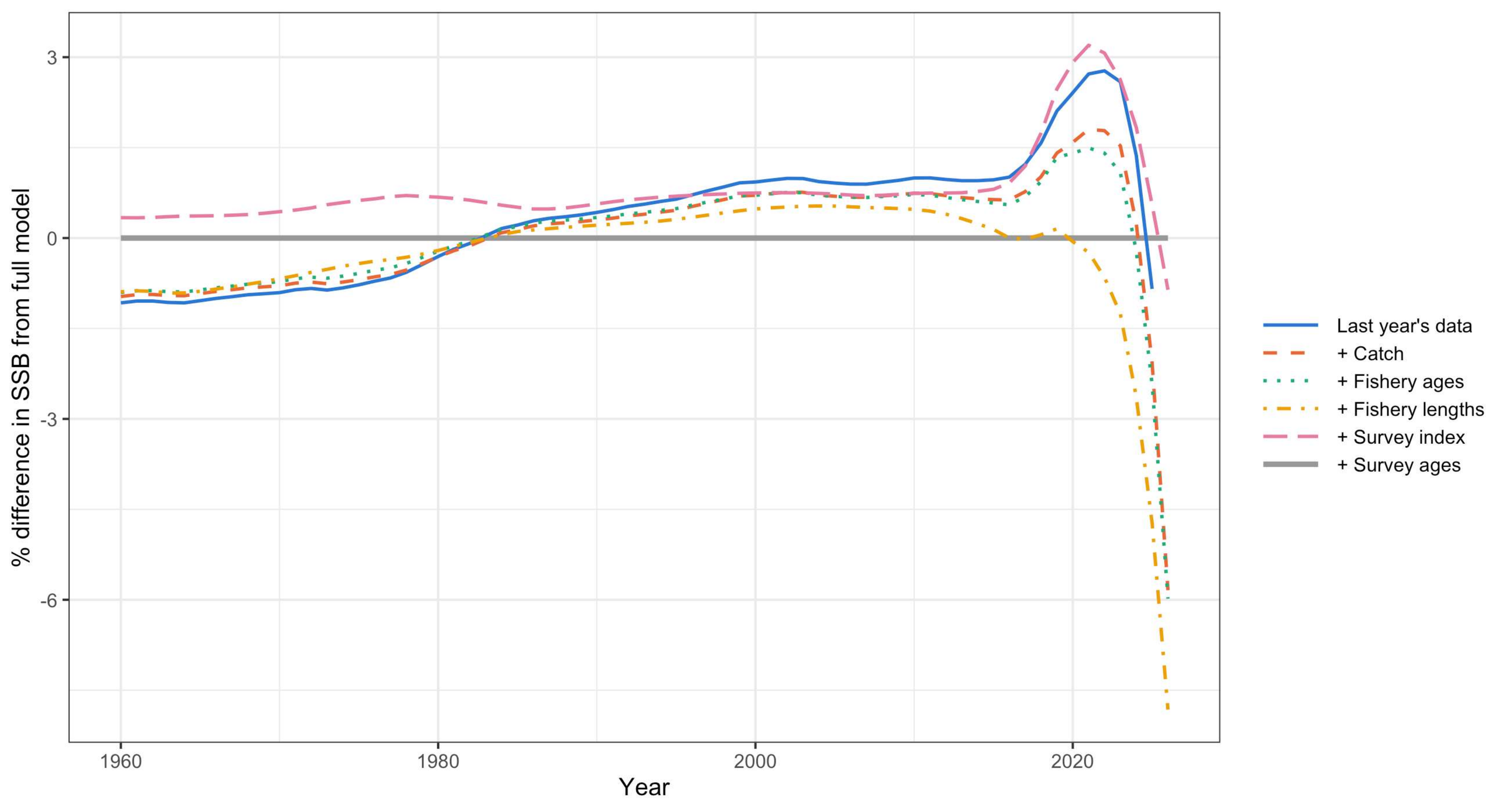
10

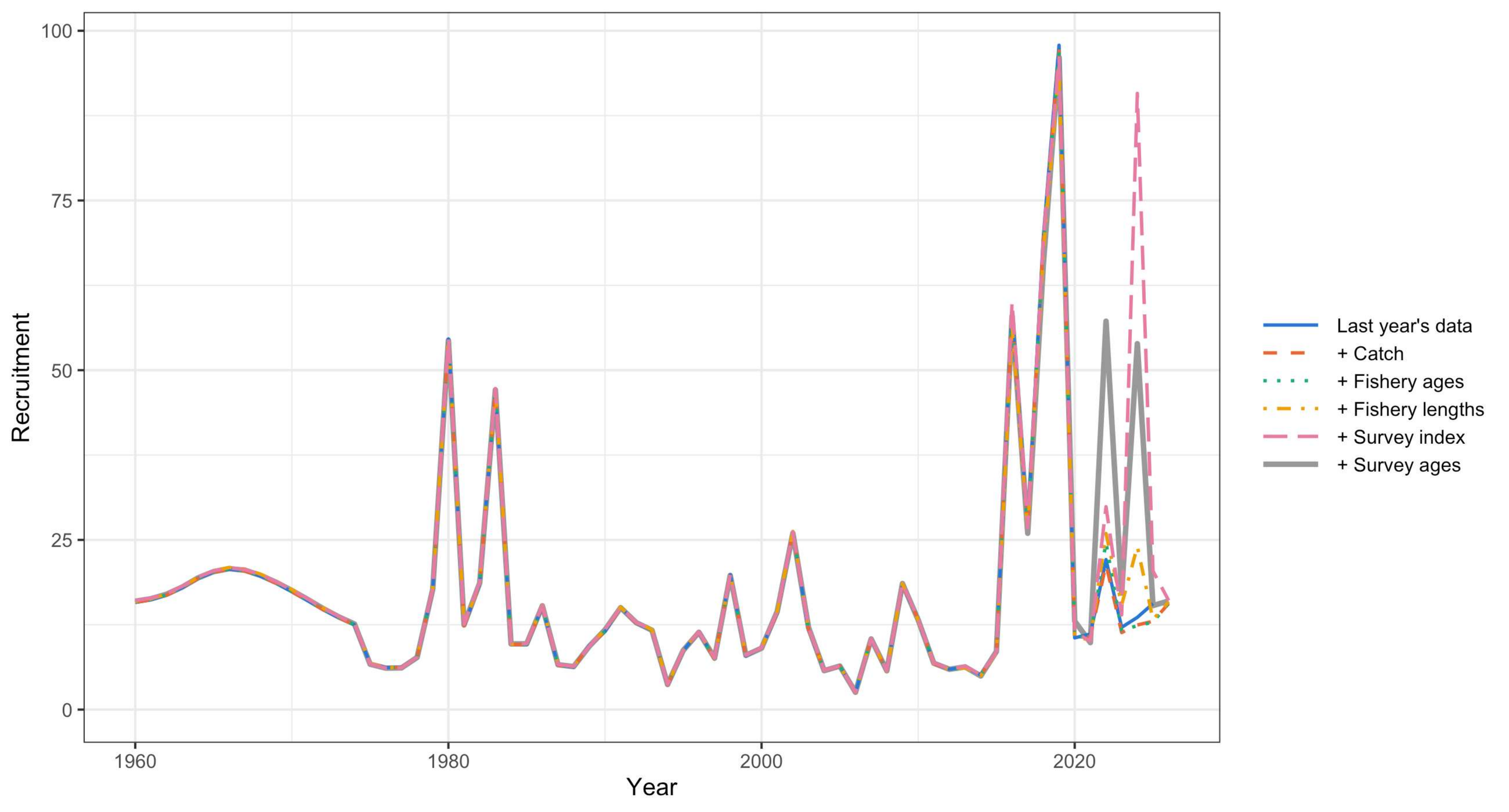
20

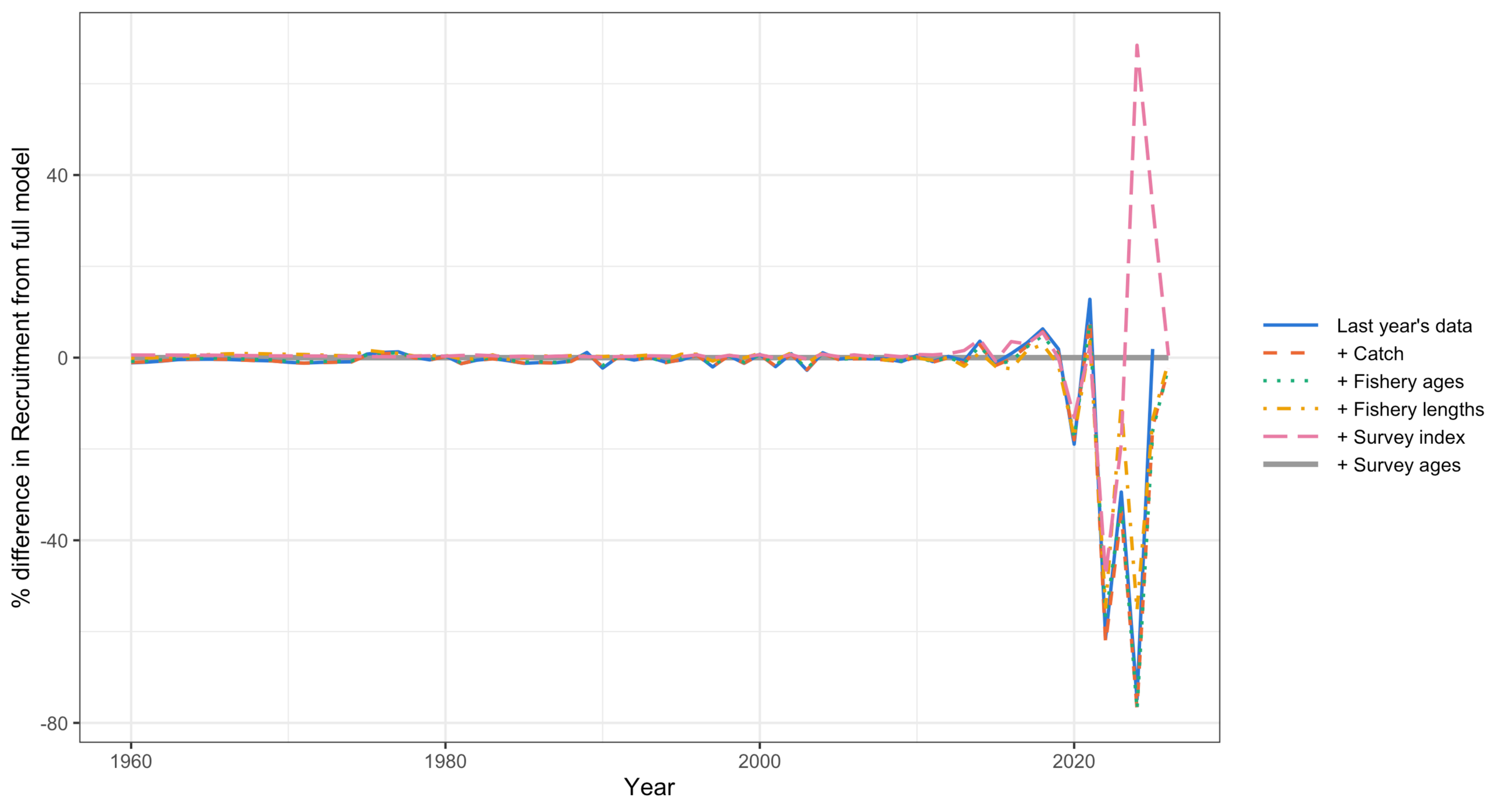
30

Age

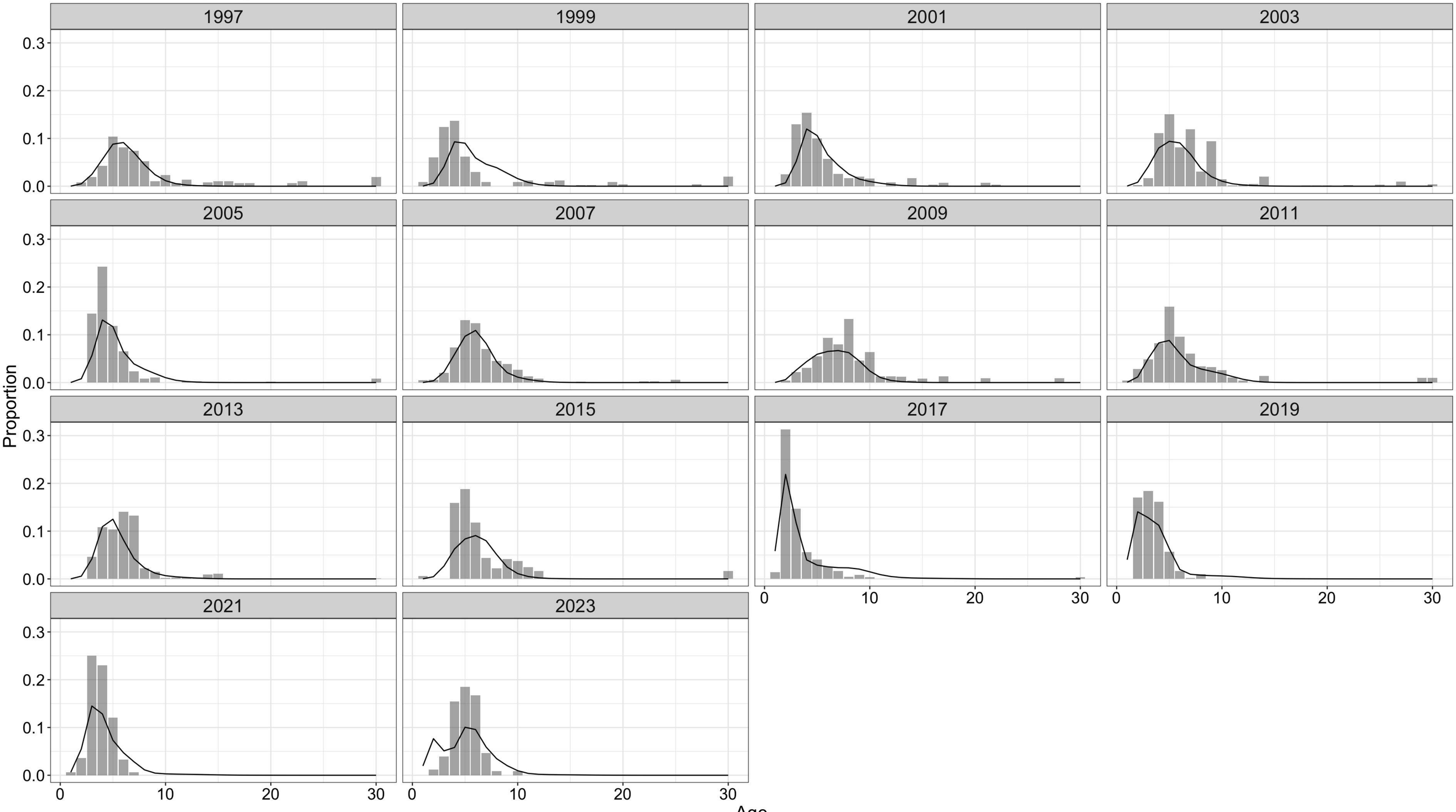

250
200
150
100
50
0
SSB
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


3
0
-3
-6
% difference in SSB from full model
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


100
75
50
25
0
Recruitment
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


% difference in Recruitment from full model
40
0
-40
-80
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages

BS LL Survey Age Comps Female AK



BS LL Survey Age Comps Male AK

1997 1999 2001 2003
2005 2007 2009 2011
2013 2015 2017 2019
2021 2023

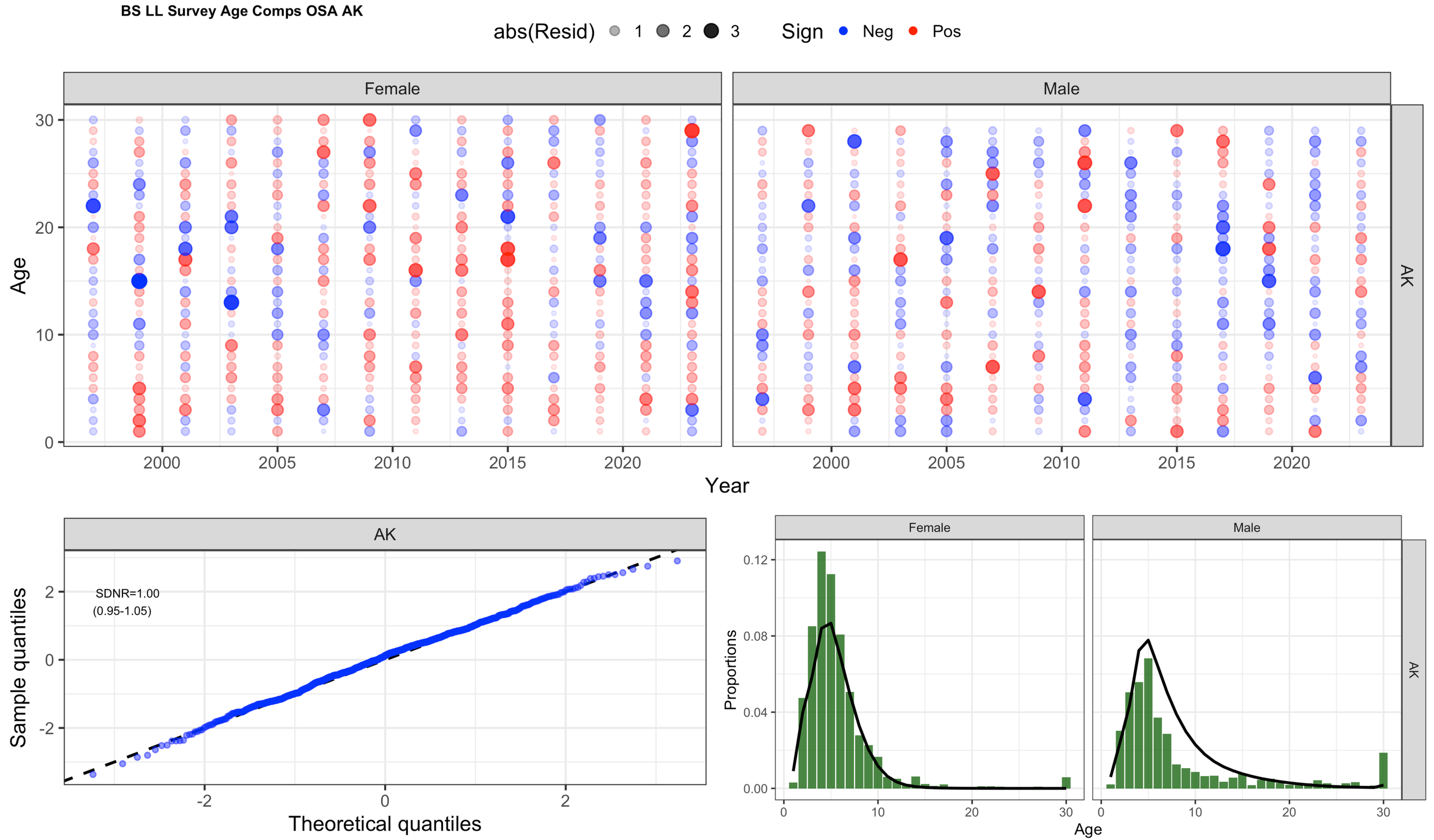
Proportion

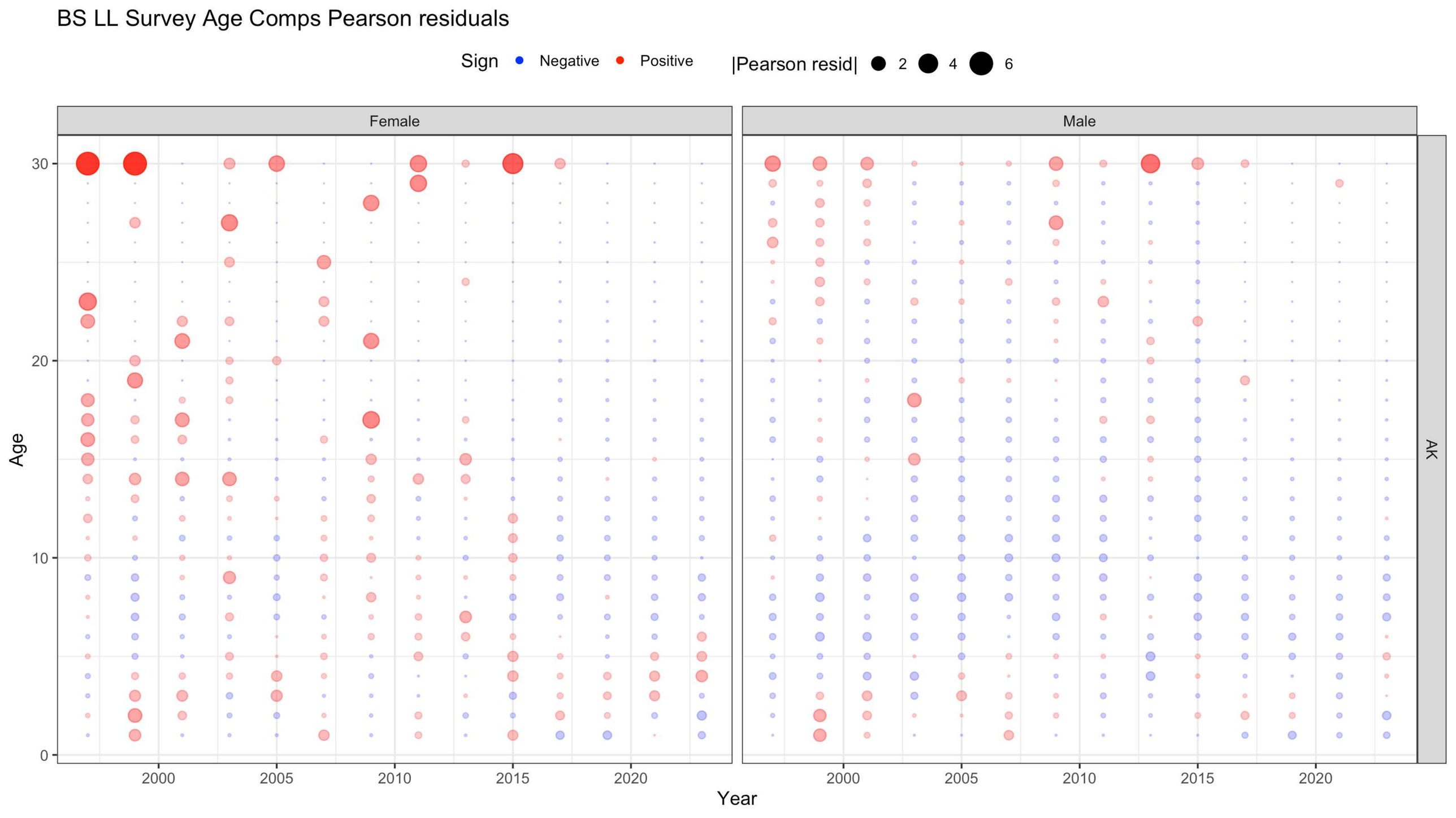
0.00 0.05 0.10 0.15

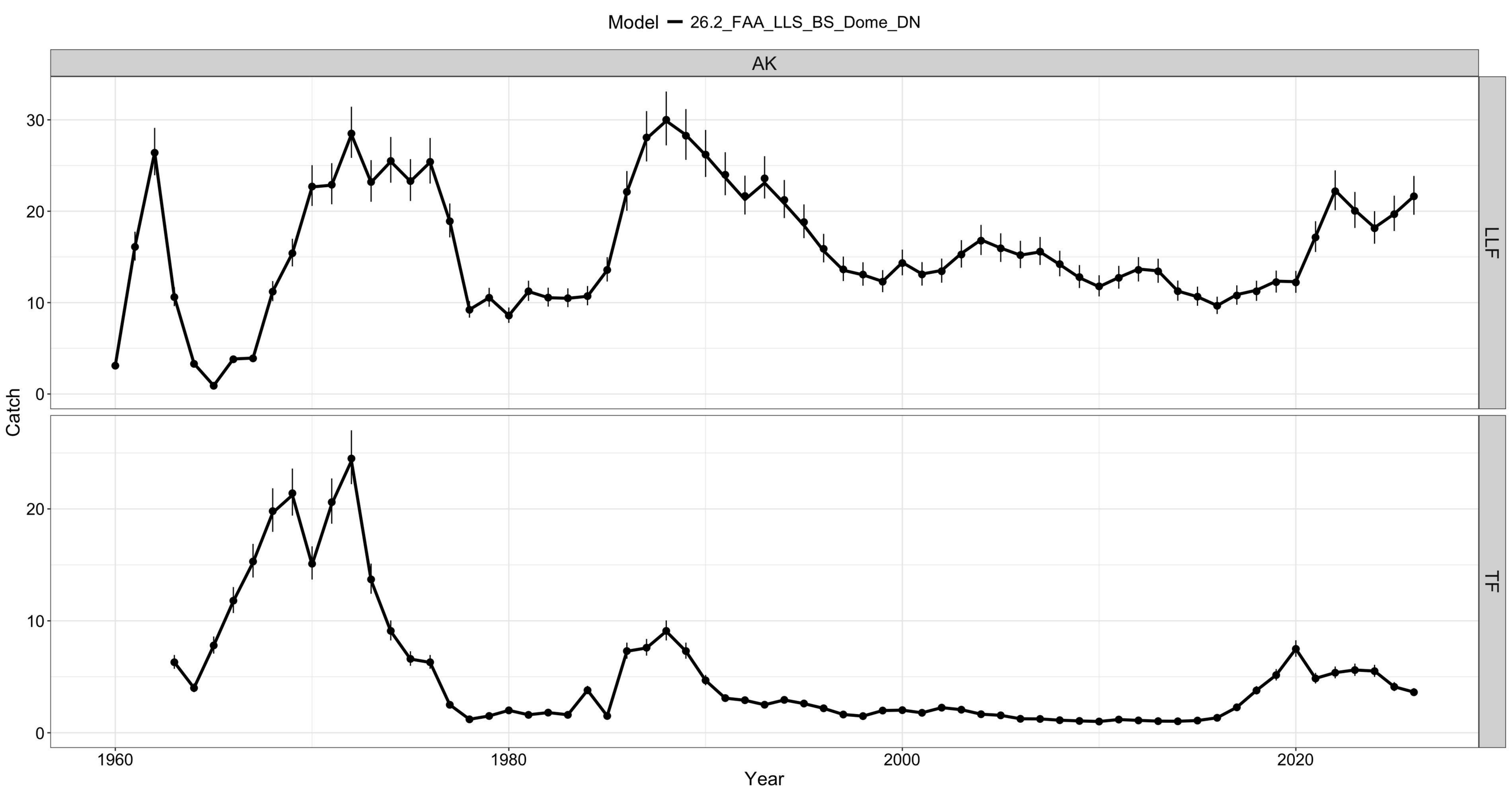
0 10 20 30

Age

BS LL Survey Age Comps OSA AK




BS LL Survey Age Comps Pearson residuals
Sign
Negative
Positive
|Pearson resid|
2
4
6
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020


Model — 26.2_FAA_LLS_BS_Dome_DN
AK
LLF
TF
Catch
Year
0
10
20
30
1960
1980
2000
2020

AK

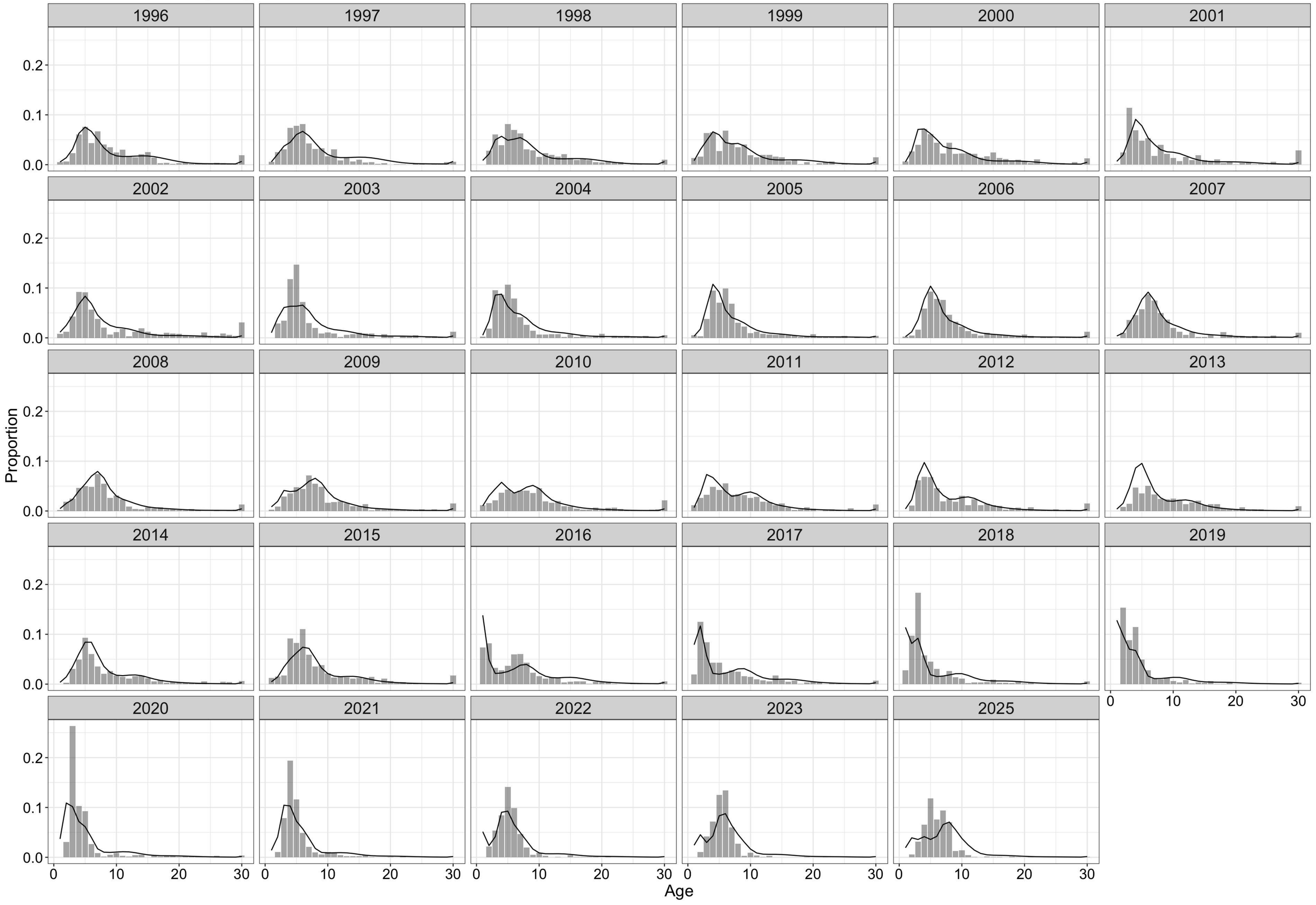
JPN LLS Lengths
JPN LLS Index
GOA LLS Index
AI LLS Index
BS LLS Index
JPN LLS Ages
GOA LLS Ages
AI LLS Ages
BS LLS Ages
TF Lengths
TF Catch
LLF Catch
LLF Ages

26.2_FAA_LLS_BS_Dome_DN

1960
1980
2000
2020

Year

GOA LL Survey Age Comps Female AK



GOA LL Survey Age Comps Male AK

1996 1997 1998 1999 2000 2001
2002 2003 2004 2005 2006 2007
2008 2009 2010 2011 2012 2013
2014 2015 2016 2017 2018 2019
2020 2021 2022 2023 2025

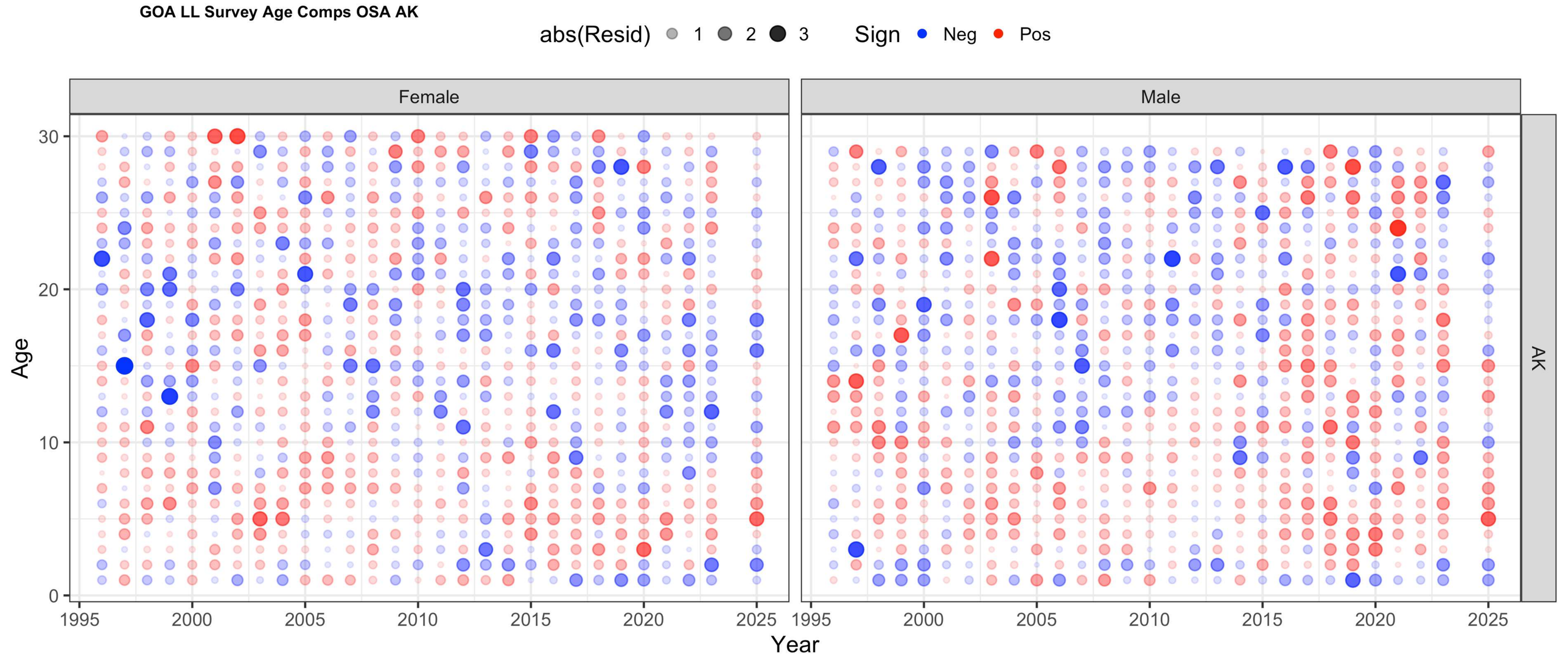
Proportion

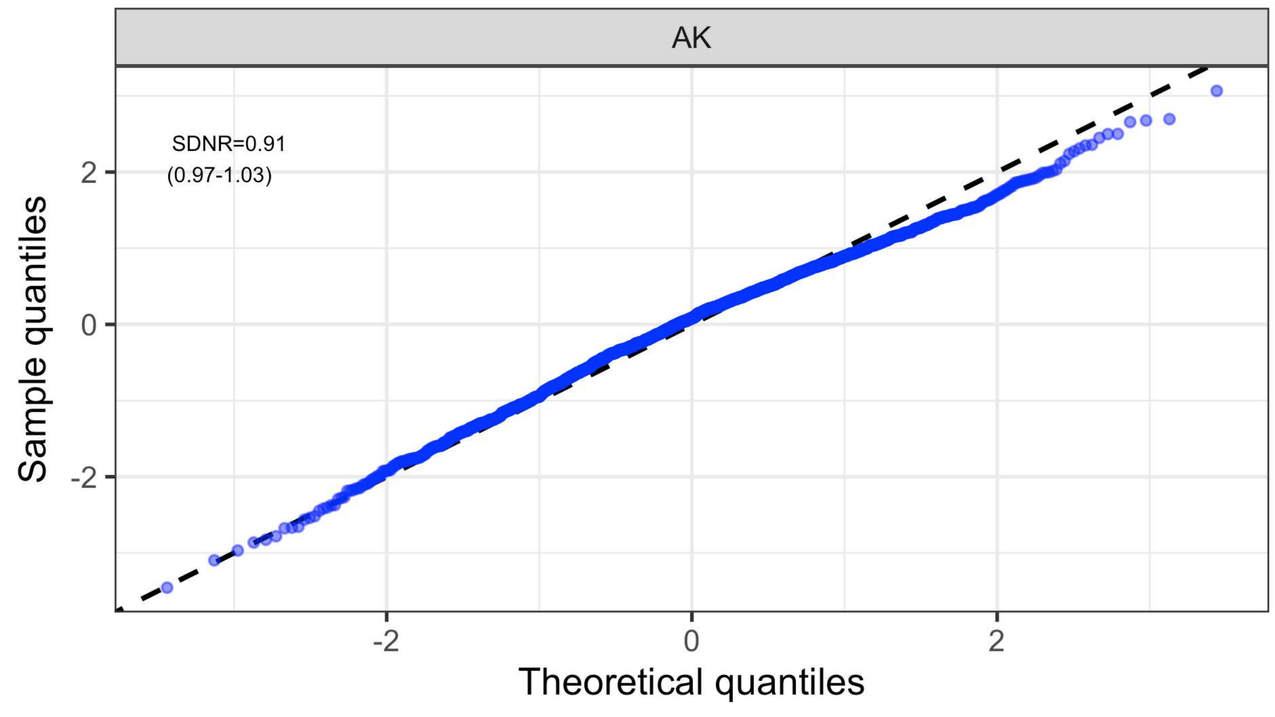
0.00 0.05 0.10 0.15

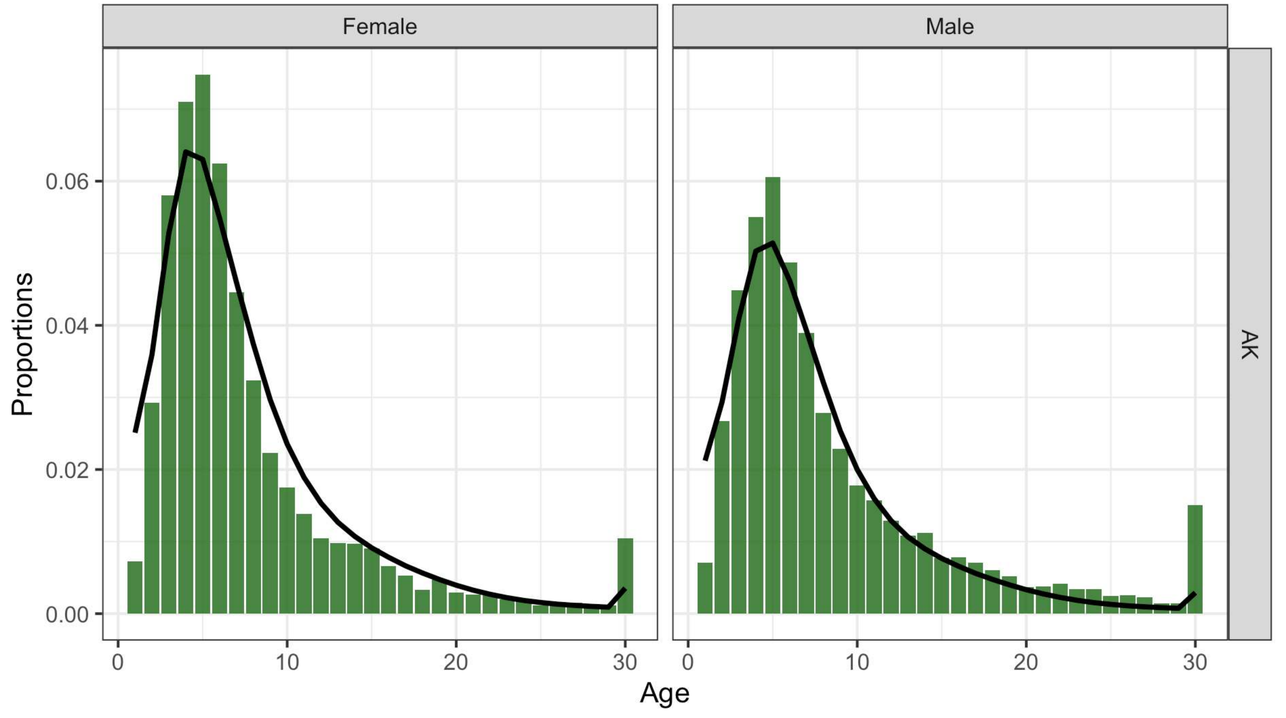
0 10 20 30

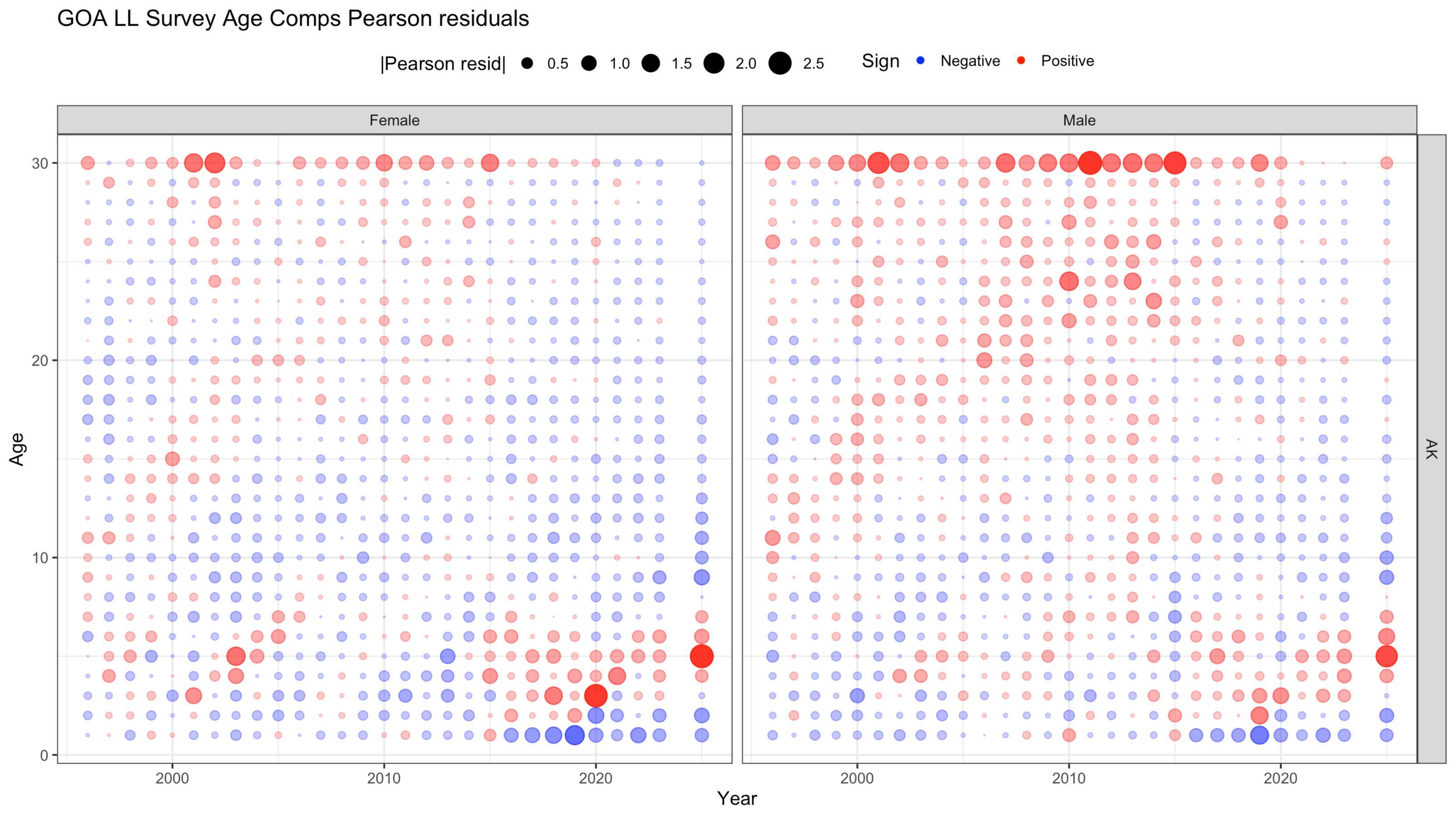
Age

GOA LL Survey Age Comps OSA AK








GOA LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
2.5
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2010
2020

Model — 26.2_FAA_LLS_BS_Dome_DN

BS LLS

AI LLS

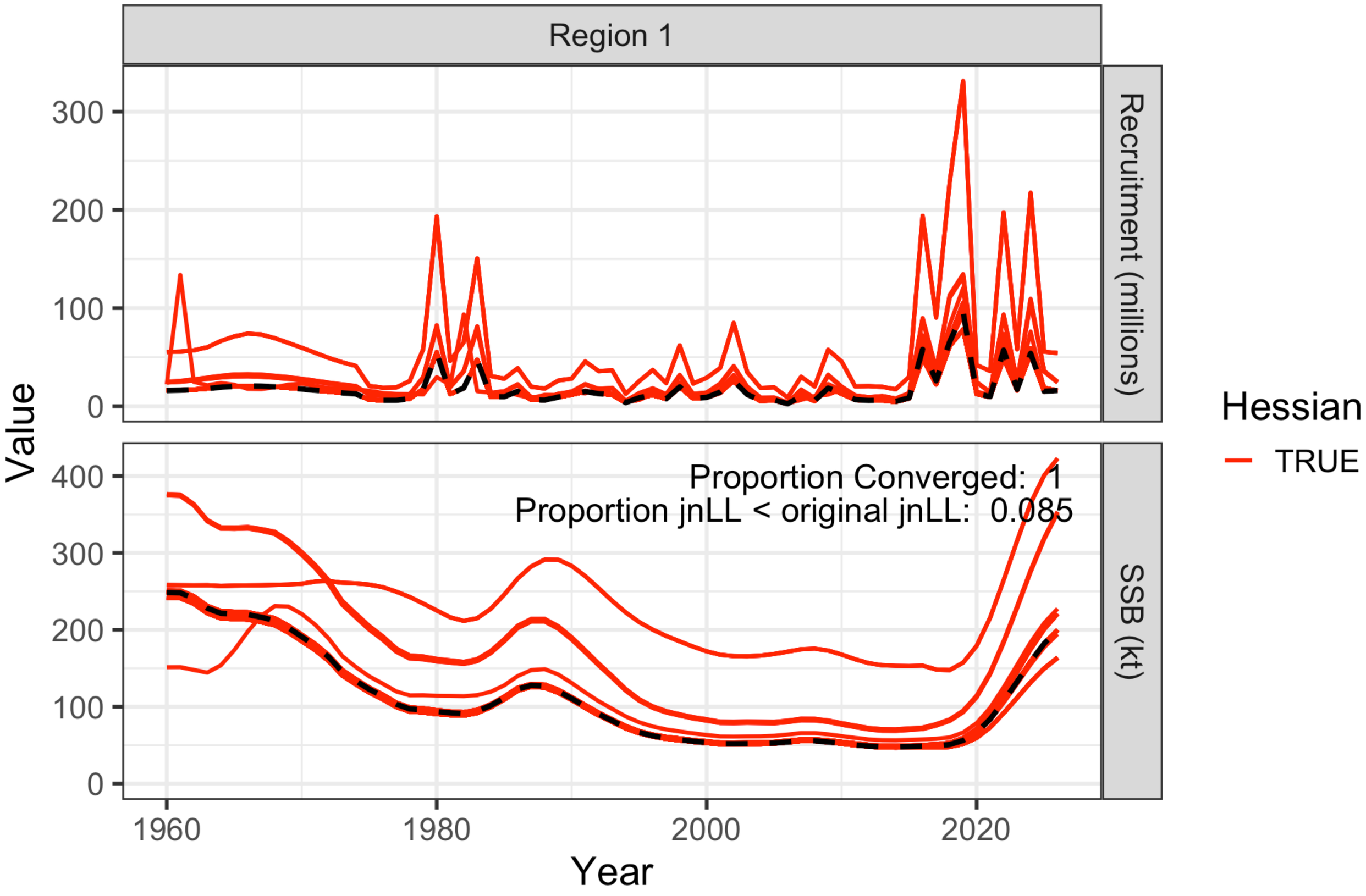
GOA LLS

JPN LLS

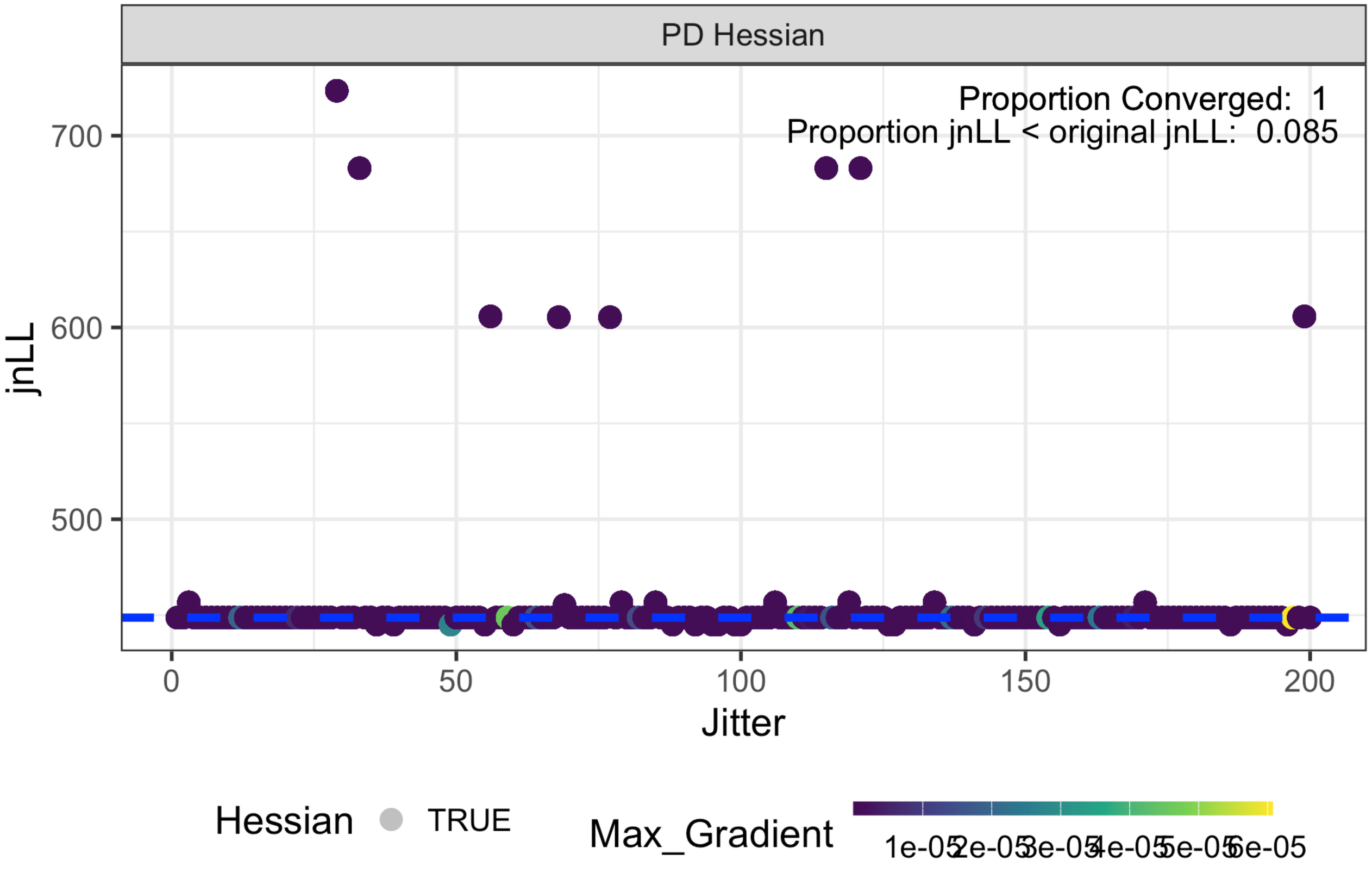
RPNs (1000s)

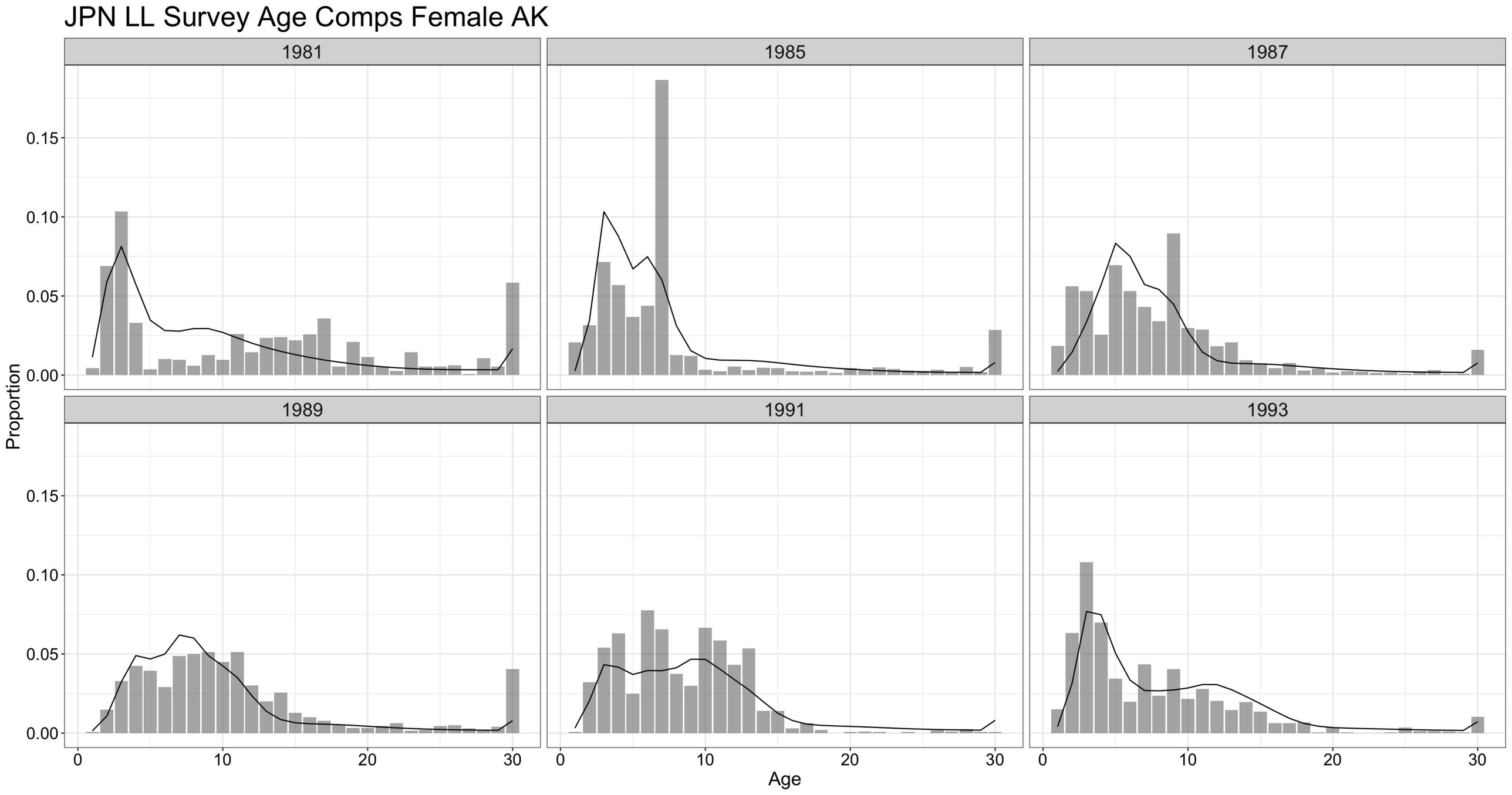
Year

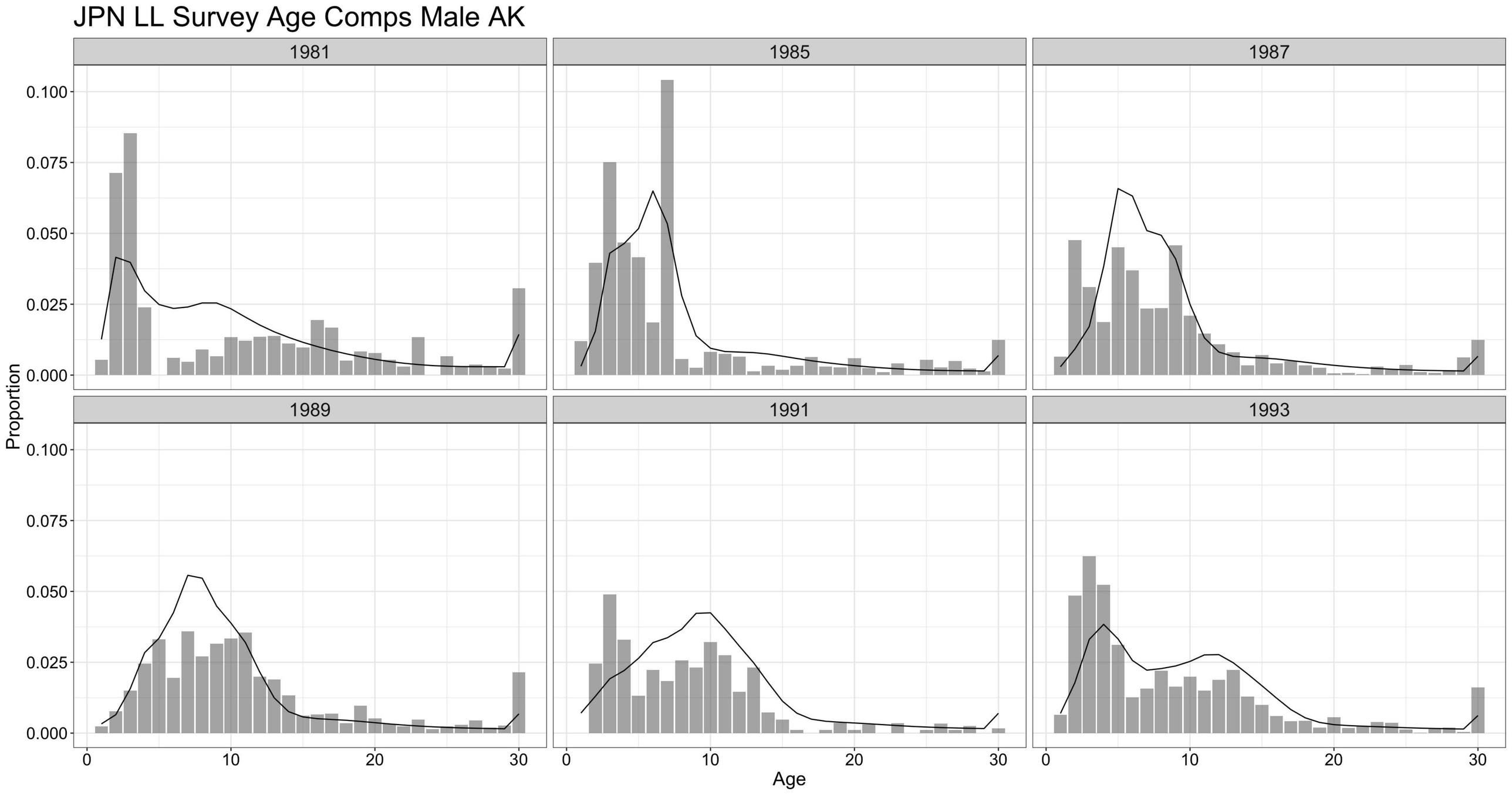
Jitter SSB and Recruitment



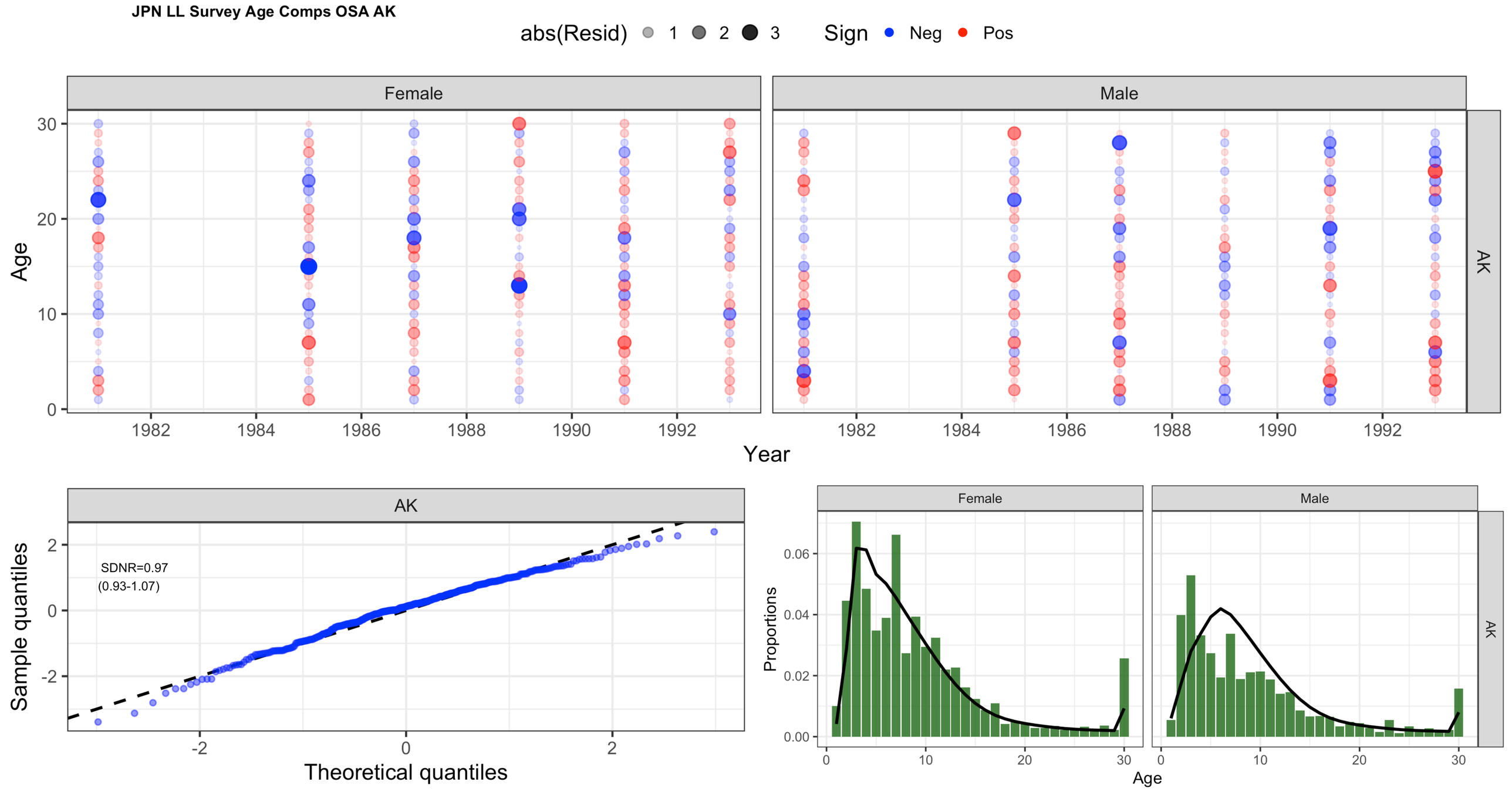
Jitter nLL and Convergence

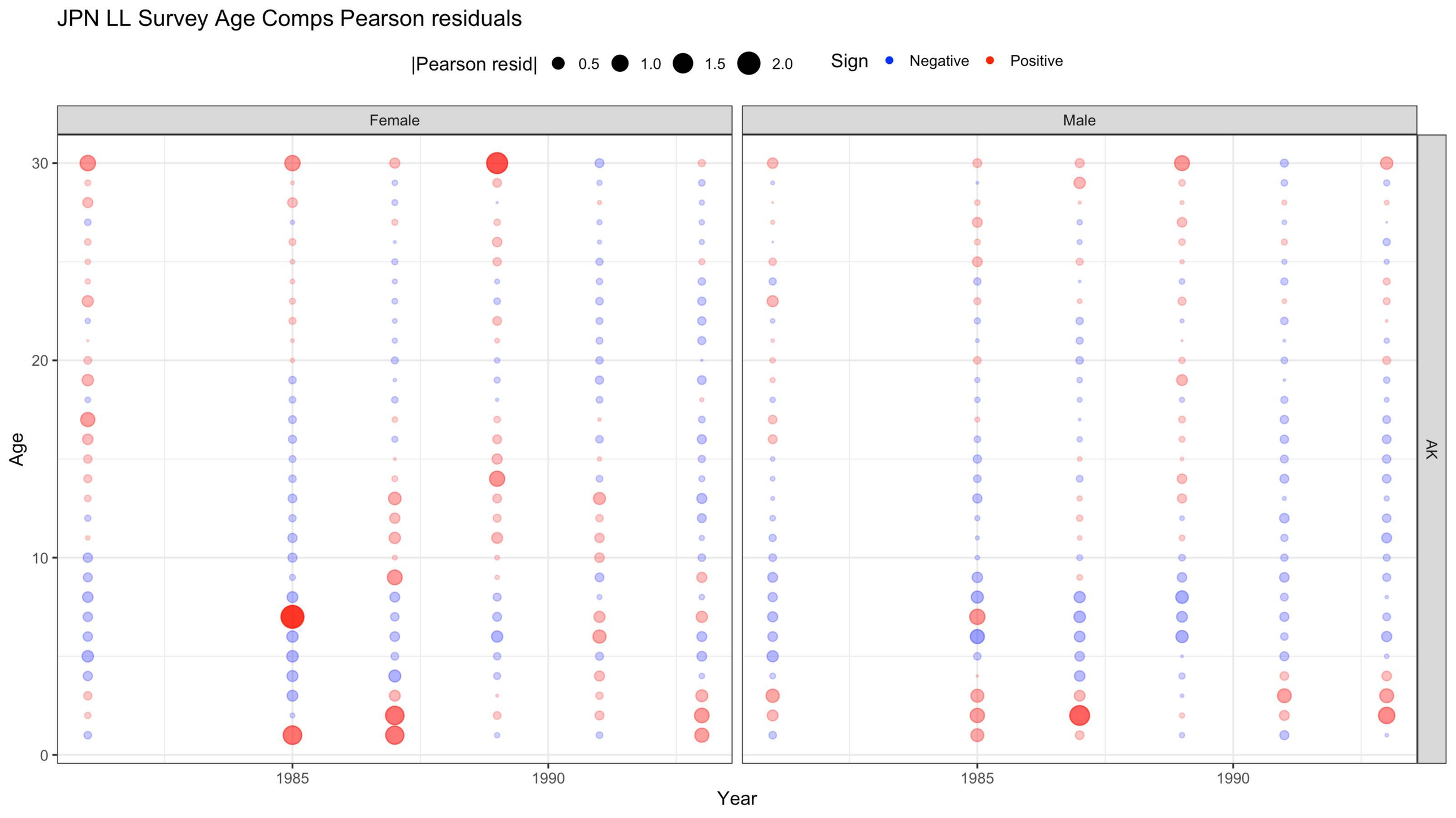



JPN LL Survey Age Comps Female AK
1981
1985
1987
1989
1991
1993
Proportion
0.00
0.05
0.10
0.15
0
10
20
30
Age


JPN LL Survey Age Comps Male AK
1981
1985
1987
1989
1991
1993
0.100
0.075
0.050
0.025
0.000
Proportion
0
10
20
30
Age

JPN LL Survey Age Comps OSA AK




JPN LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
1985
1990
1985
1990

JPN LL Survey Length Comps Female AK

1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

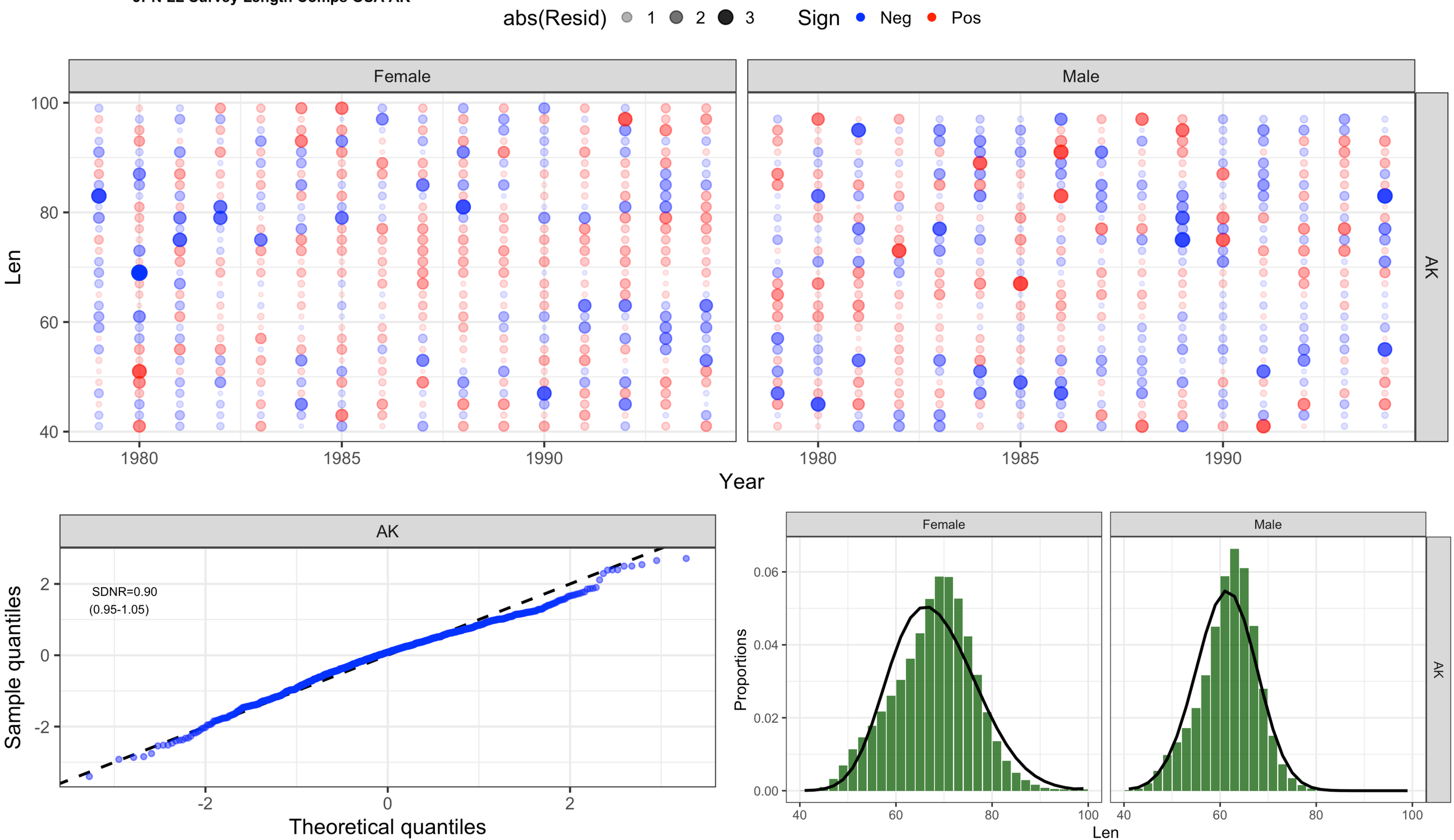
Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

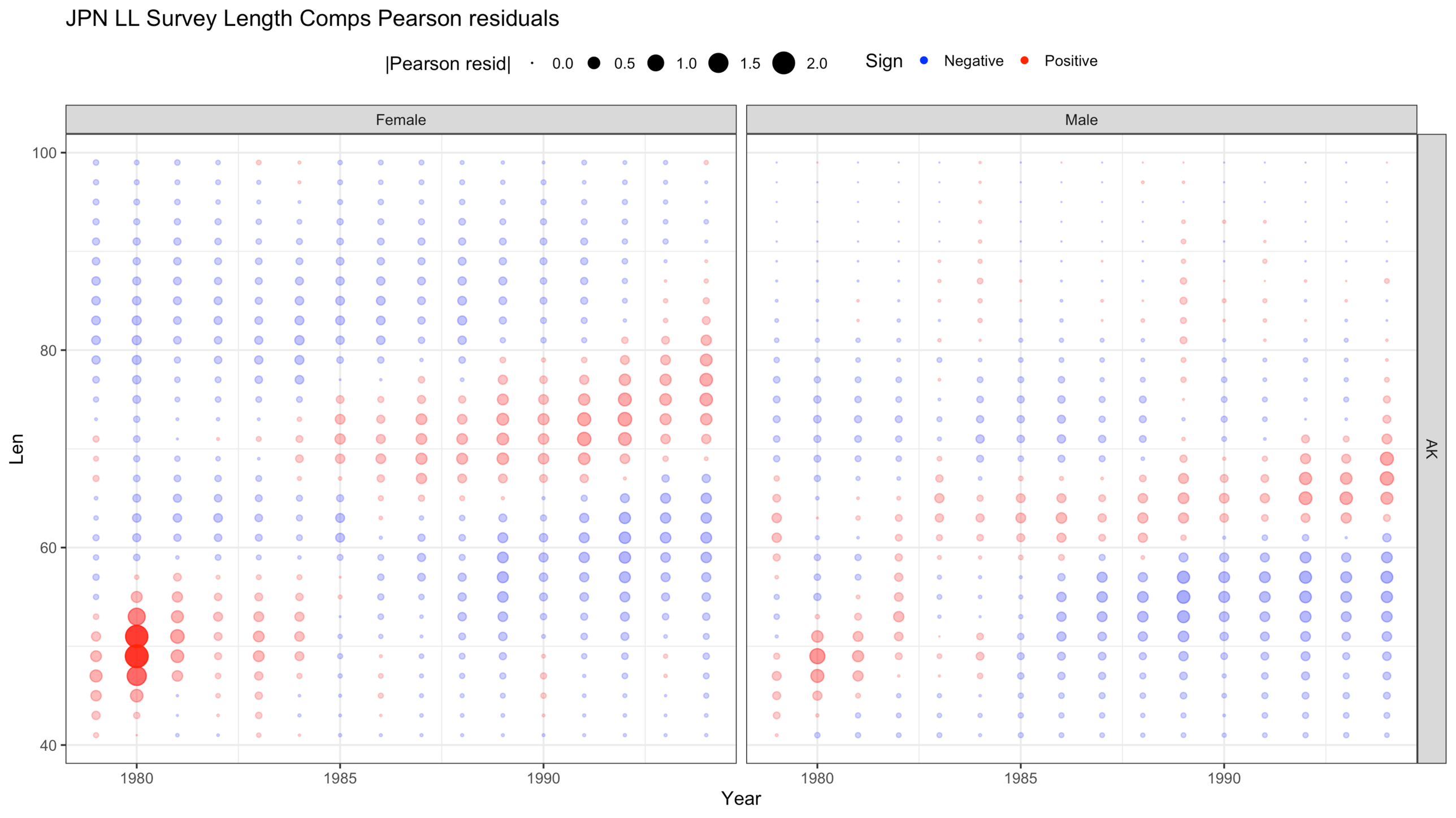
JPN LL Survey Length Comps Male AK

1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

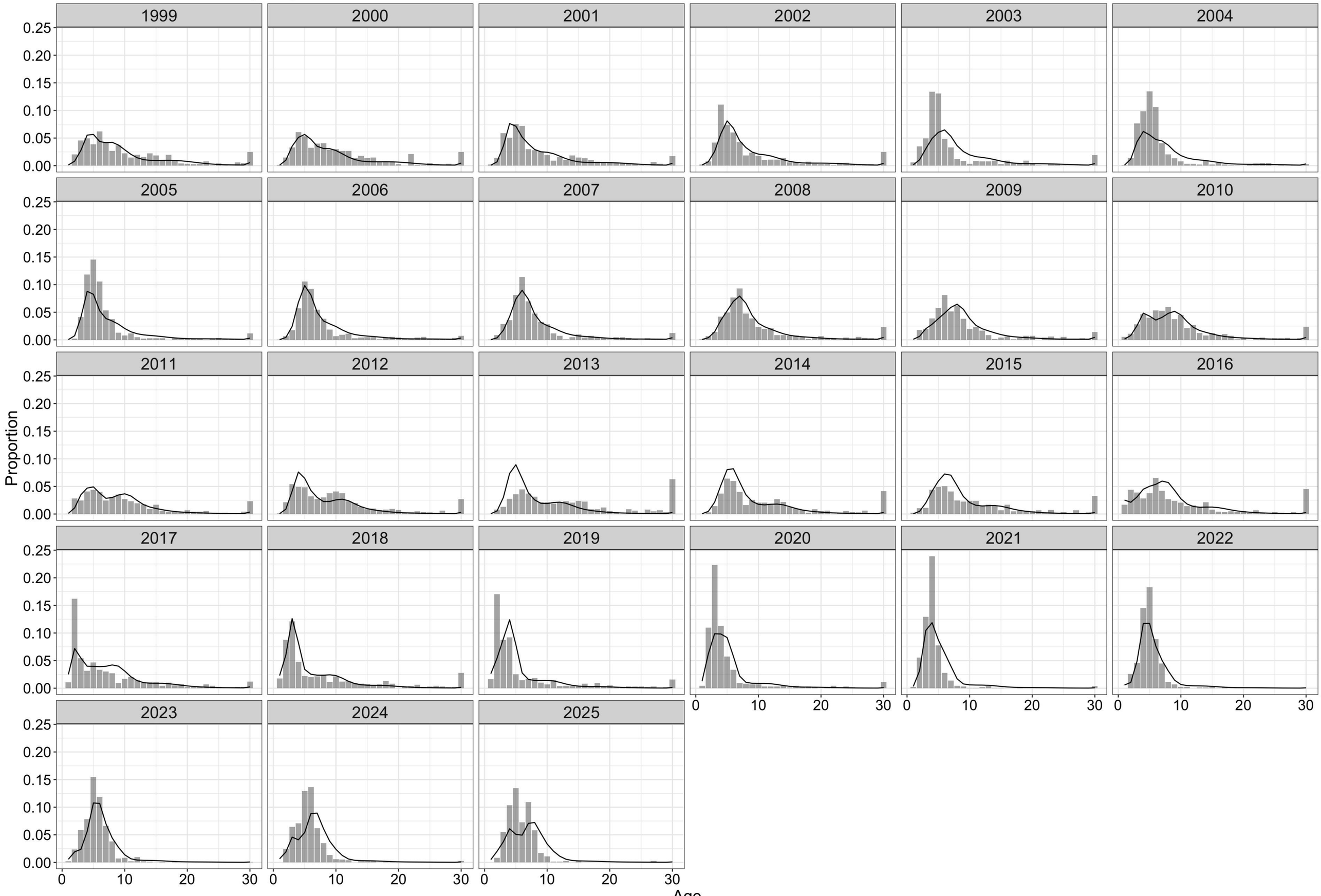
Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

JPN LL Survey Length Comps OSA AK

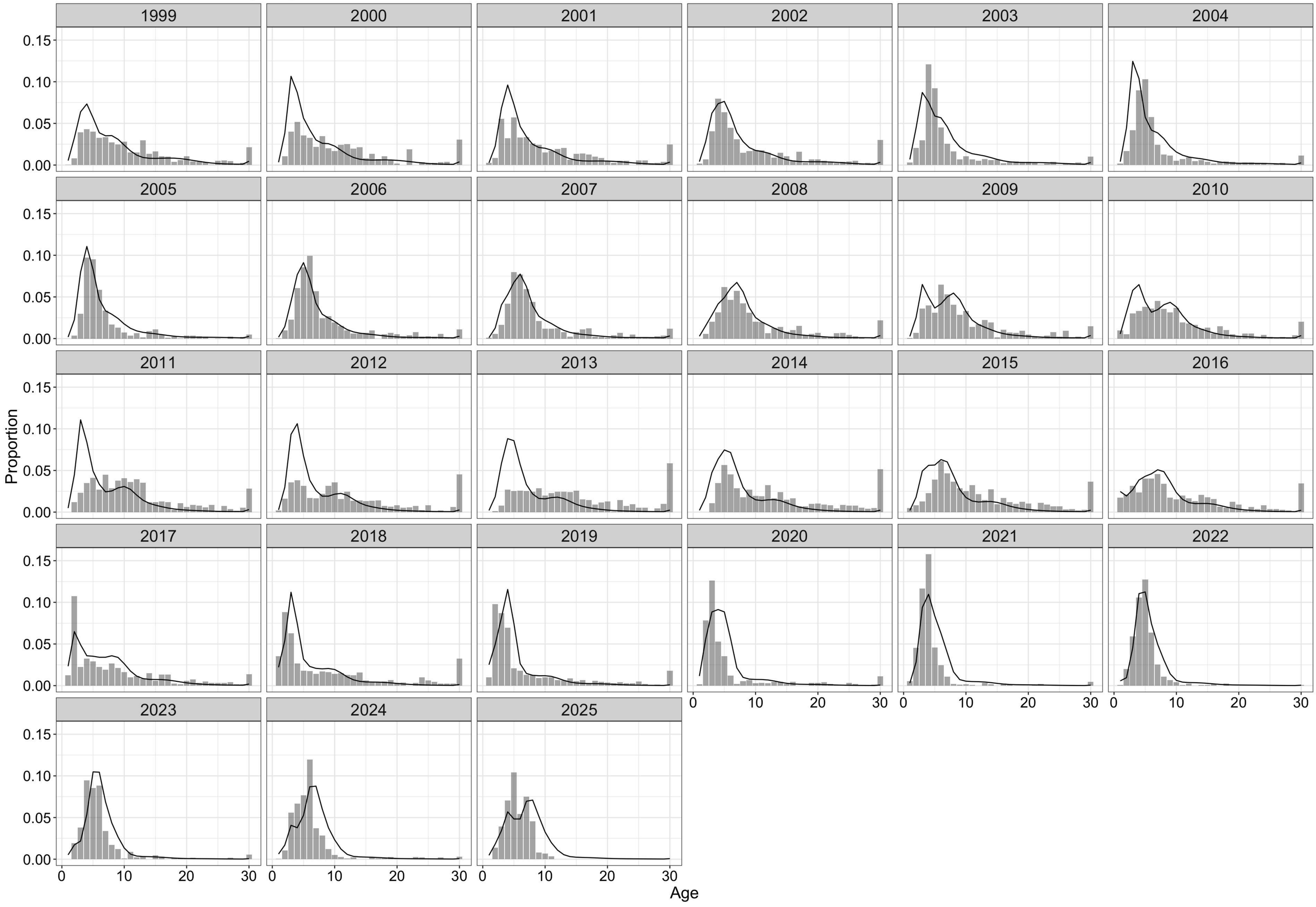



JPN LL Survey Length Comps Pearson residuals
|Pearson resid|
0.0
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1980
1985
1990
Year

Fixed Gear Fishery Age Comps Female AK



Fixed Gear Fishery Age Comps Male AK



Fixed Gear Fishery Age Comps OSA AK

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

AK

Age

Year

AK

SDNR=1.00
(0.97-1.03)

Sample quantiles

Theoretical quantiles

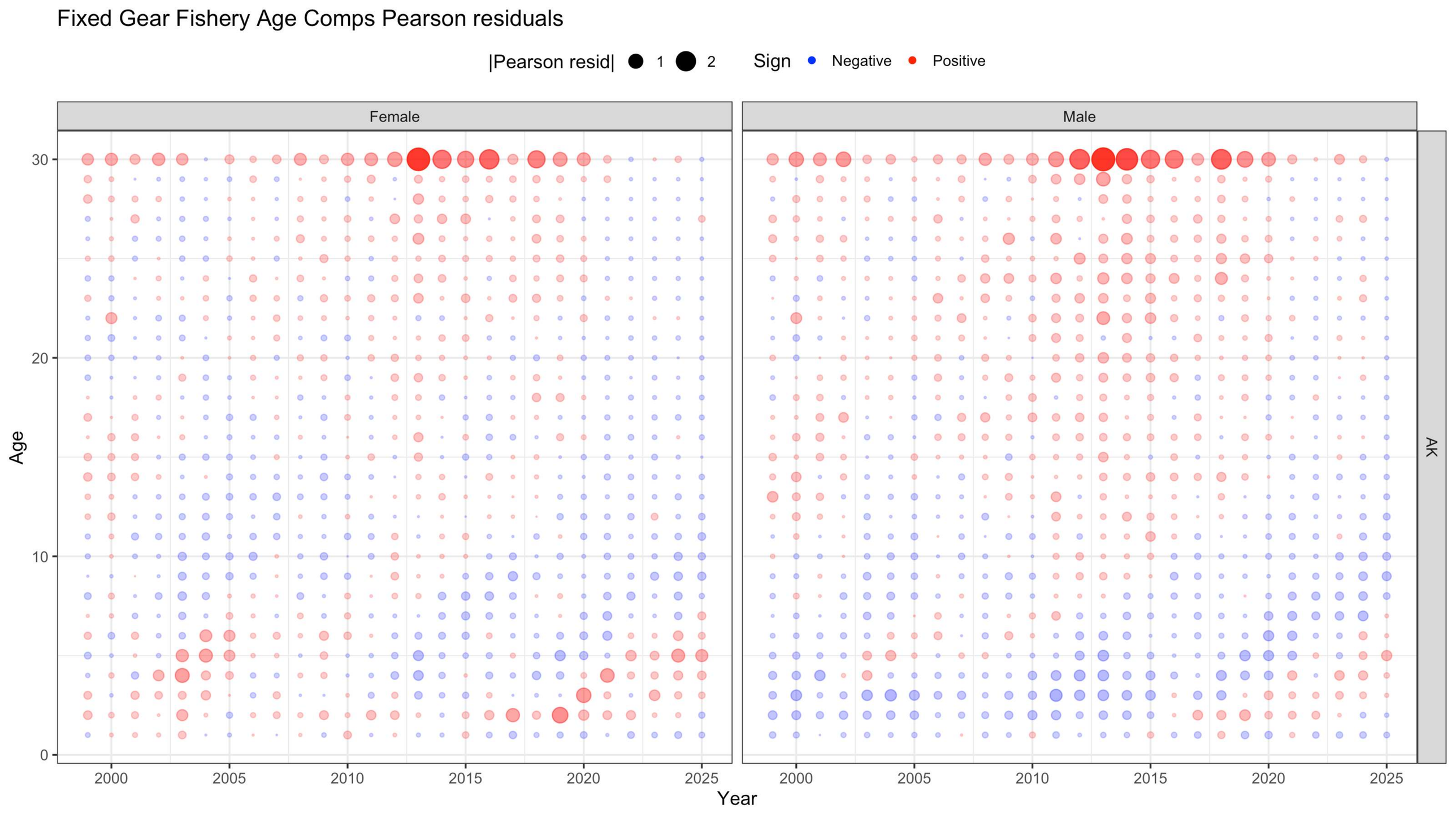
Female

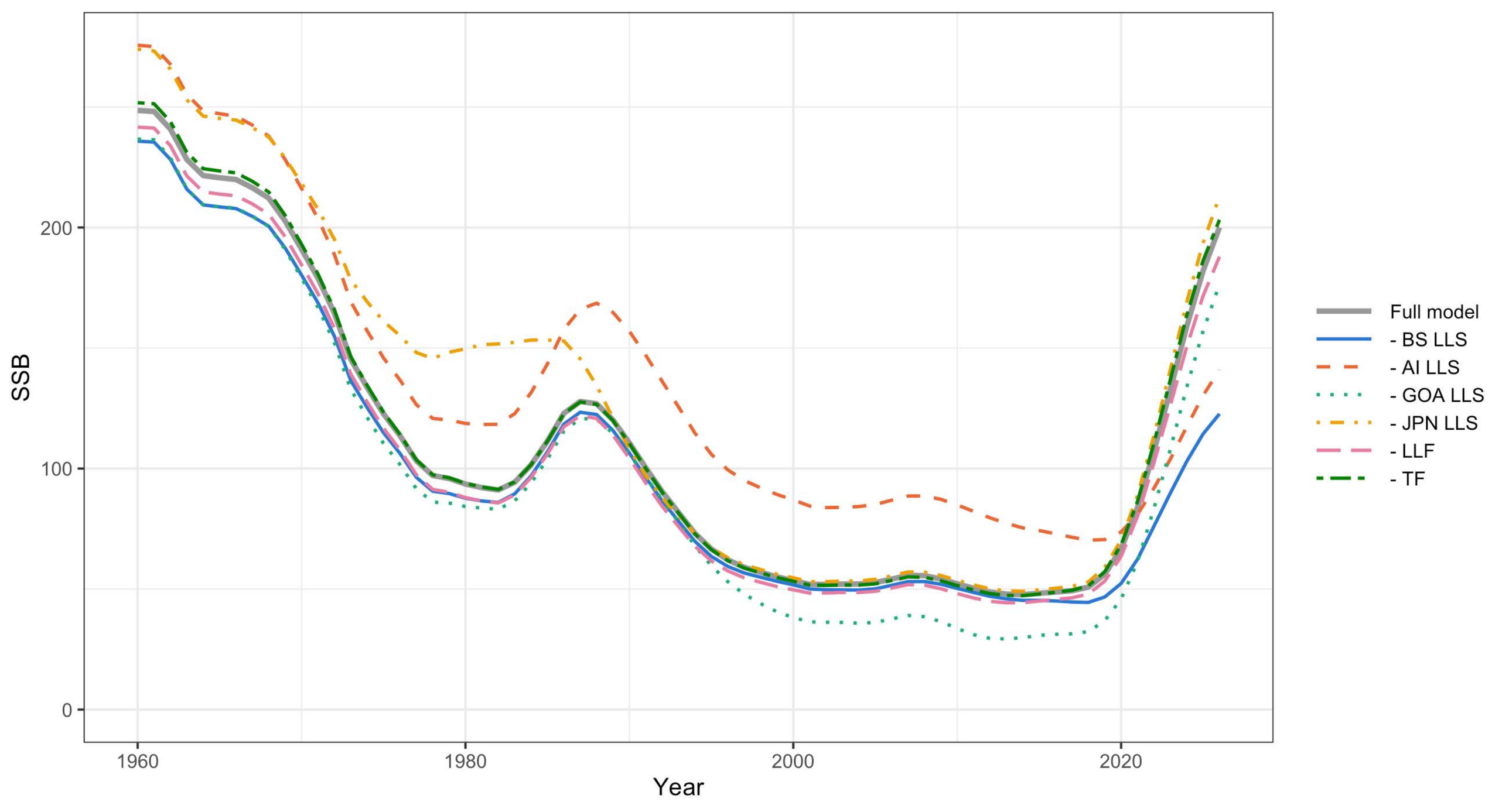
Male

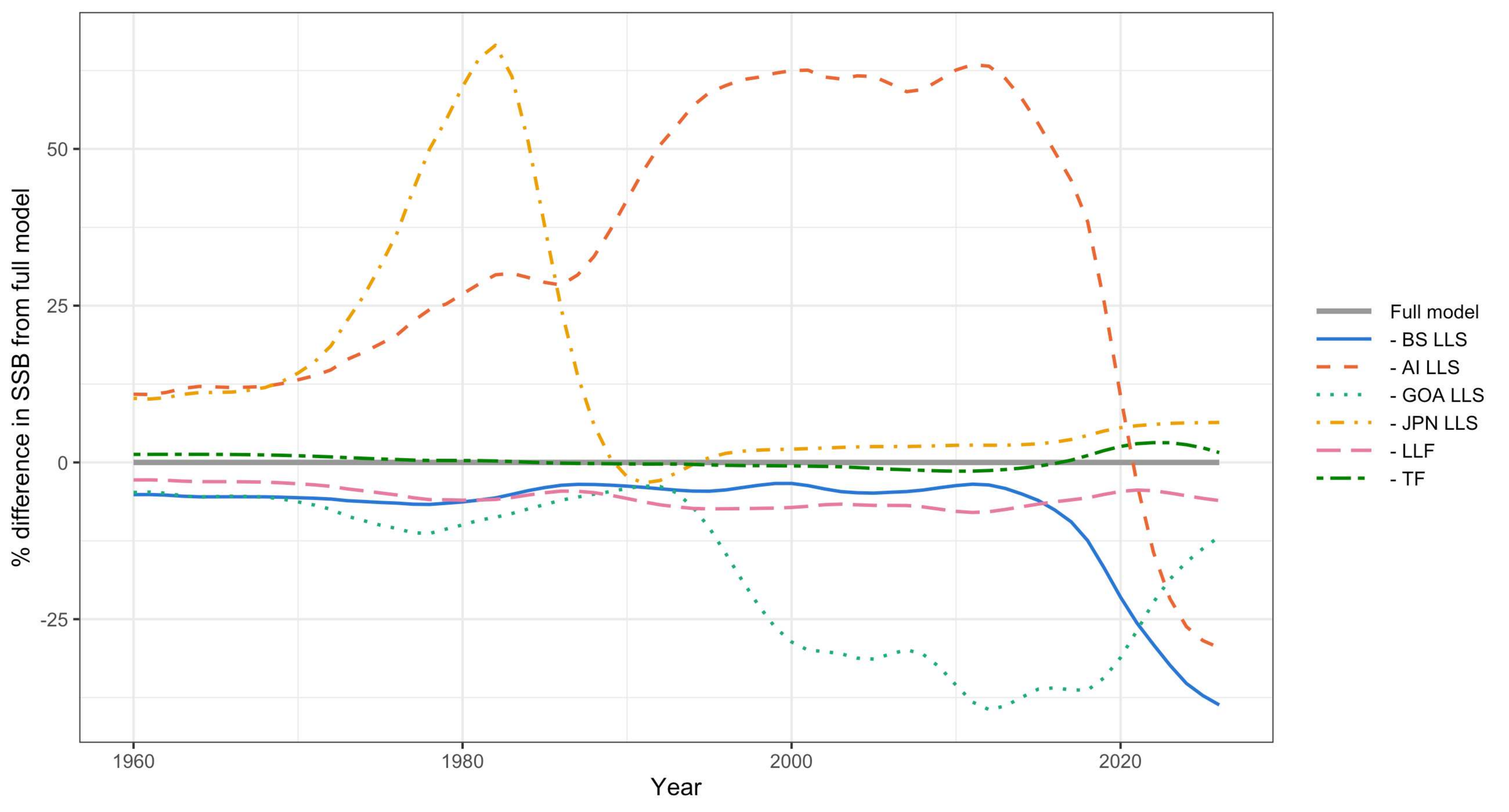
AK

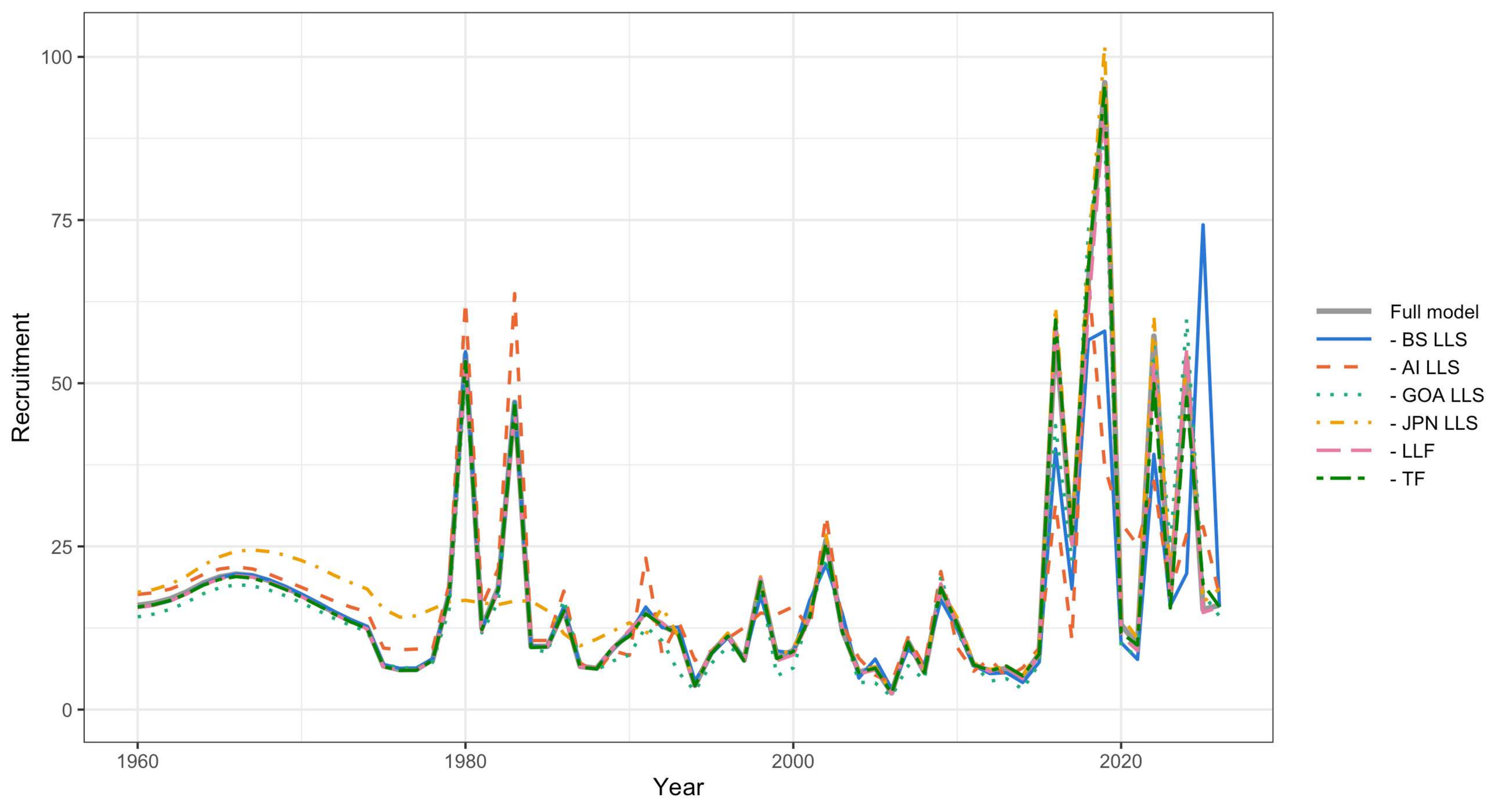
Proportions

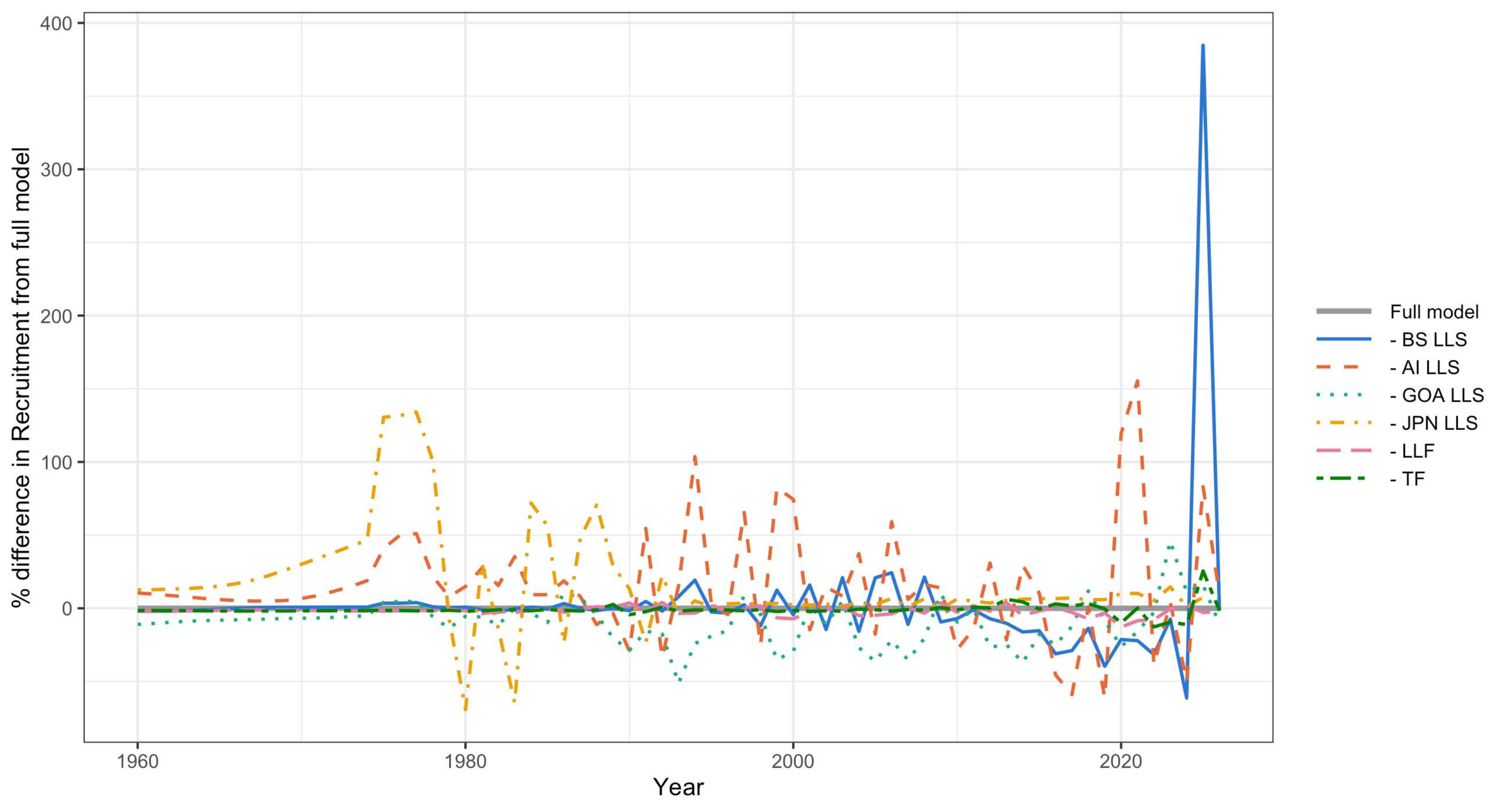
Age


Fixed Gear Fishery Age Comps Pearson residuals
|Pearson resid|
1
2
Sign
Negative
Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025


SSB
200
100
0
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF

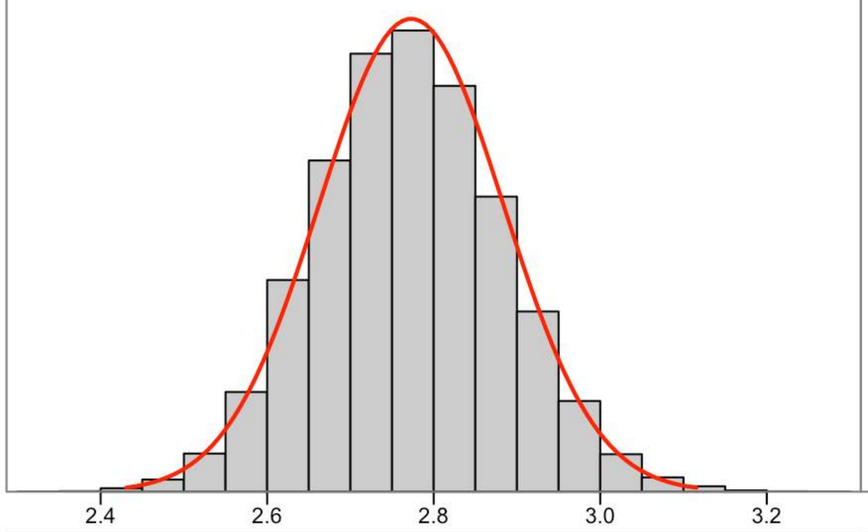

% difference in SSB from full model
50
25
0
-25
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF


Recruitment
100
75
50
25
0
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF


400
300
200
100
0
% difference in Recruitment from full model
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF

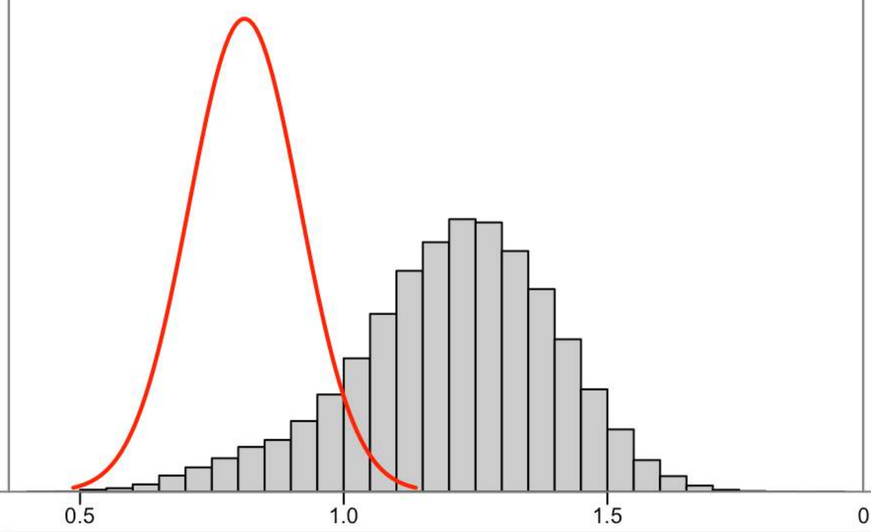
ln_global_R0

ESS=78531
Rhat=1



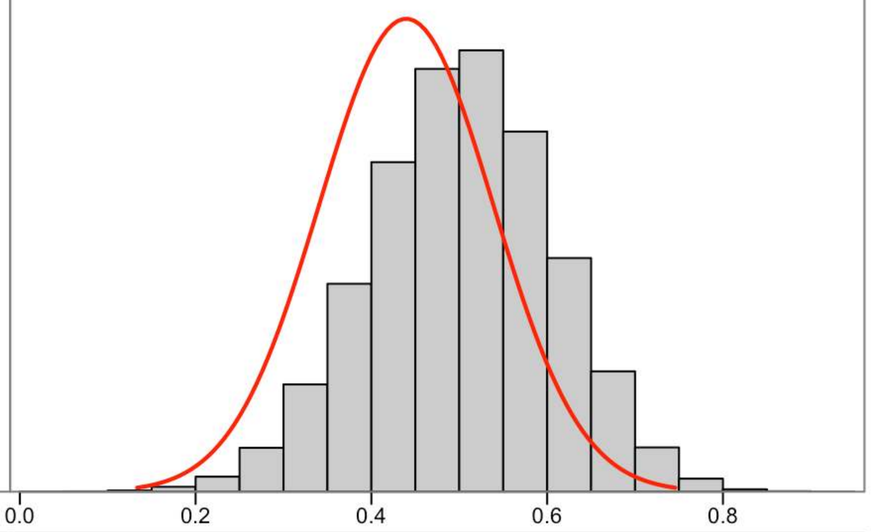
ln_srv_q[1]

ESS=14
Rhat=1.19



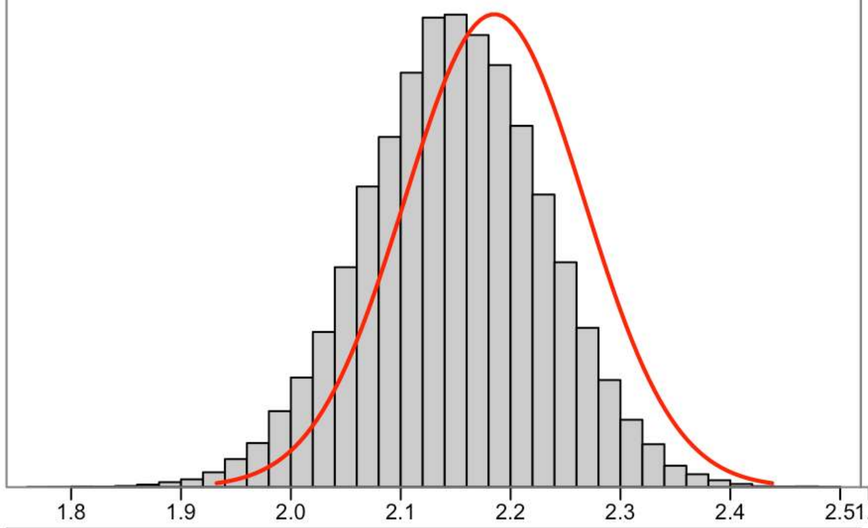
ln_srv_q[2]

ESS=261
Rhat=1.01



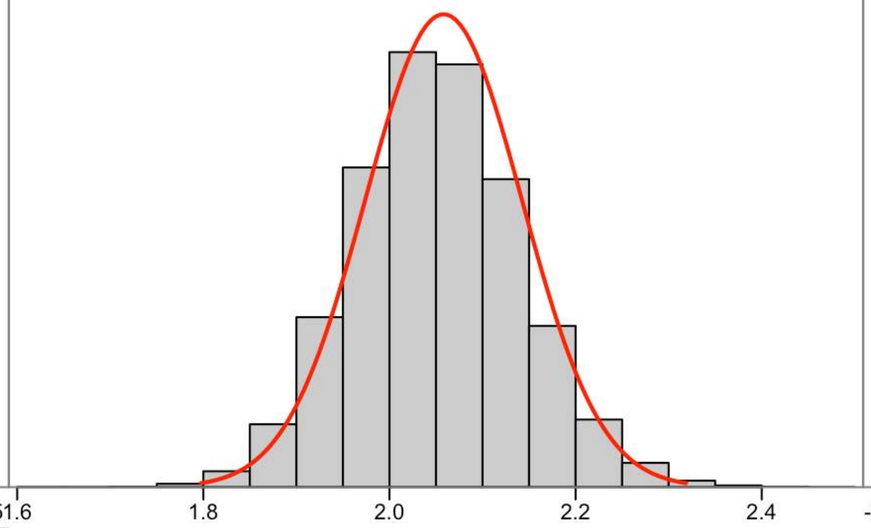
ln_srv_q[3]

ESS=31403
Rhat=1



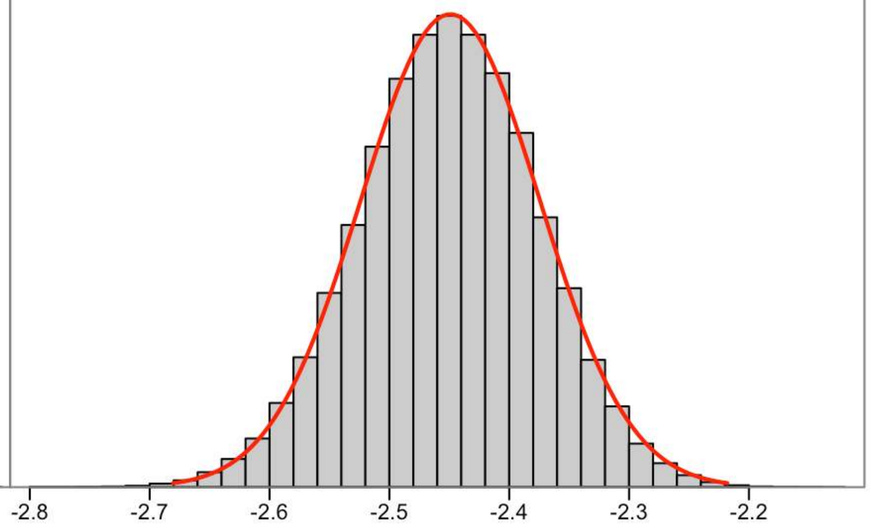
ln_srv_q[4]

ESS=5803
Rhat=1



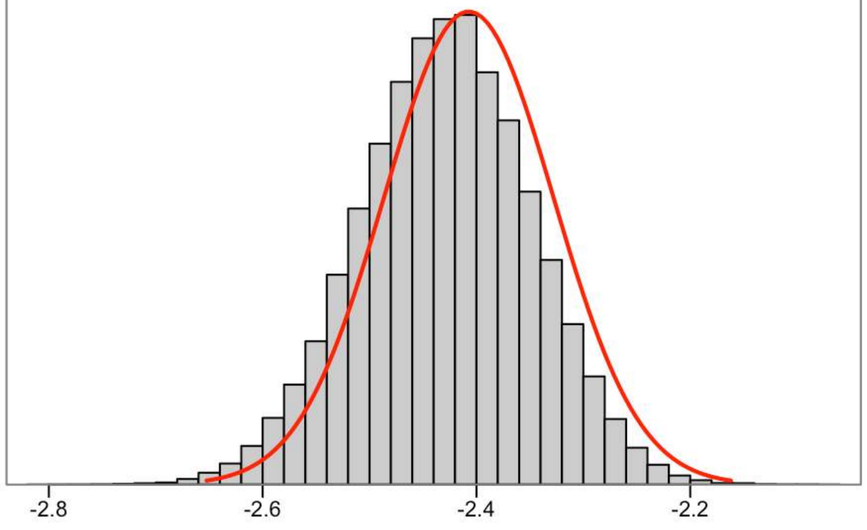
ln_M[1]

ESS=76919
Rhat=1

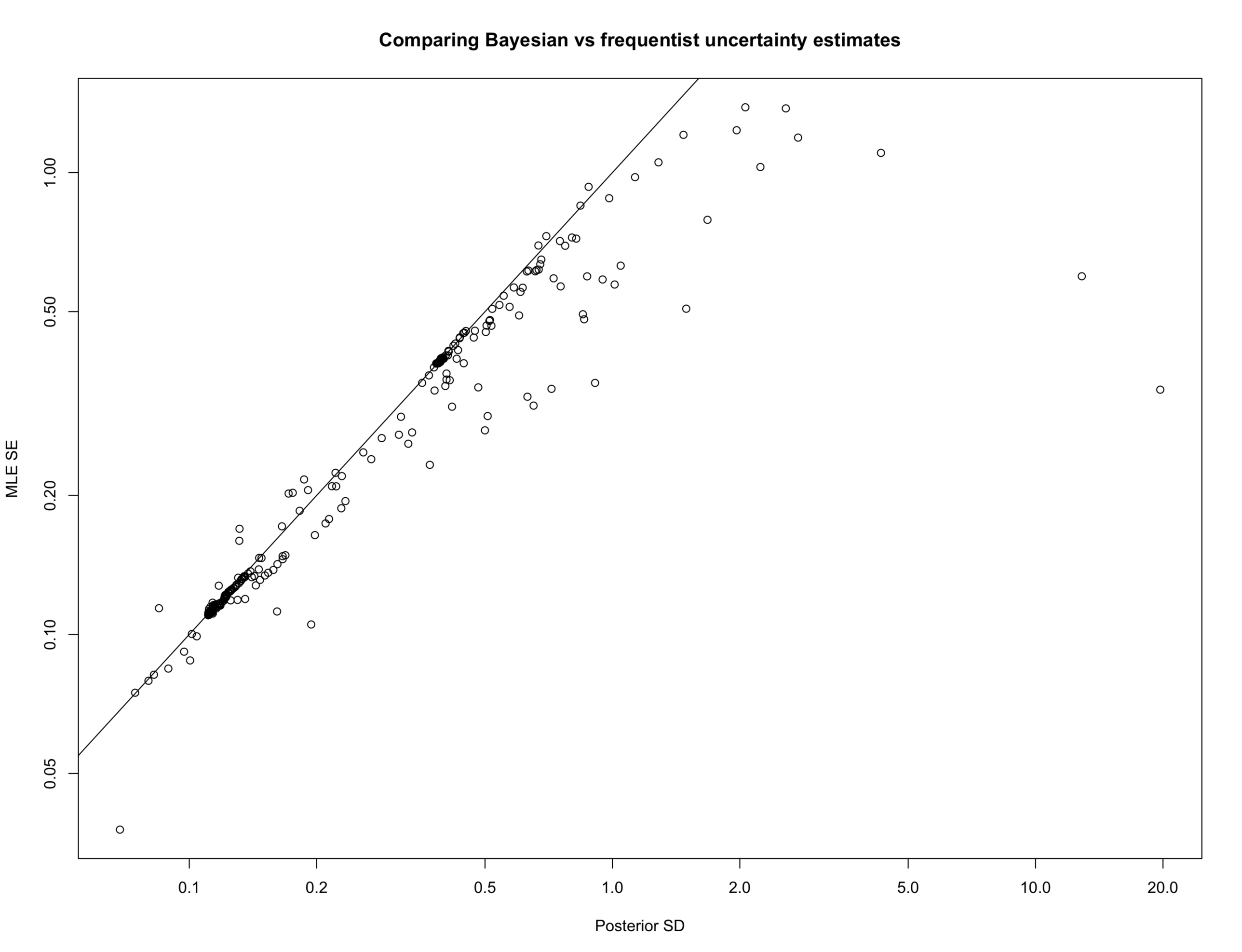


ln_M[2]

ESS=64042
Rhat=1



ln_global_R0	ln_srv_q[1]	ln_srv_q[2]	ln_srv_q[3]	ln_srv_q[4]	ln_M[1]	ln_M[2]	
ESS=78531 Rhat=1.00	-0.24	-0.46	-0.53	-0.48	0.7	0.66	ln_global_R0
	ESS=14 Rhat=1.19	0.5	0.26	0.25	-0.2	-0.23	ln_srv_q[1]
		ESS=261 Rhat=1.01	0.61	0.51	-0.41	-0.43	ln_srv_q[2]
			ESS=31403 Rhat=1.00	0.7	-0.49	-0.49	ln_srv_q[3]
				ESS=5803 Rhat=1.00	-0.39	-0.43	ln_srv_q[4]
					ESS=76919 Rhat=1.00	0.59	ln_M[1]
						ESS=64042 Rhat=1.00	ln_M[2]


Comparing Bayesian vs frequentist uncertainty estimates
MLE SE
1.00
0.50
0.20
0.10
0.05
0.1
0.2
0.5
1.0
2.0
5.0
10.0
20.0
Posterior SD

accept_stat
1.00
0.75
0.50
0.25
0.00
divergent
1.00
0.75
0.50
0.25
0.00
energy
50000
40000
30000
20000
10000
0
value
log_stepsize
2.5
0.0
-2.5
-5.0
n_leapfrog
1000
750
500
250
0
0
5000
10000
15000
20000
iteration
chain
1
2
3
4

88 divergent transitions after warmup

Standardized value (posterior SDs)

4

0

-4

Divergent

Non-divergent

fish_fixed_sel_pars[12]

srv_fixed_sel_pars[14]

srv_fixed_sel_pars[8]

ln_srv_q[1]

srv_fixed_sel_pars[3]

srv_fixed_sel_pars[6]

srv_fixed_sel_pars[7]

fish_fixed_sel_pars[2]

fish_fixed_sel_pars[4]

srv_fixed_sel_pars[12]

srv_fixed_sel_pars[8]

ESS=8
Rhat=1.57

-10
-5
0
5

srv_fixed_sel_pars[7]

ESS=8
Rhat=1.49

0
20
40
60
80

ln_srv_q[1]

ESS=14
Rhat=1.19

0.5
1.0
1.5

srv_fixed_sel_pars[3]

ESS=30
Rhat=1.08

-4
-2
0
2
4
6

srv_fixed_sel_pars[10]

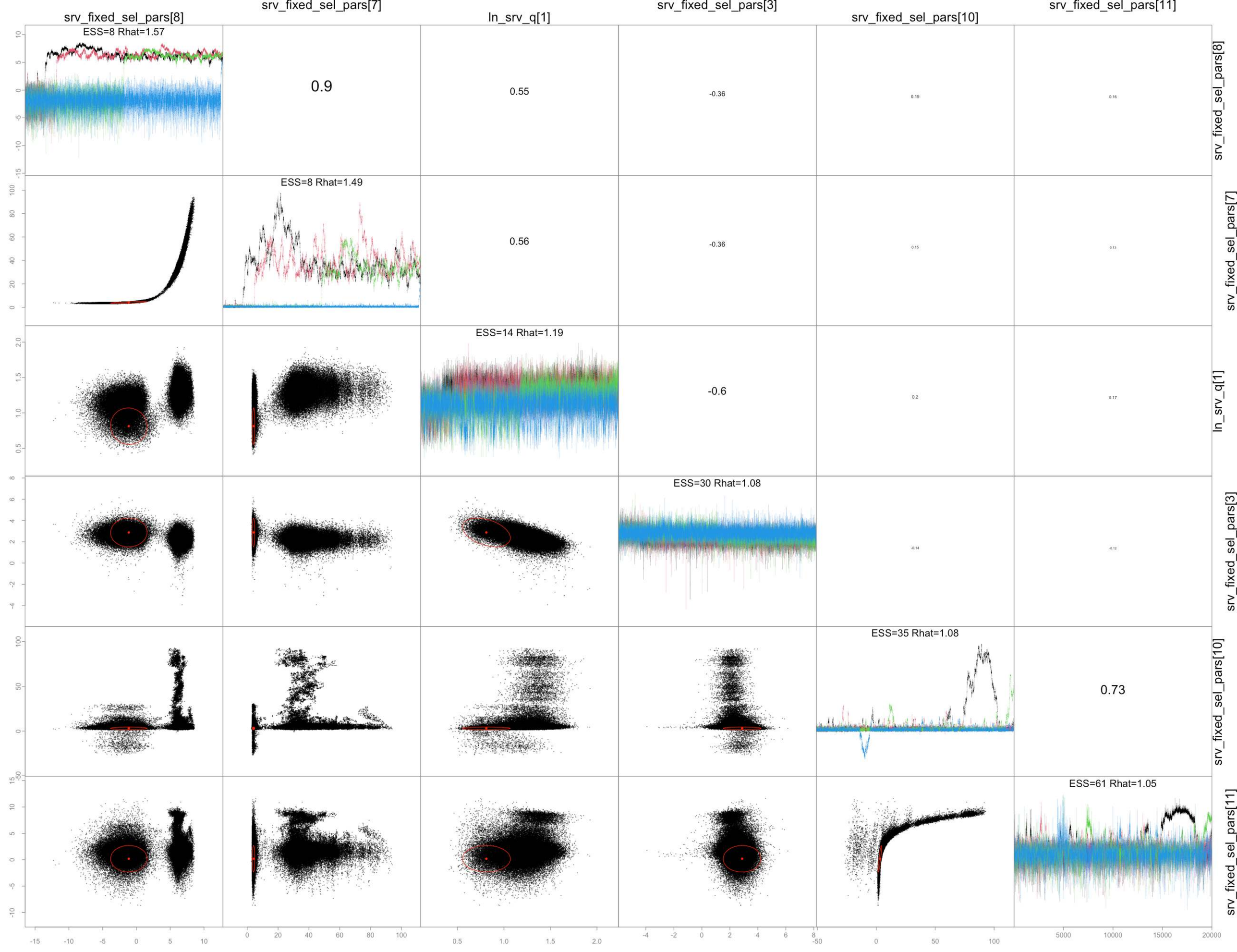
ESS=35
Rhat=1.08

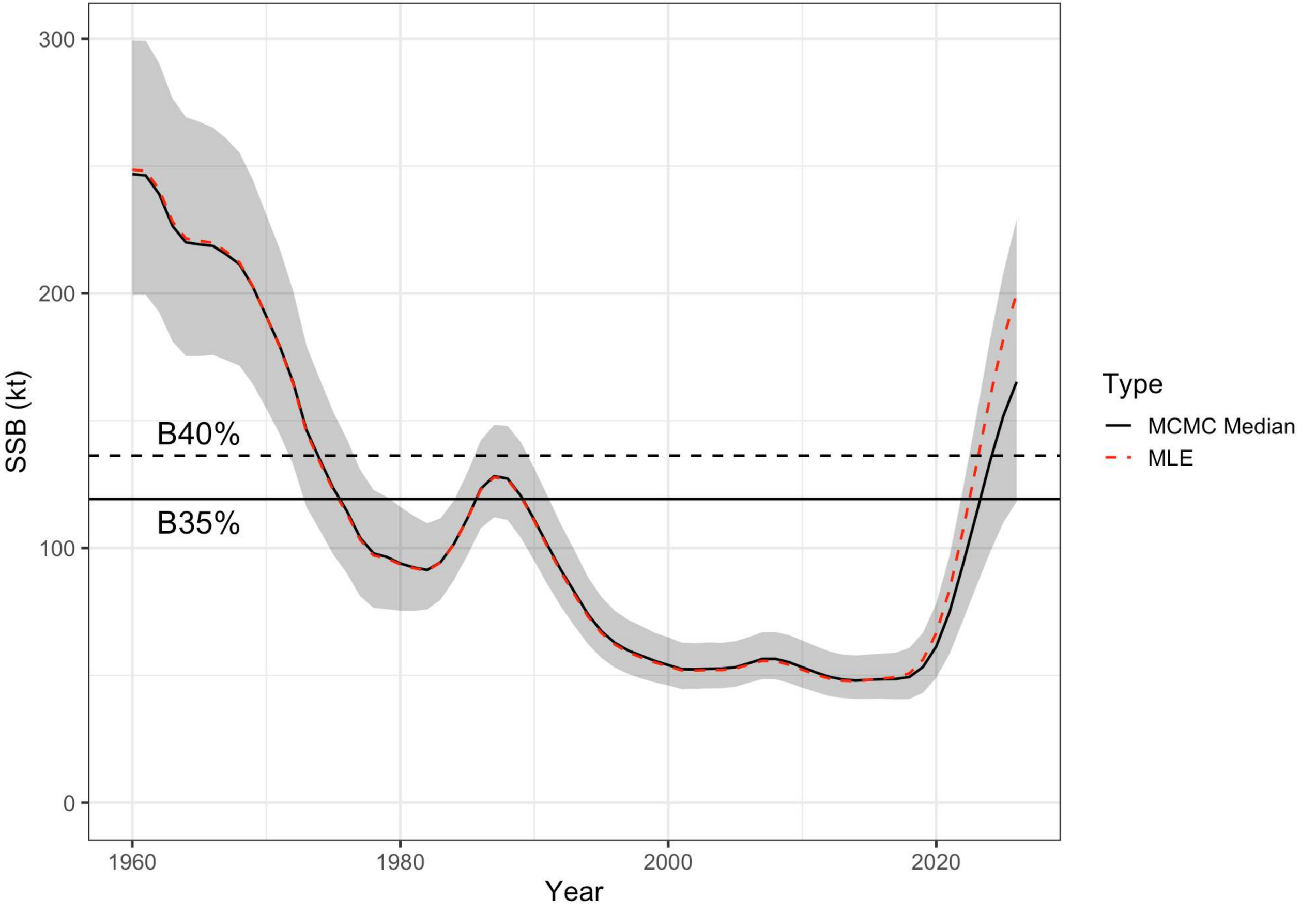
-20
0
20
40
60
80

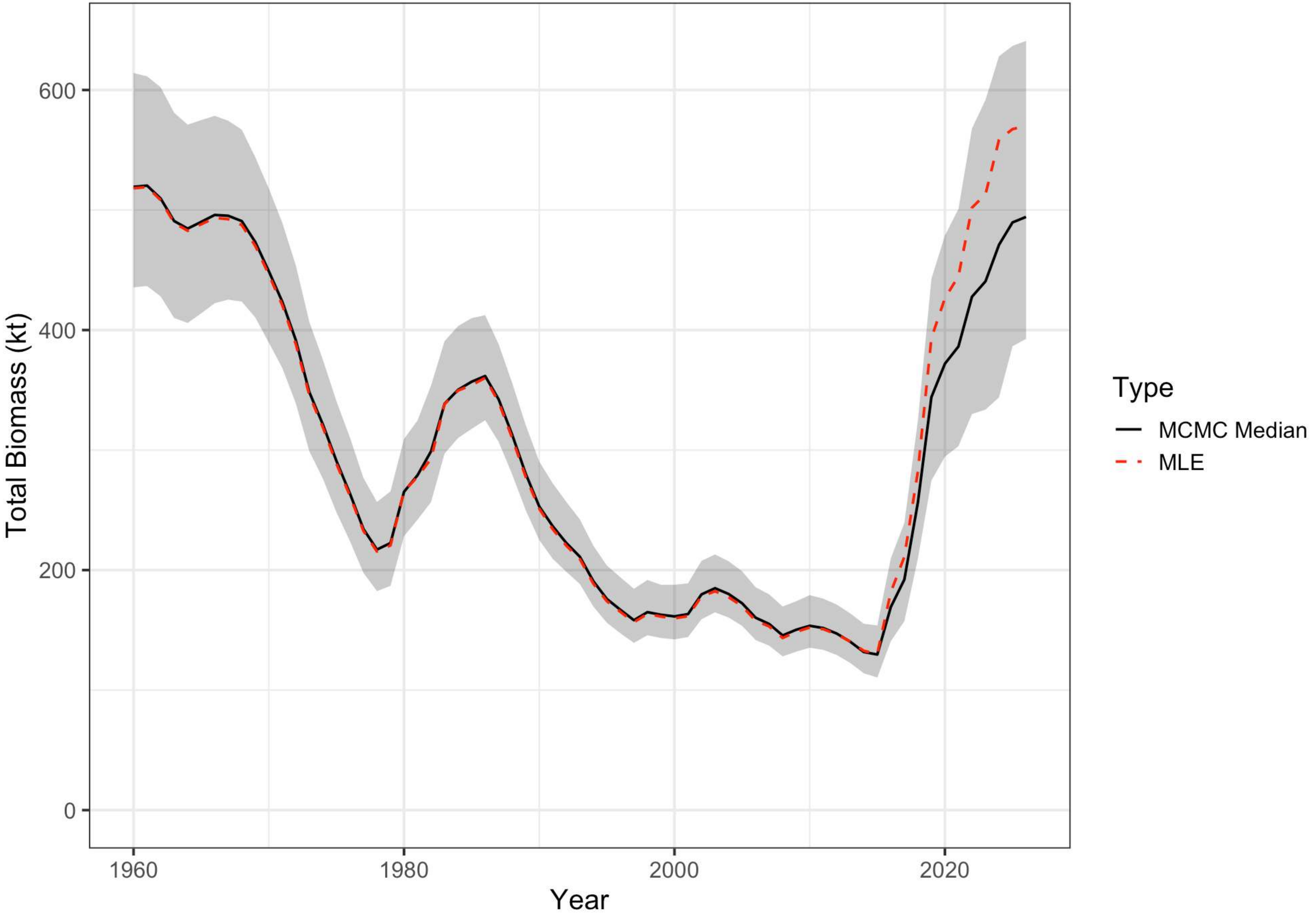
srv_fixed_sel_pars[11]

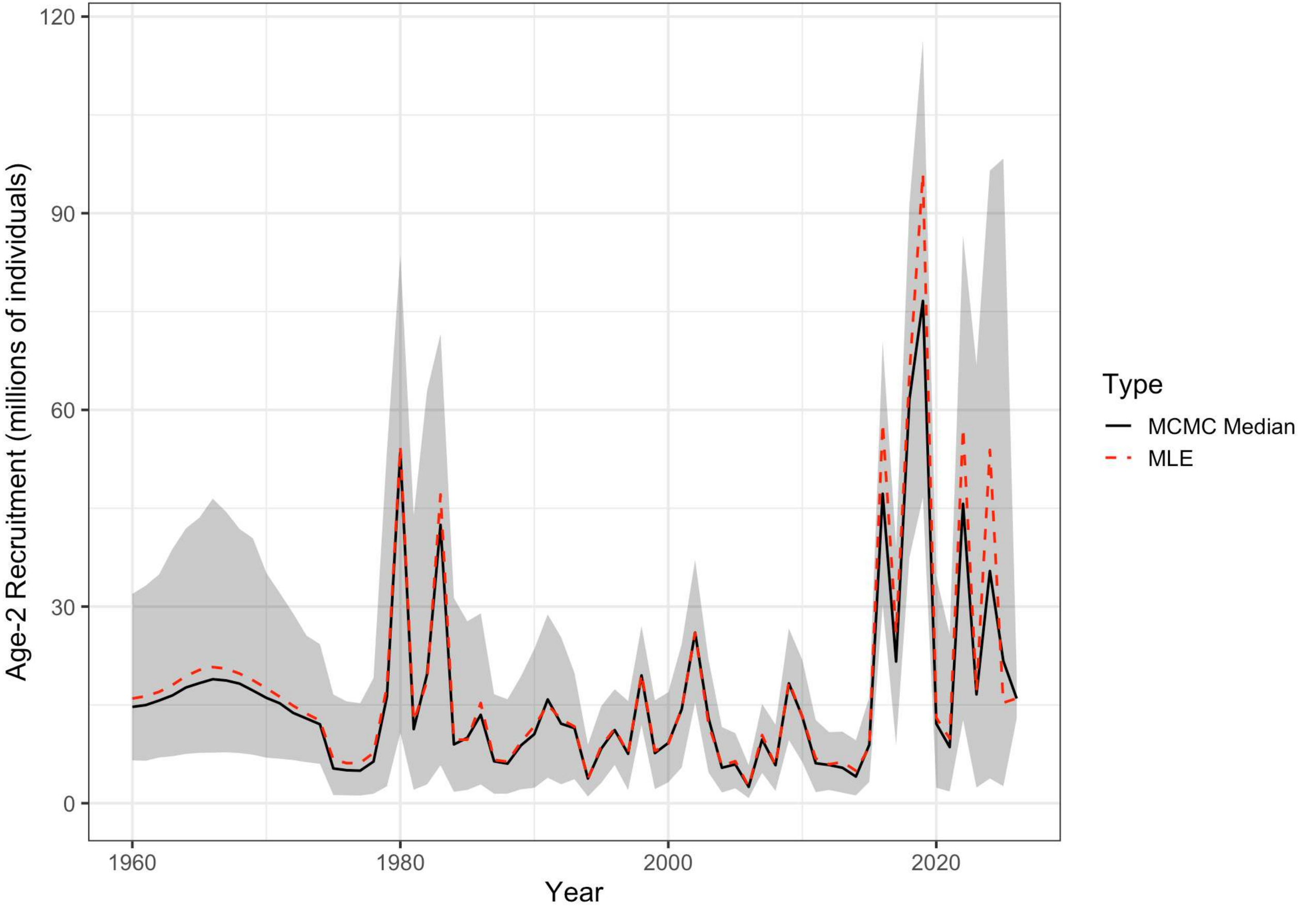
ESS=61
Rhat=1.05

-5
0
5
10


srv_fixed_sel_pars[8]
srv_fixed_sel_pars[7]
ln_srv_q[1]
srv_fixed_sel_pars[3]
srv_fixed_sel_pars[10]
srv_fixed_sel_pars[11]
ESS=8 Rhat=1.57
0.9
0.55
-0.36
0.19
0.16
ESS=8 Rhat=1.49
0.56
-0.36
0.15
0.13
ESS=14 Rhat=1.19
-0.6
0.2
0.17
ESS=30 Rhat=1.08
-0.14
-0.12
ESS=35 Rhat=1.08
0.73
ESS=61 Rhat=1.05


300
200
100
0
SSB (kt)
B40%
B35%
1960
1980
2000
2020
Year
Type
MCMC Median
MLE


600
400
200
0
Total Biomass (kt)
1960
1980
2000
2020
Year
Type
MCMC Median
MLE


120
90
60
30
0
Age-2 Recruitment (millions of individuals)
1960
1980
2000
2020
Year
Type
MCMC Median
MLE

Type: Age, Catch, Index, jnLL, Length, Penalty, Prior

26.2_FAA_LLS_BS_Dome_DN

Likelihood

250

0

-250

Catch nLL

Fishery Age nLL

Fishery Length nLL

Fishing Mortality Penalty

Initial Age Penalty

jnLL

M Prior

Recruitment Penalty

Selectivity Penalty

Survey Age nLL

Survey Index nLL

Survey Length nLL

Likelihood Component

Numbers-at-Age

Sex — Female ···· Male

Age- 1 | Age- 2 | Age- 3 | Age- 4 | Age- 5 | Age- 6

Age- 7 | Age- 8 | Age- 9 | Age- 10 | Age- 11 | Age- 12

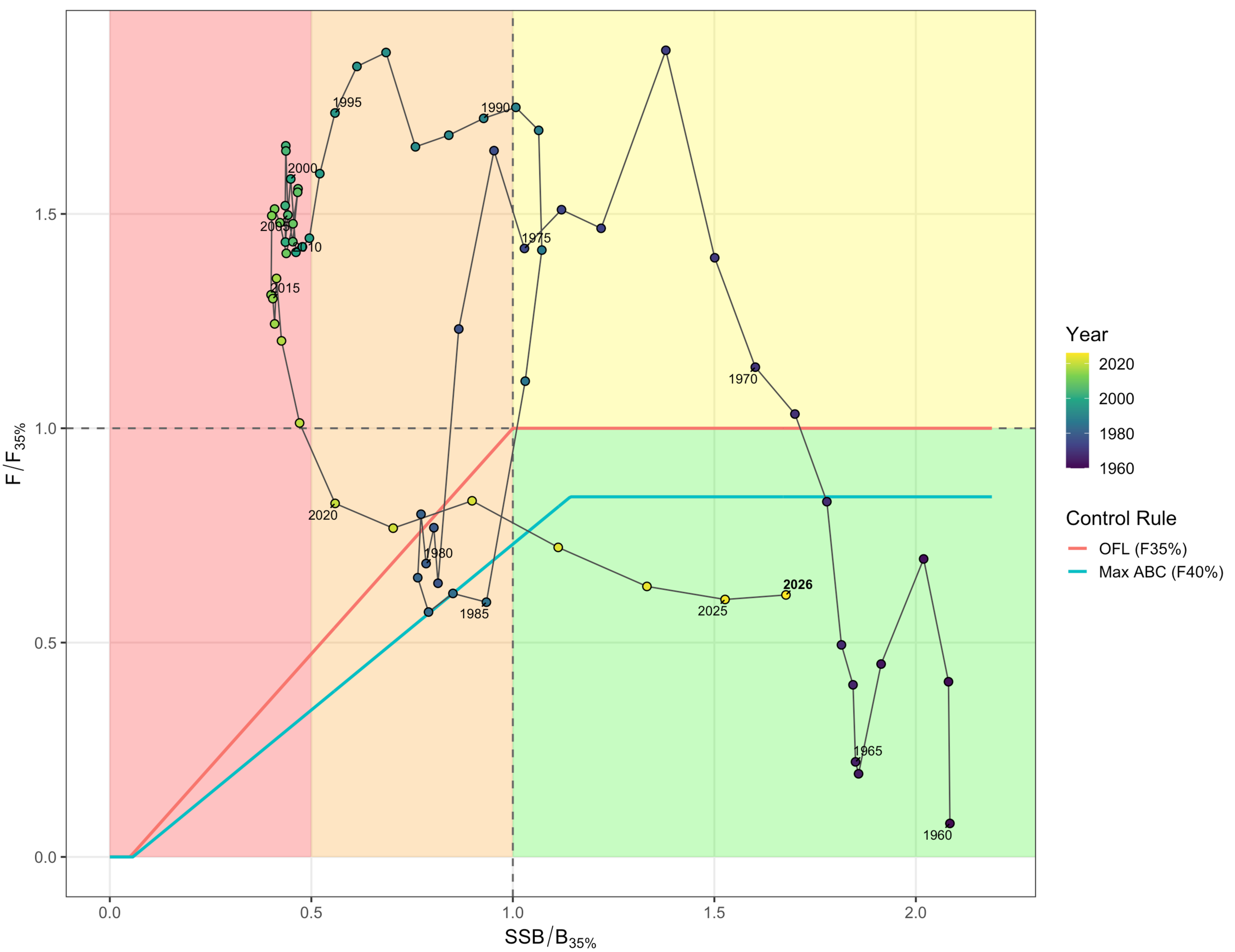
Age- 13 | Age- 14 | Age- 15 | Age- 16 | Age- 17 | Age- 18

Age- 19 | Age- 20 | Age- 21 | Age- 22 | Age- 23 | Age- 24

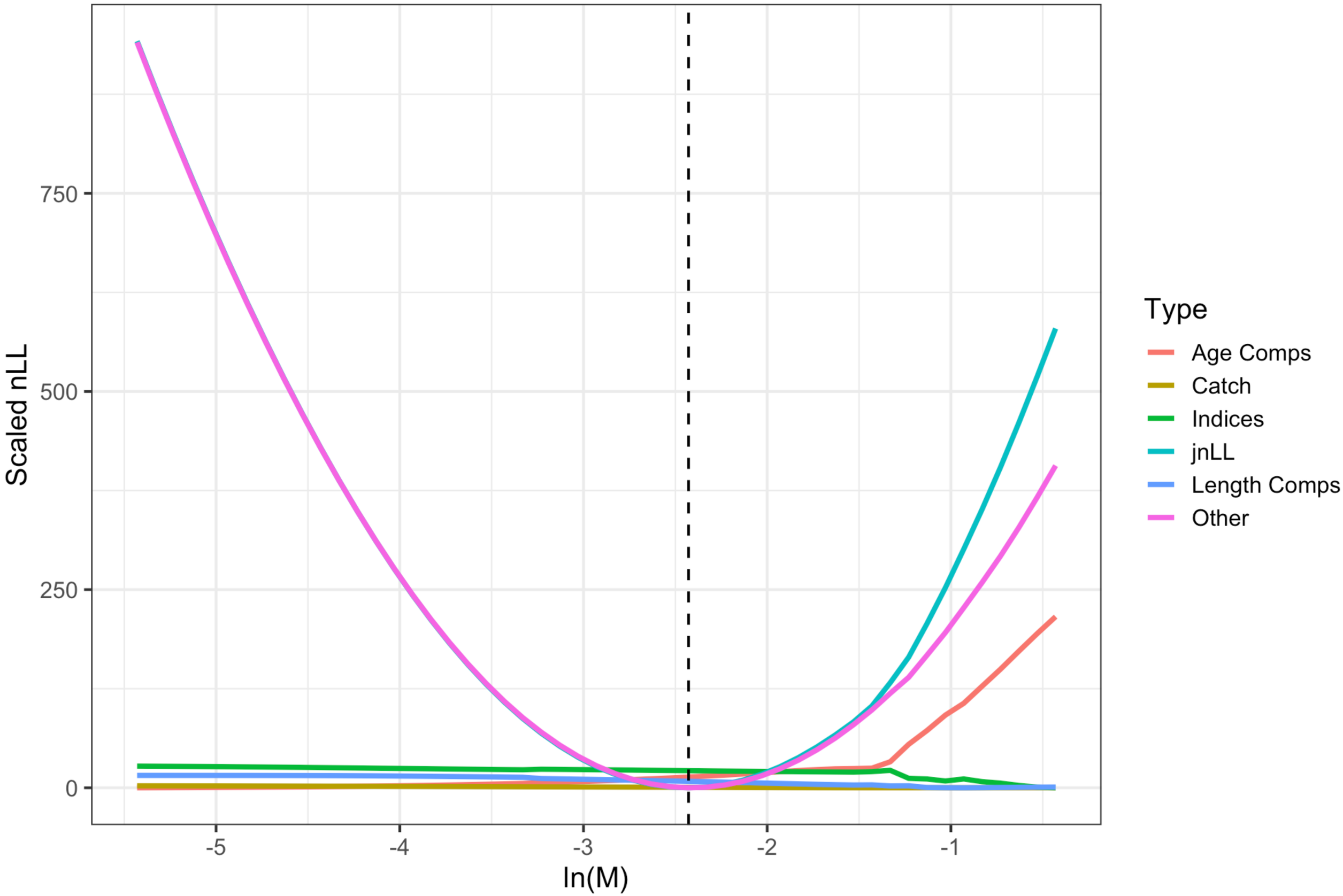
Age- 25 | Age- 26 | Age- 27 | Age- 28 | Age- 29 | Age- 30

Numbers-at-Age (millions)

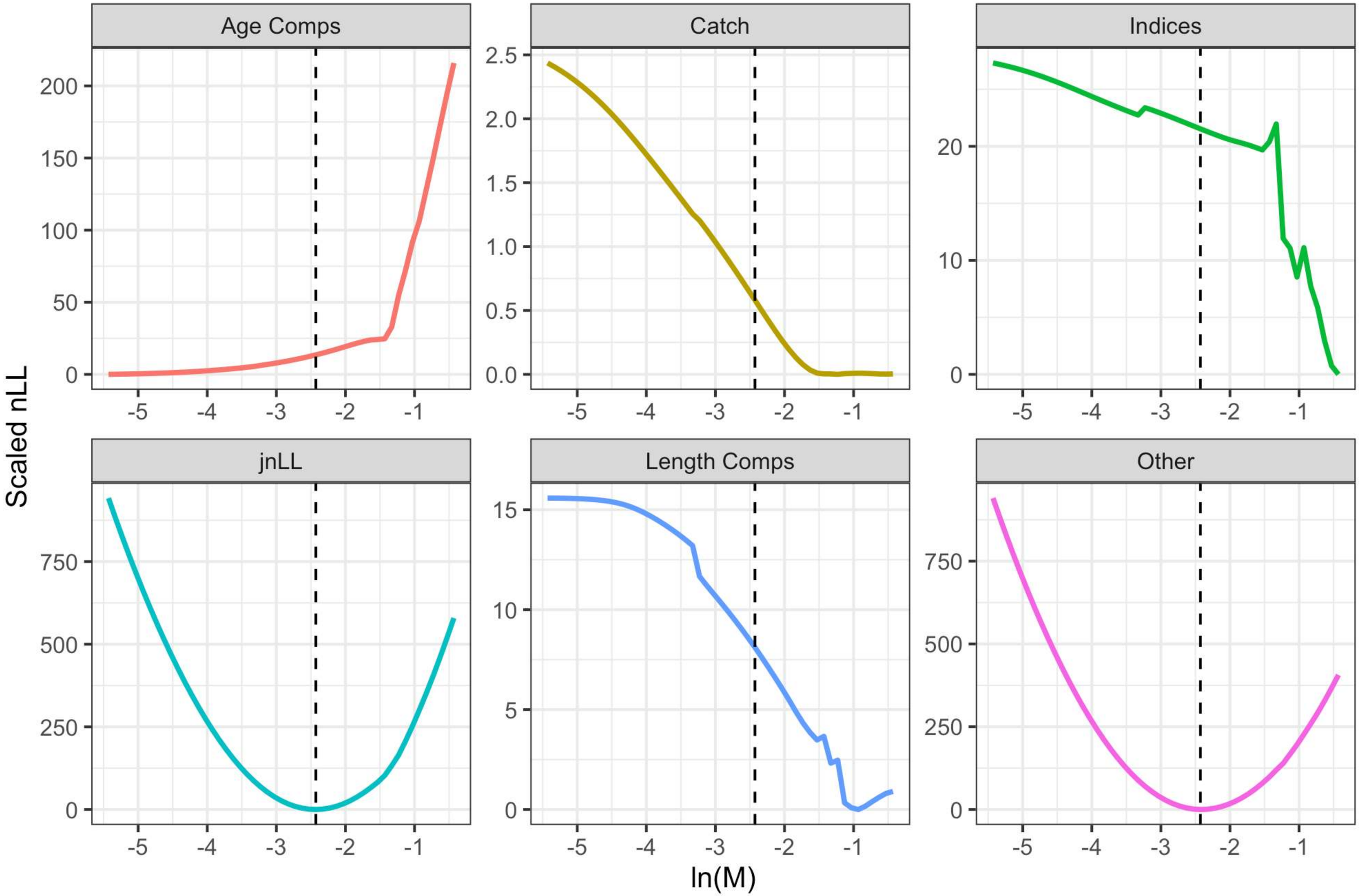
Year


F/F35%
SSB/B35%
0.0
0.5
1.0
1.5
2.0
1995
1990
2000
2005
2010
2015
1975
1970
2020
1980
1985
2025
2026
1965
1960
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)

Natural mortality profile



Natural mortality profile



Recruitment (R0) profile

Scaled nLL

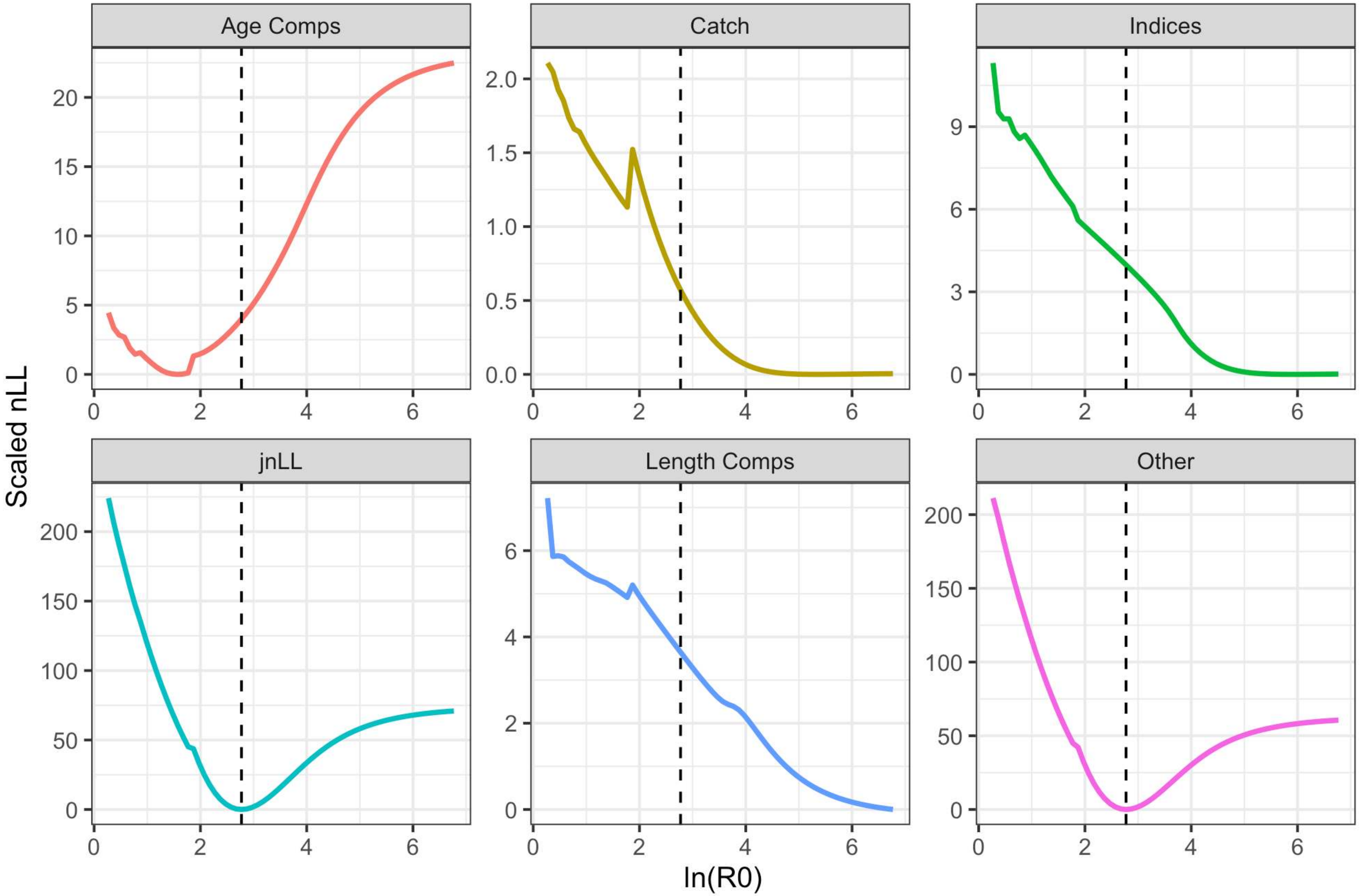
200
150
100
50
0

0
2
4
6

ln(R0)

Type
Age Comps
Catch
Indices
jnLL
Length Comps
Other

Recruitment (R0) profile



Retrospective Year
1 2 3 4 5 6 7

Recruitment

SSB

Mohns Rho: -0.2354

Mohns Rho: 0.0479

AK

0.5

0.0

-0.5

Relative Difference from Terminal Year

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

Year

AK

Recruitment (millions)

100
75
50
25

0
1
2
3
4
5
6
7

Years since cohort was last estimated

Cohort
2016
2017
2018
2019
2020
2021
2022
2023

Retrospective Year
1 2 3 4 5 6 7

Recruitment

SSB

Mohns Rho: -0.2359

Mohns Rho: 0.0688

0.5

0.0

-0.5

Relative Difference from Terminal Year

AK

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

Year

AK

100
75
50
25

Recruitment (millions)

0
1
2
3
4
5
6
7

Years since cohort was last estimated

Cohort
2016
2017
2018
2019
2020
2021
2022
2023

Index Residuals Runs Test

BS LLS

p = 0.092

AI LLS

p = 0.272

GOA LLS

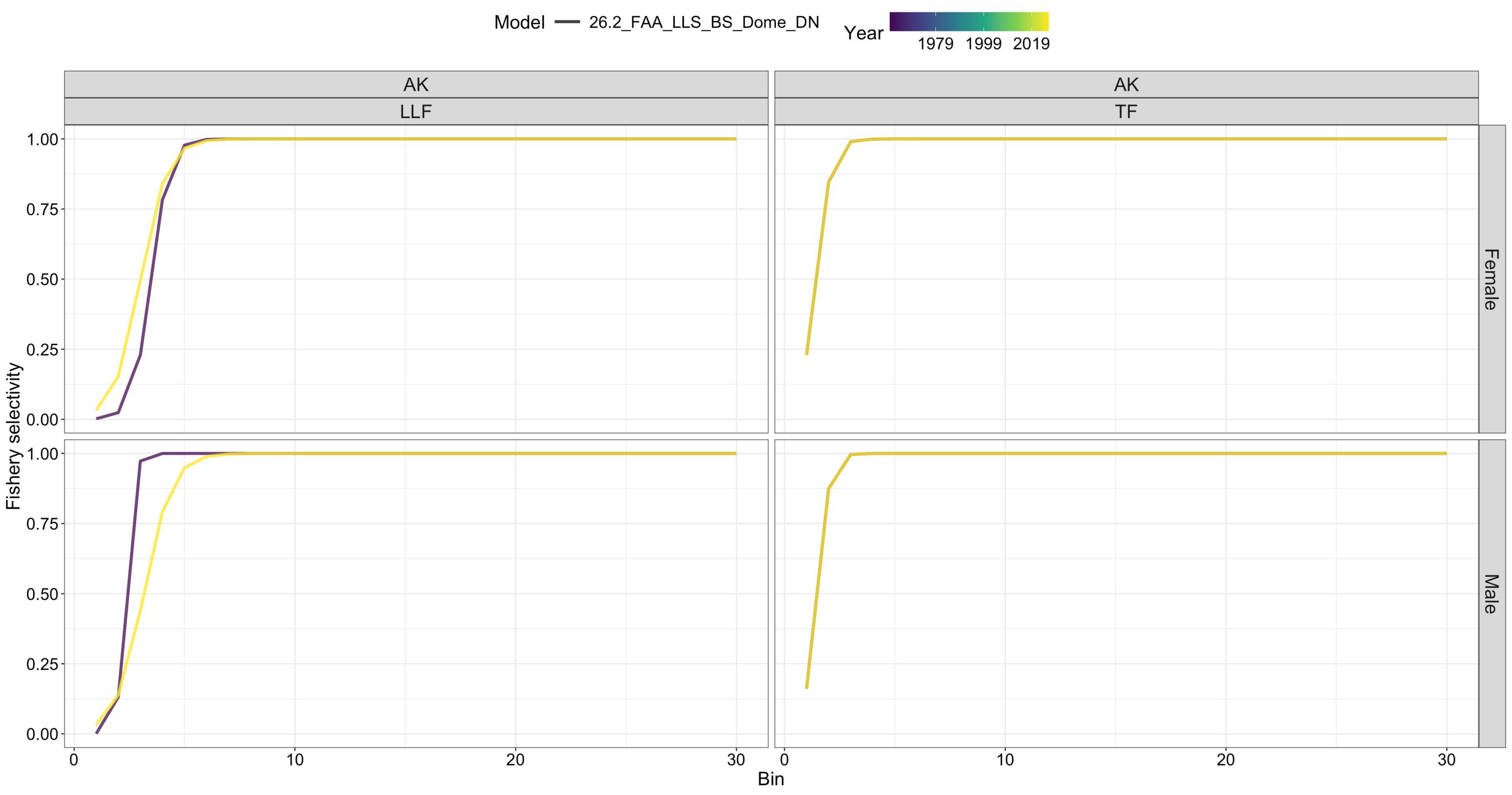
p = 0.032

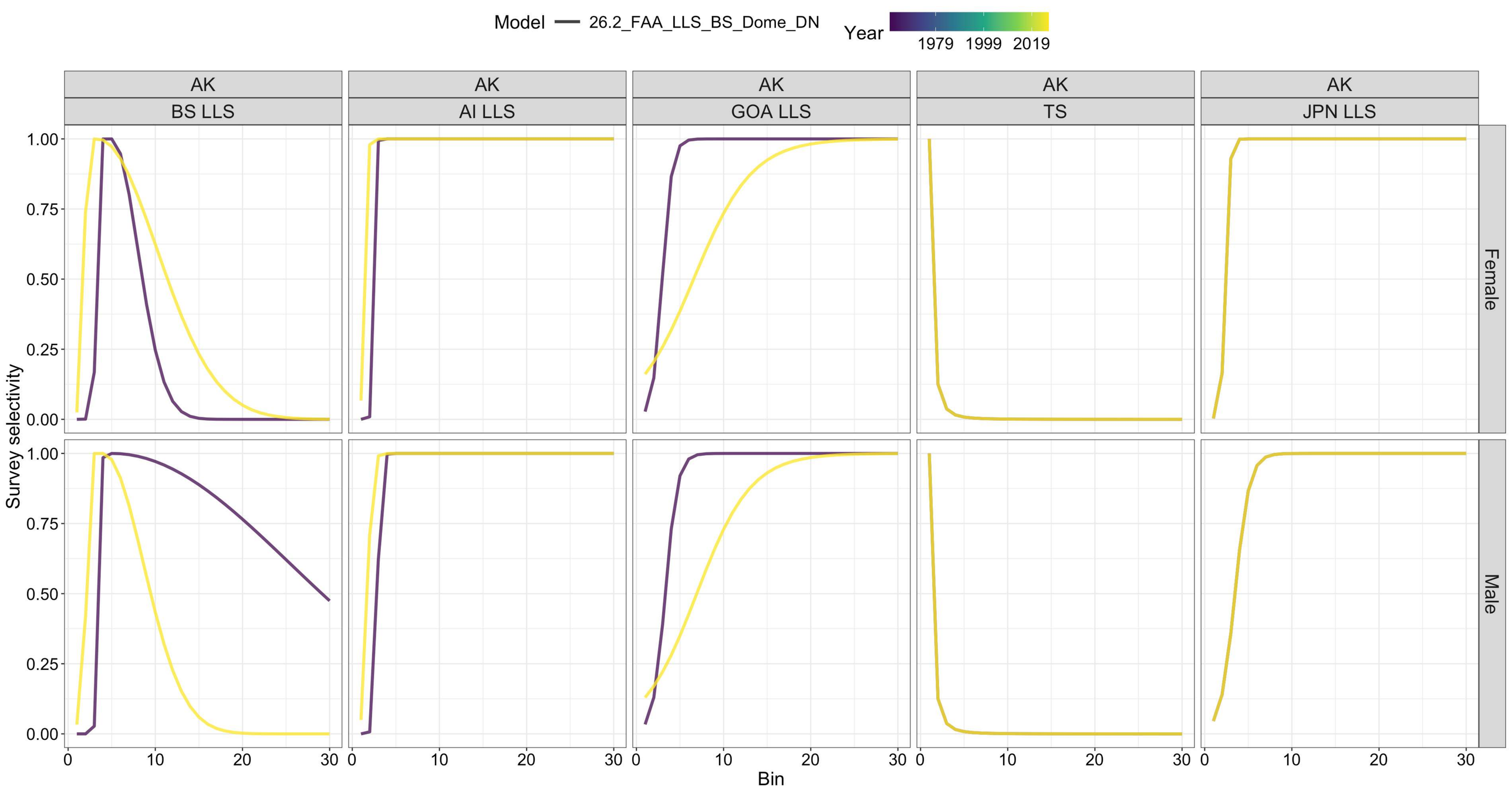
JPN LLS

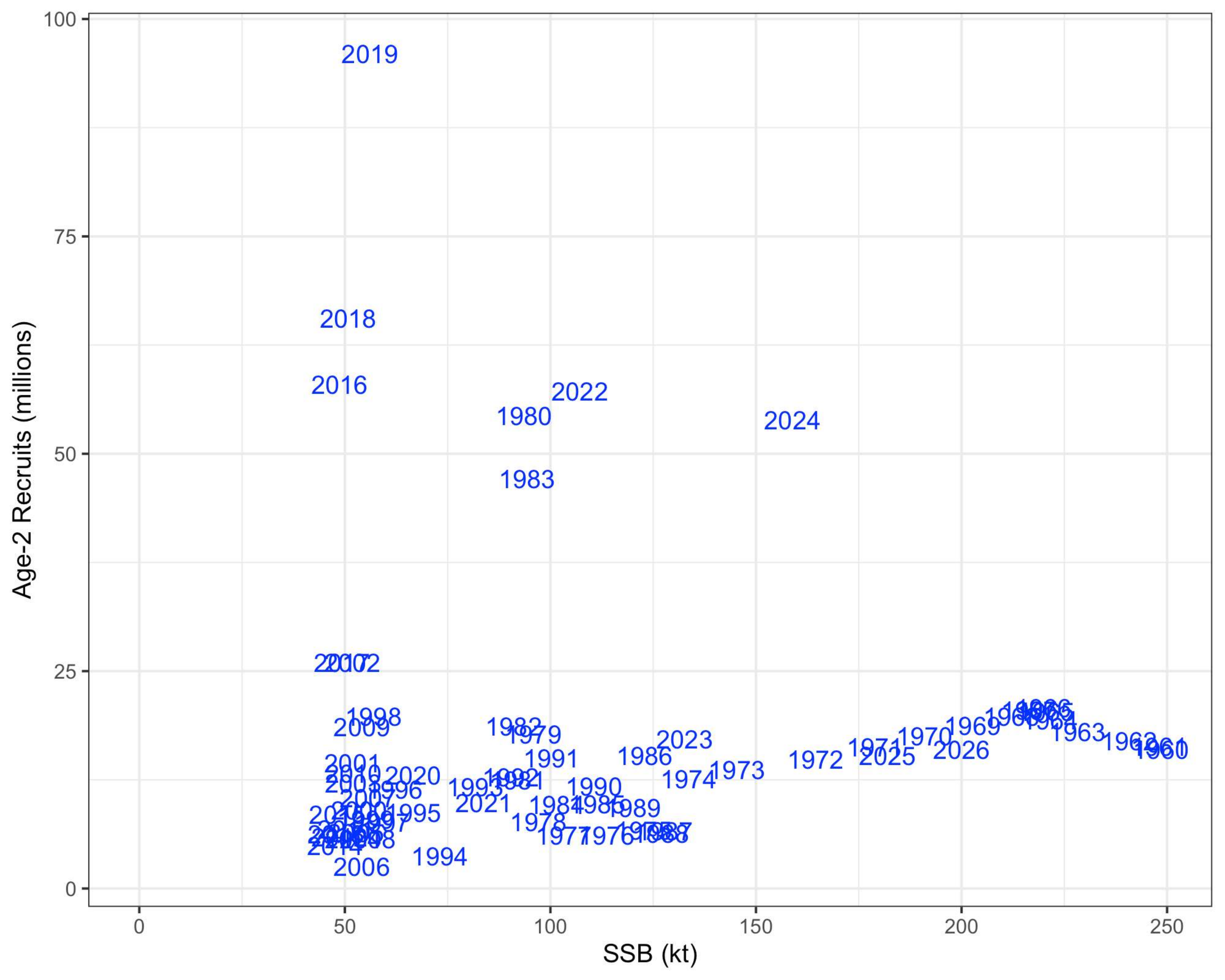
p = 0.723

Residuals

Year


Model
26.2_FAA_LLS_BS_Dome_DN
Year
1979
1999
2019
AK
LLF
AK
TF
Female
Male
Fishery selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Model
26.2_FAA_LLS_BS_Dome_DN
Year
1979
1999
2019
AK
BS LLS
AK
AI LLS
AK
GOA LLS
AK
TS
AK
JPN LLS
Female
Male
Survey selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Age-2 Recruits (millions)
SSB (kt)
100
75
50
25
0
0
50
100
150
200
250
2019
2018
2016
2022
1980
2024
1983
2009
1998
2006
1994
2023
1991
1986
1974
1973
1972
2025
2026
1971
1970
1969
1963
2020
2021
1990
1989
1995
1979
1982

Population and Catch Trends

Recruitment (millions of fish)

SSB or Catch (kt)

100
75
50
25
0

300
200
100
0

1960
1980
2000
2020

Year

Catch
Recruitment
SSB

Model 26.2_FAA_LLS_BS_Dome_DN

AK

Dynamic SSB0

Recruitment

SSB

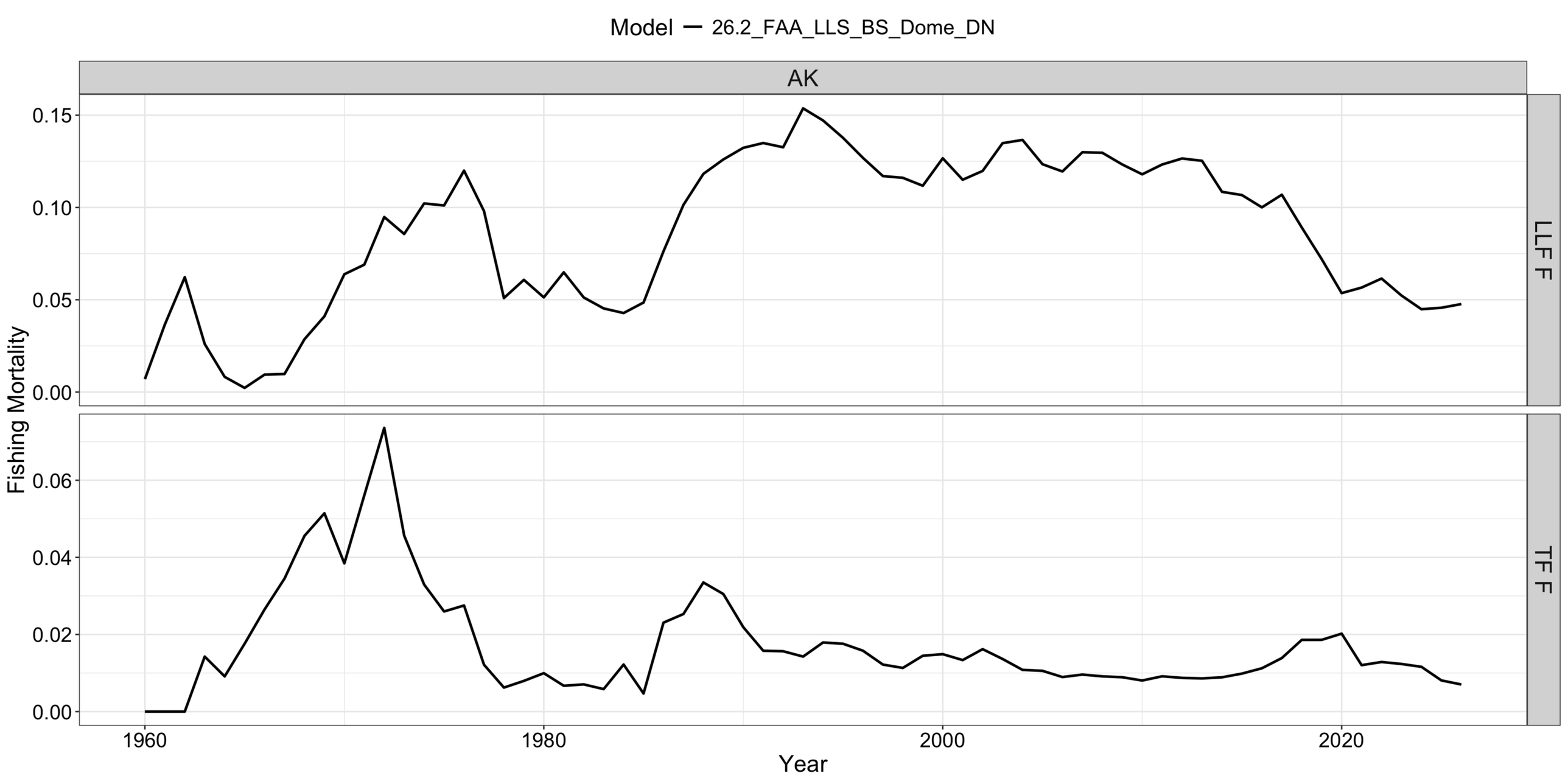
Total Biom

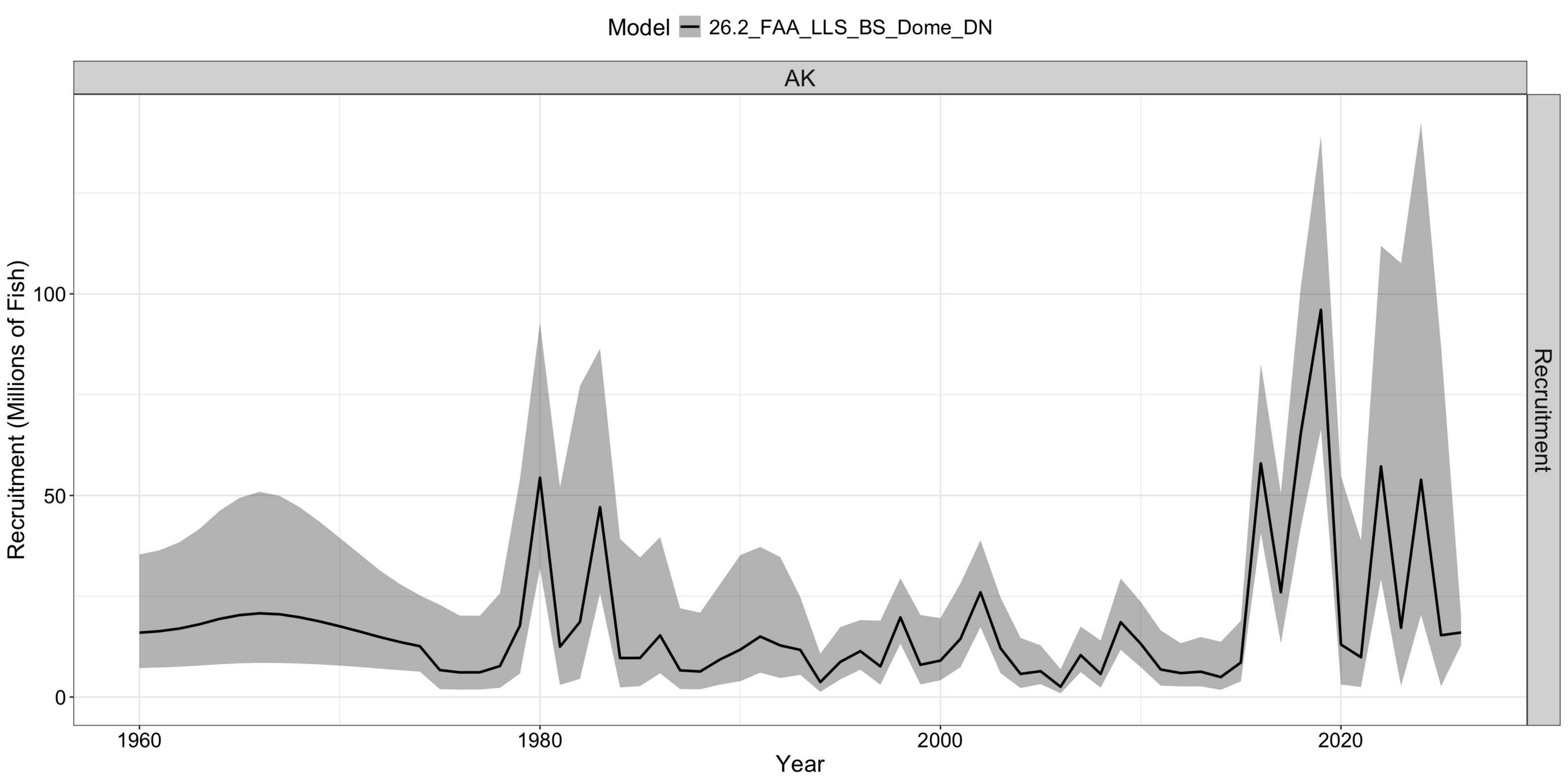
LLF F

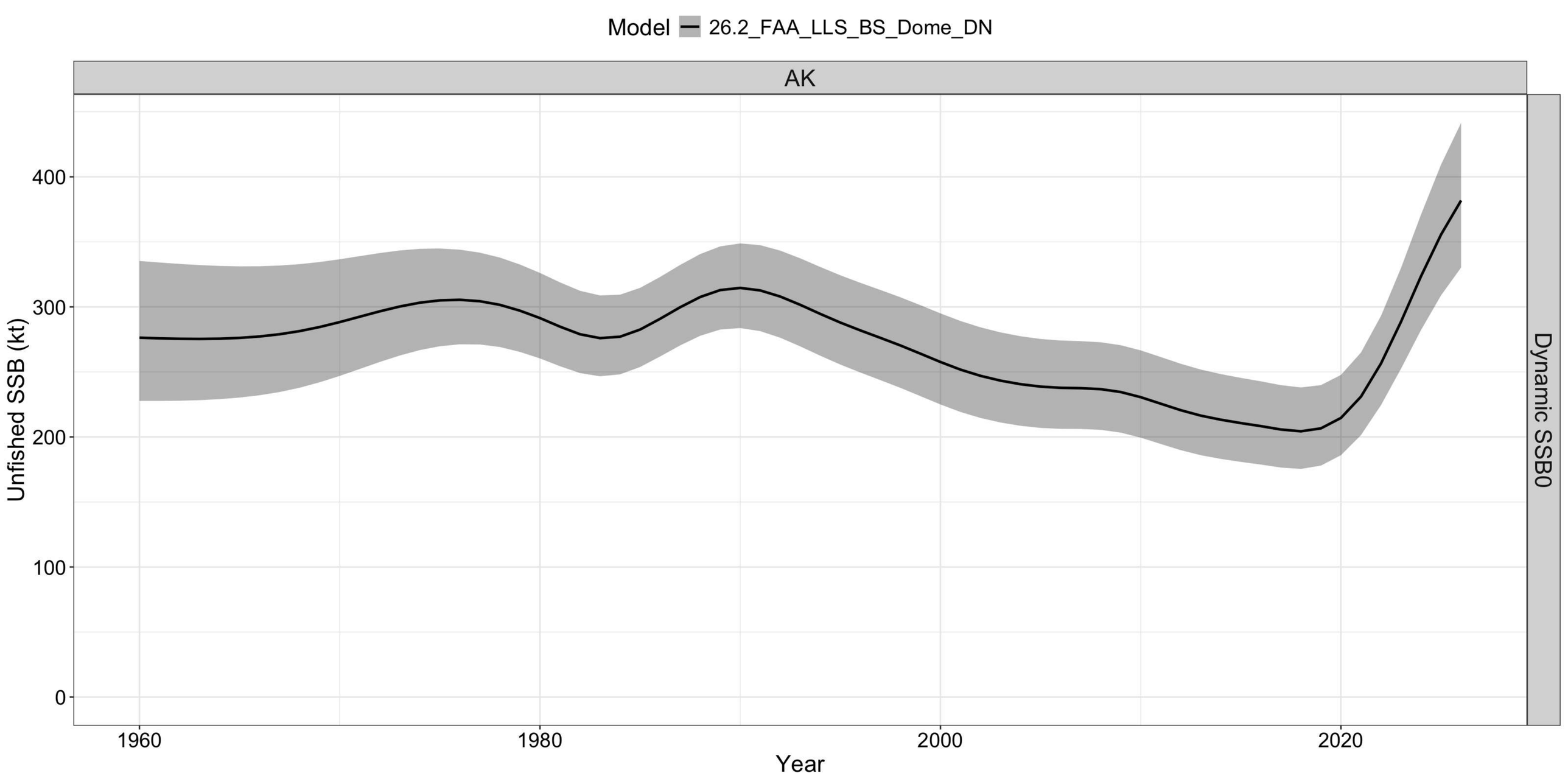
TF F

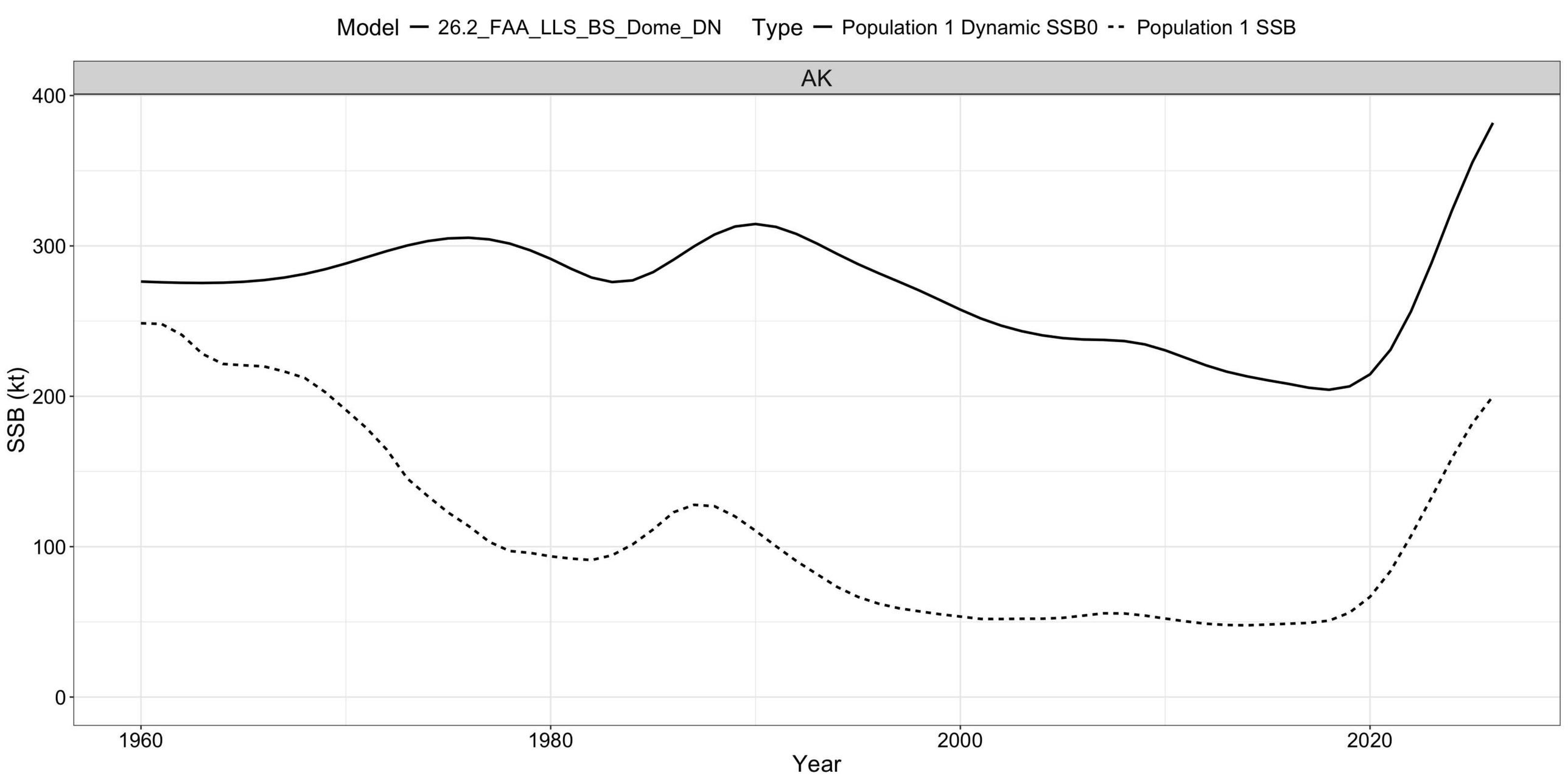
Value

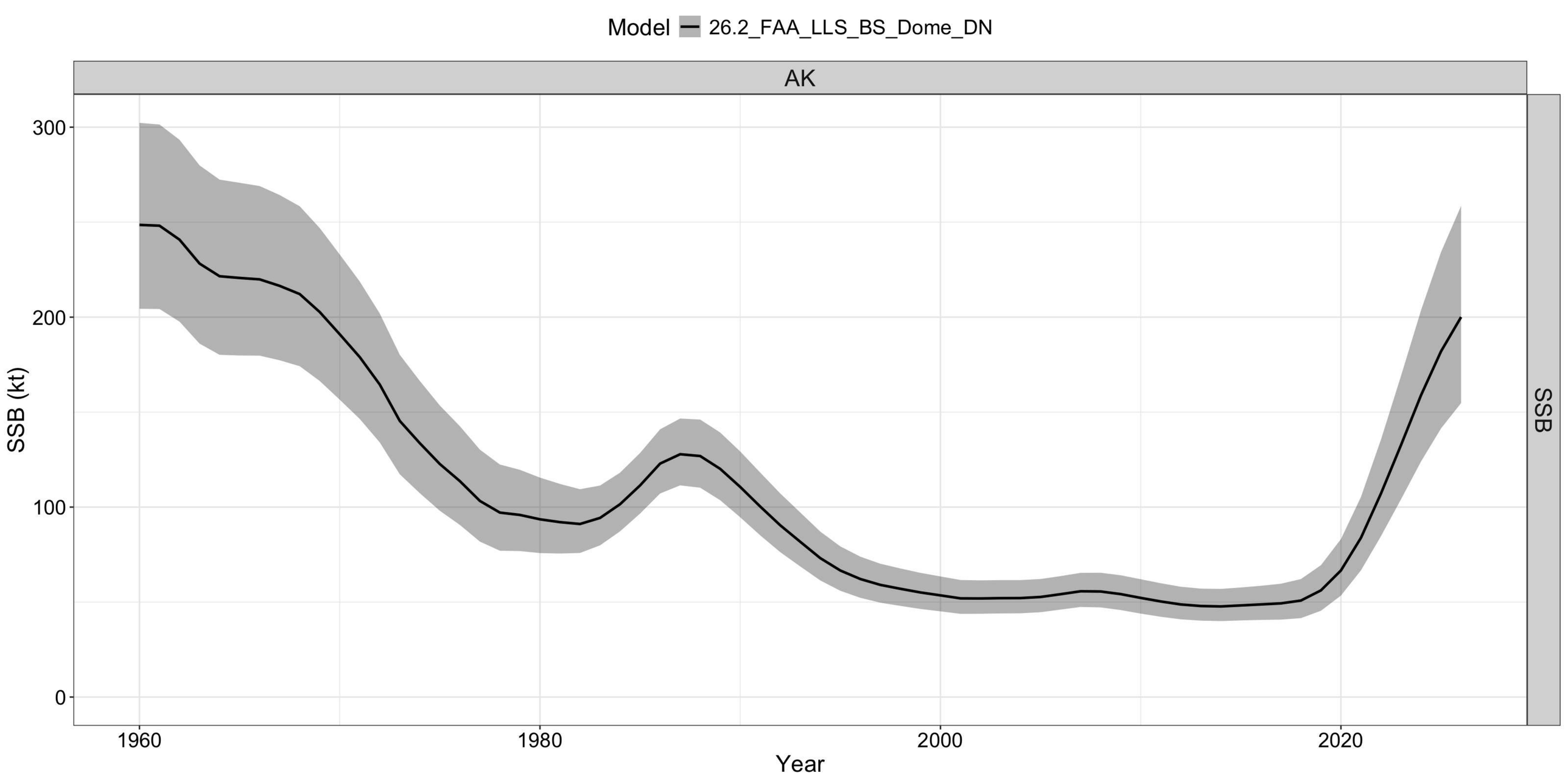
Year

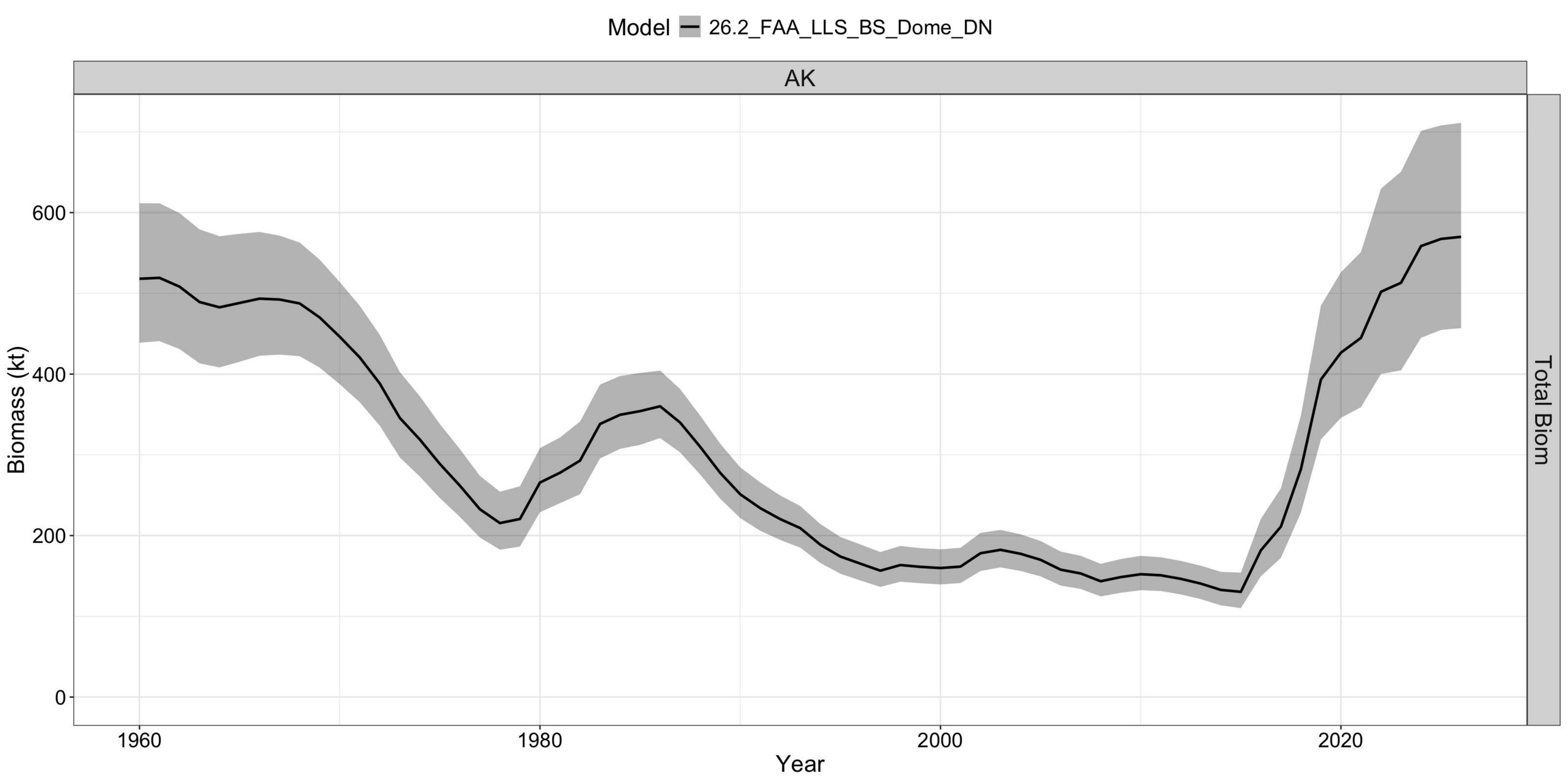

Model — 26.2_FAA_LLS_BS_Dome_DN
AK
LLF F
TF F
Fishing Mortality
0.15
0.10
0.05
0.00
0.06
0.04
0.02
0.00
1960
1980
2000
2020
Year


Model
26.2_FAA_LLS_BS_Dome_DN
AK
Recruitment (Millions of Fish)
100
50
0
1960
1980
2000
2020
Year
Recruitment

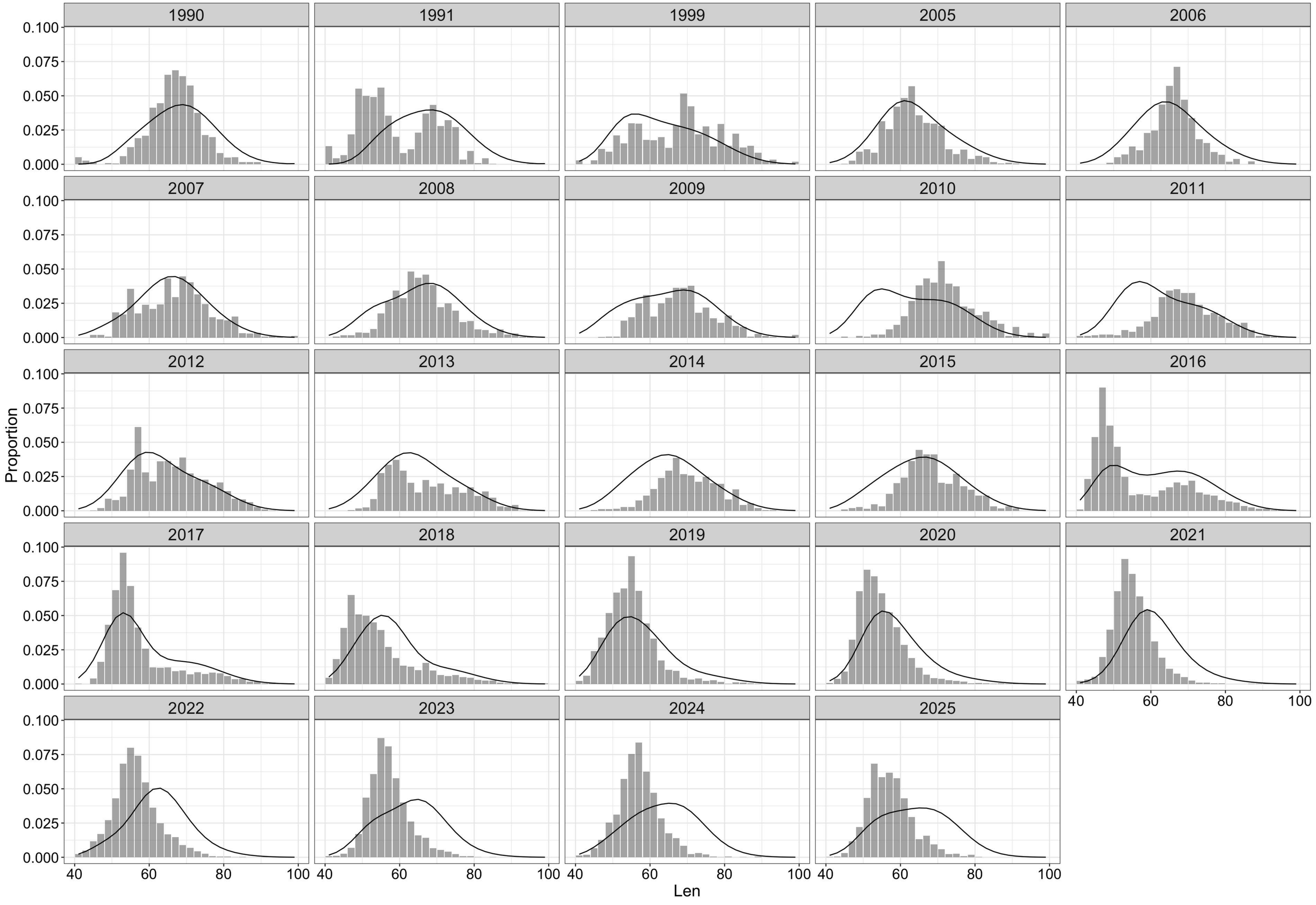

Model
26.2_FAA_LLS_BS_Dome_DN
AK
Dynamic SSB0
Unfished SSB (kt)
400
300
200
100
0
1960
1980
2000
2020
Year


Model — 26.2_FAA_LLS_BS_Dome_DN
Type — Population 1 Dynamic SSB0 -- Population 1 SSB
AK
400
300
200
100
0
SSB (kt)
1960
1980
2000
2020
Year

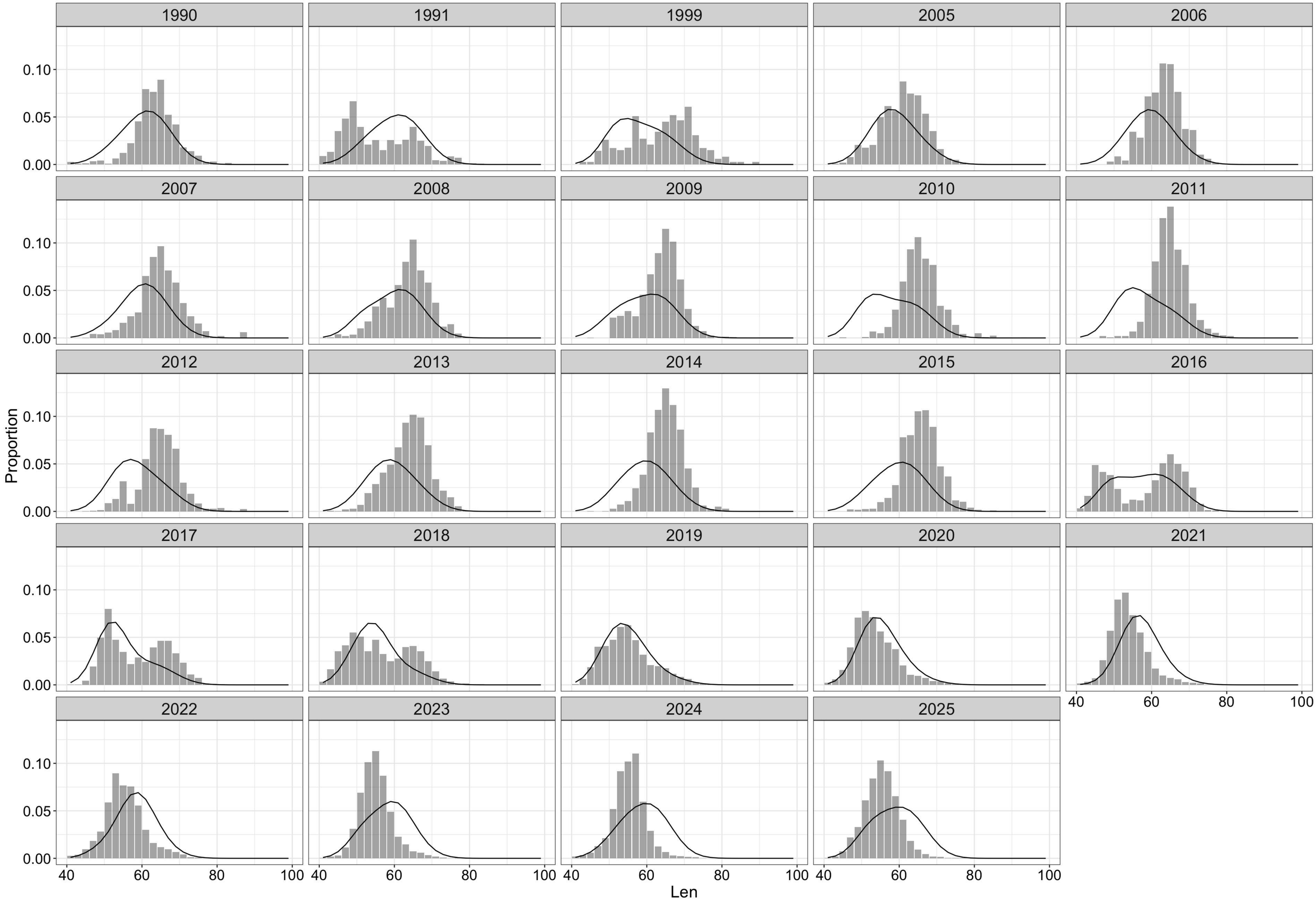

Model
26.2_FAA_LLS_BS_Dome_DN
AK
SSB
SSB (kt)
300
200
100
0
1960
1980
2000
2020
Year


Model
26.2_FAA_LLS_BS_Dome_DN
AK
Total Biom
Biomass (kt)
600
400
200
0
1960
1980
2000
2020
Year

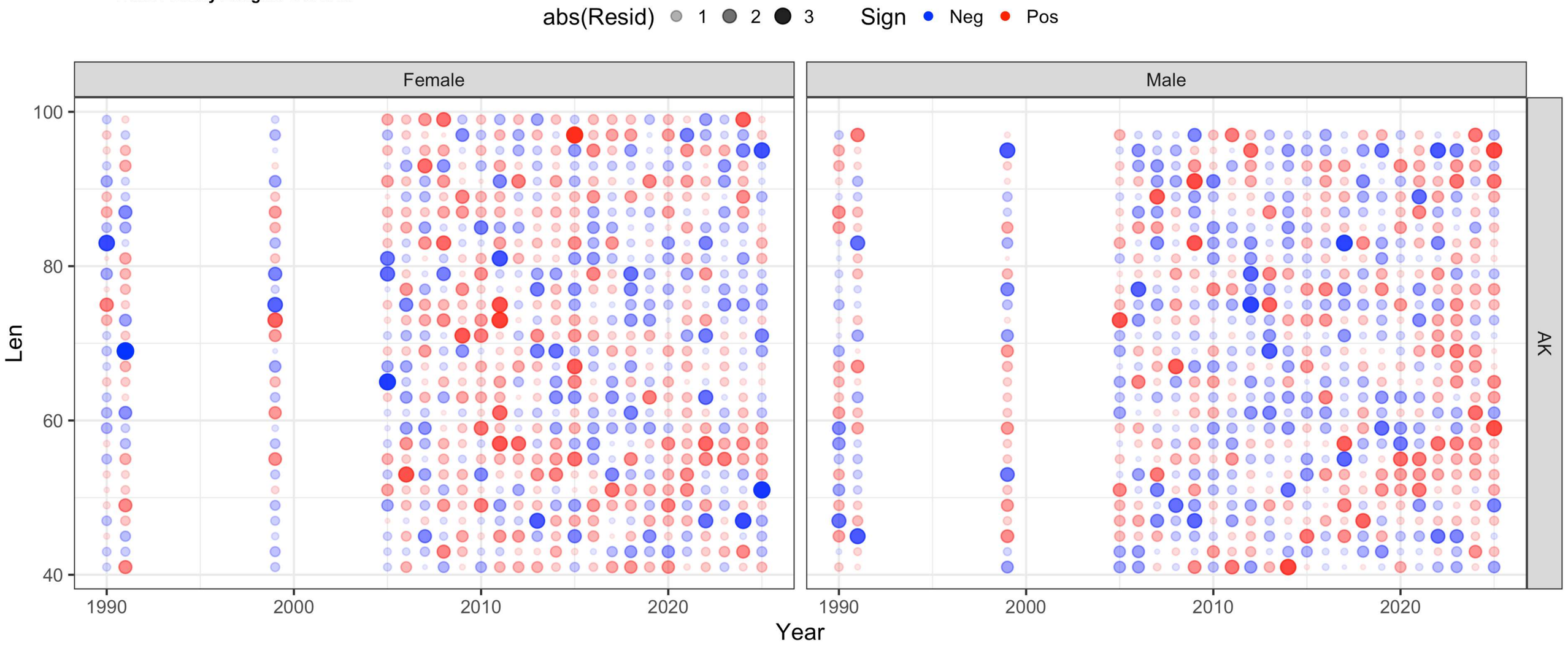
Trawl Fishery Lengths Female AK

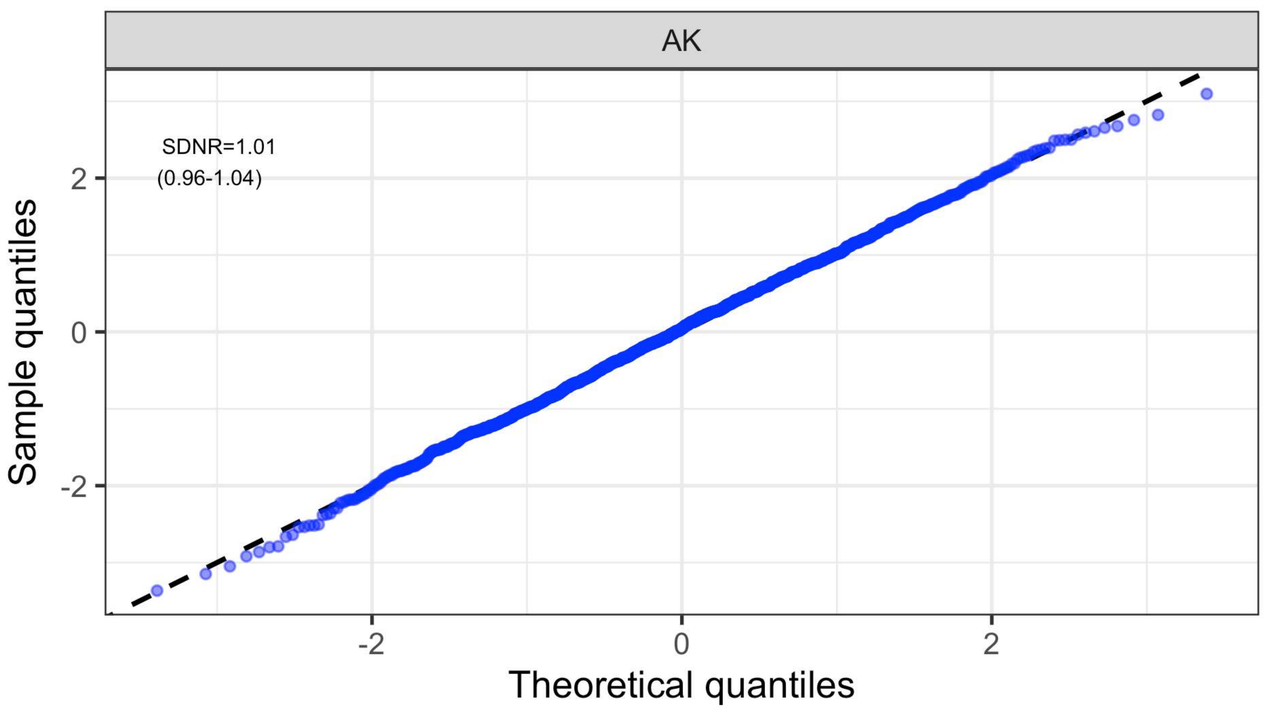


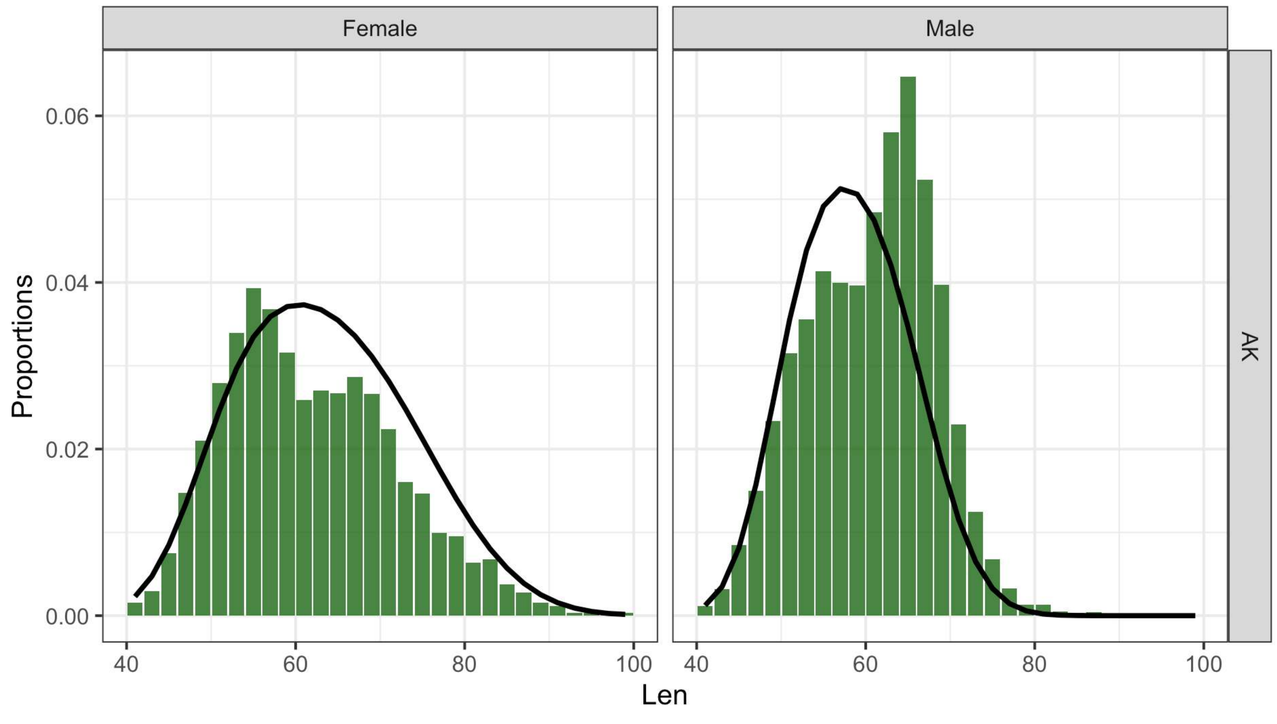
Trawl Fishery Lengths Male AK



Trawl Fishery Lengths OSA AK

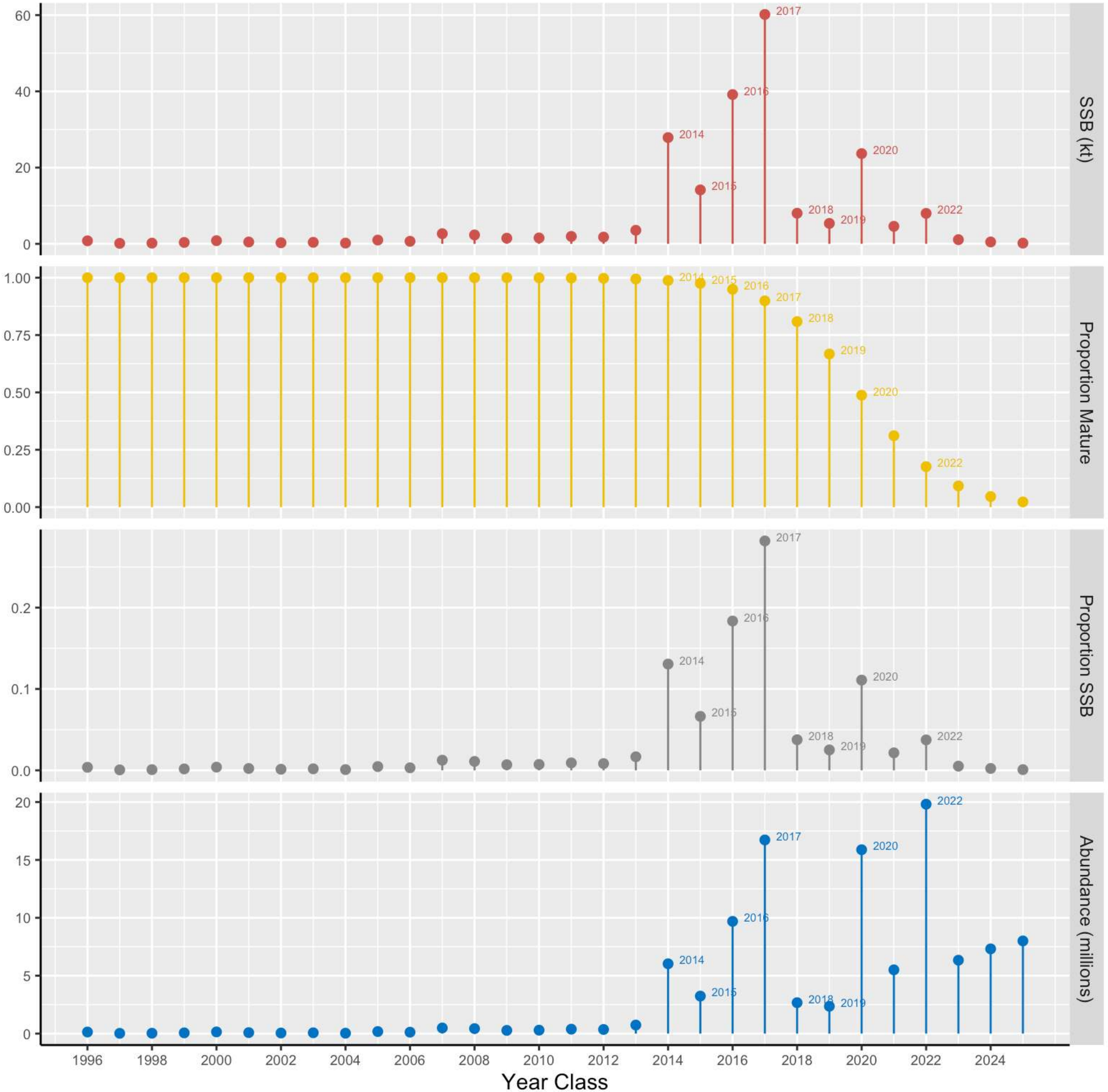






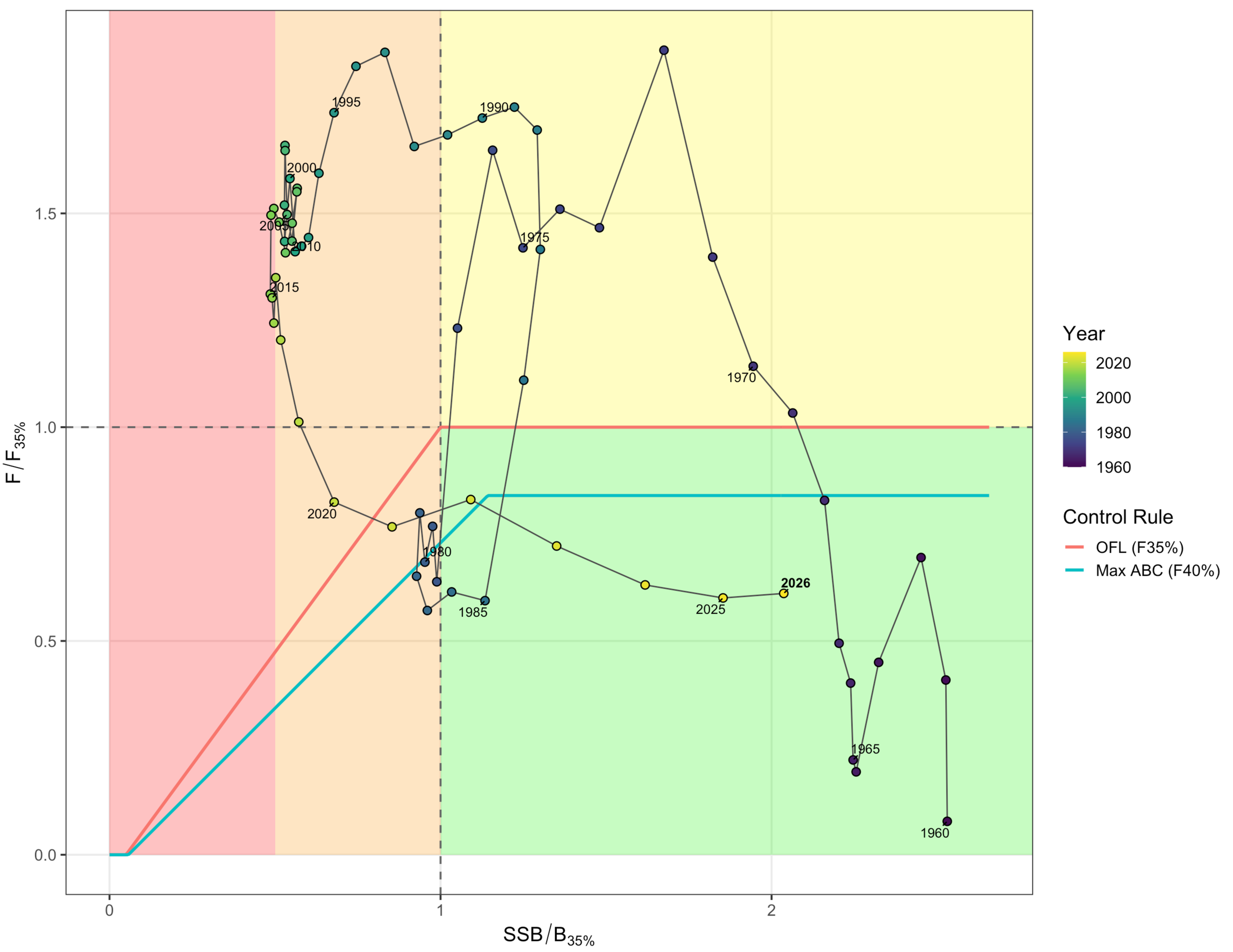
Trawl Fishery Length Comps Pearson residuals
|Pearson resid|
0.3
0.6
0.9
1.2
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1990
2000
2010
2020
Year

Contribution to 2027 SSB by Year Class



Source	SE	Data_Wt	ISS	ESS
BS LL Survey Index	0.20	1.00	NA	NA
AI LL Survey Index	0.23	1.00	NA	NA
GOA LL Survey Index	0.08	1.00	NA	NA
Trawl Survey Index	NA	NA	NA	NA
JPN LL Survey Index	0.07	1.00	NA	NA
Fixed Gear Fishery CPUE	NA	NA	NA	NA
BS LL Survey Ages	NA	0.31	34.07	10.42
AI LL Survey Ages	NA	2.18	35.57	77.68
GOA LL Survey Ages	NA	0.37	67.17	25.15
JPN LL Survey Ages	NA	0.19	83.33	15.76
JPN LL Survey Lengths	NA	0.33	54.62	17.87
Fixed Gear Fishery Ages	NA	0.04	142.85	6.42
Trawl Fishery Lengths	NA	0.10	47.62	4.54

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_
FAA_LLS_BS_Dome_DN	1	200.048	381.8275	0.05469878	35.34834	136.2785	0.07519	1.46794	0.52392	0.727


1995
1990
2000
2005
2010
2015
1975
1970
2020
1980
1985
2025
2026
1965
1960
1.5
1.0
0.5
0.0
0
1
2
F/F35%
SSB/B35%
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)

Quantity	2025	2026	2027	2028
M Female (estimated)	0.114	0.114	0.086	0.086
M Male (estimated)	0.114	0.114	0.09	0.09
Average Recruitment (millions)	25.9	25.9	18.9	18.9
Tier	3a	3a	3a	3a
Projected Total Biomass (Age-2+, t)	704,713	695,681	568274	569188
Projected Female Spawning Biomass (t)	219,714	241,217	213447	226763
B100% (t)	302,672	302,672	340696	340696
B40% (t)	121,069	121,069	136278	136278
B35% (t)	105,935	105,935	119244	119244
FOFL	0.102	0.102	0.089	0.089
max FABC	0.087	0.087	0.075	0.075
Total Depredation (t)	171	169	131	131
OFL (t)	58,731	57,993	41796	42192
OFLw (t)	58,532	57,797	41665	42061
max ABC (t)	50,283	49,651	35347	35685
ABC (t)	50,283	49,651	35347	35685
ABCw (t)	50,111	49,482	35216	35554
Status	2023	2024	2025	2026
Overfishing	No	NA	No	NA
Overfished	NA	No	NA	No
Approaching Overfished	NA	No	NA	No

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.00

0.05

0.10

0.15

1960

1980

2000

2020

2040

Year

S1: FABC (F40)
S1a: Author's F (FABC Ratio) Year 1, FABC After
S2: Author's F (FABC Ratio) Years 1+2
S3: 5-year Average F
S4: Half max FABC (1/2 * F40)
S5: No Fishing
S6: Overfished (FOFL)
S6a: Author's F (FABC Ratio) Year 1, FOFL After
S7: Approaching Overfishing (FABC -> FOFL)
Fishing Mortality
0.00
0.05
0.10
2010
2020
2030
2040
Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0
250
500
750

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Biomass (kt)

0

250

500

750

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0

20

40

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30
40

1960
1980
2000
2020
2040

Year

Fleet
LLF
TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30
40

2010
2020
2030
2040

Year

Fleet
LLF
TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30
40

2010
2020
2030
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0

25

50

75

100

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0

25

50

75

100

2010

2020

2030

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0

100

200

300

400

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0

100

200

300

400

2010

2020

2030

2040

Year

Year	Apportion_Type	BS	AI	WGOA	CGOA	WY	SEO
2026	Carryover_No_wAI	0.30	0.20	0.11	0.21	0.06	0.12
2026	REMA_LLS_No_wAI	0.28	0.23	0.10	0.21	0.06	0.12
2026	Carryover	0.27	0.26	0.10	0.20	0.06	0.11
2026	REMA_LLS	0.26	0.30	0.09	0.19	0.05	0.11
2026	GOA_Fix	0.30	0.24	0.09	0.20	0.05	0.12

AI
BS
CGOA
SEO
WGOA
WY
Apportionment Value
0.0
0.1
0.2
0.3
2014
2016
2018
2020
2022
2024
2026
Year
Apportion_Type
Carryover
Carryover_No_wAI
GOA_Fix
REMA_LLS
REMA_LLS_No_wAI

AI
BS
CGOA
SEO
WGOA
WY
Apportionment Value
0.0
0.1
0.2
0.3
2021
2022
2023
2024
2025
2026
Year
Apportion_Type
Carryover
Carryover_No_wAI
GOA_Fix
REMA_LLS
REMA_LLS_No_wAI

Year	Apportion_Type	BS	AI	GOA	WGOA	CGOA	WY	SEO	ABCw
2026	GOA_Fix	13037	11421	22550	4687	9622	2652	5589	47008
2027	Carryover_No_wAI	10565	7043	17608	3874	7395	2113	4226	35216
2027	REMA_LLS_No_wAI	9860	8100	17256	3522	7395	2113	4226	35216
2027	Carryover	9508	9156	16552	3522	7043	2113	3874	35216
2027	REMA_LLS	9156	10565	15495	3169	6691	1761	3874	35216
2027	GOA_Fix	10565	8452	16199	3169	7043	1761	4226	35216
2028	Carryover_No_wAI	10666	7111	17777	3911	7466	2133	4267	35554
2028	REMA_LLS_No_wAI	9955	8177	17421	3555	7466	2133	4268	35554
2028	Carryover	9600	9244	16710	3555	7111	2133	3911	35554
2028	REMA_LLS	9244	10666	15644	3200	6755	1778	3911	35554
2028	GOA_Fix	10666	8533	16355	3200	7111	1778	4266	35554

Year	Apportion_Type	BS	AI	GOA	WGOA	CGOA	WY	SEO	ABCw
2026	GOA_Fix	0	0	0	0	0	0	0	0
2027	Carryover_No_wAI	-19	-38	-22	-17	-23	-20	-24	-25
2027	REMA_LLS_No_wAI	-24	-29	-23	-25	-23	-20	-24	-25
2027	Carryover	-27	-20	-27	-25	-27	-20	-31	-25
2027	REMA_LLS	-30	-7	-31	-32	-30	-34	-31	-25
2027	GOA_Fix	-19	-26	-28	-32	-27	-34	-24	-25
2028	Carryover_No_wAI	-18	-38	-21	-17	-22	-20	-24	-24
2028	REMA_LLS_No_wAI	-24	-28	-23	-24	-22	-20	-24	-24
2028	Carryover	-26	-19	-26	-24	-26	-20	-30	-24
2028	REMA_LLS	-29	-7	-31	-32	-30	-33	-30	-24
2028	GOA_Fix	-18	-25	-27	-32	-26	-33	-24	-24

Year	Region	OFLw	ABCw	TAC	Catch	Year2	Region	OFLw	ABCw	Year3	Region	OFLw	ABCw
2026	BS	NA	13037	8996	3339	2027	BS	NA	10565	2028	BS	NA	10666
2026	AI	NA	11421	7440	632	2027	AI	NA	7043	2028	AI	NA	7111
2026	GOA	NA	22550	22550	18191	2027	GOA	NA	17608	2028	GOA	NA	17777
2026	WGOA	NA	4687	4687	3458	2027	WGOA	NA	3874	2028	WGOA	NA	3911
2026	CGOA	NA	9622	9622	7597	2027	CGOA	NA	7395	2028	CGOA	NA	7466
2026	WY	NA	2652	2652	2271	2027	WY	NA	2113	2028	WY	NA	2133
2026	SEO	NA	5589	5589	4865	2027	SEO	NA	4226	2028	SEO	NA	4267
2026	Total	57797	47008	38986	22161	2027	Total	41665	35216	2028	Total	42061	35554

Region	Year	Biomass (4+)	OFLw	ABCw	TAC	Catch
GOA	2025	305000	NA	24038	22836	18387
GOA	2026	299000	NA	23737	22550	18191
GOA	2027	245363	NA	17608	NA	NA
GOA	2028	244047	NA	17777	NA	NA
BS	2025	175000	NA	13898	8496	3909
BS	2026	172000	NA	13723	8996	3339
BS	2027	134356	NA	10565	NA	NA
BS	2028	133635	NA	10666	NA	NA
AI	2025	153000	NA	12175	7940	948
AI	2026	151000	NA	12022	7440	632
AI	2027	147914	NA	7043	NA	NA
AI	2028	147121	NA	7111	NA	NA